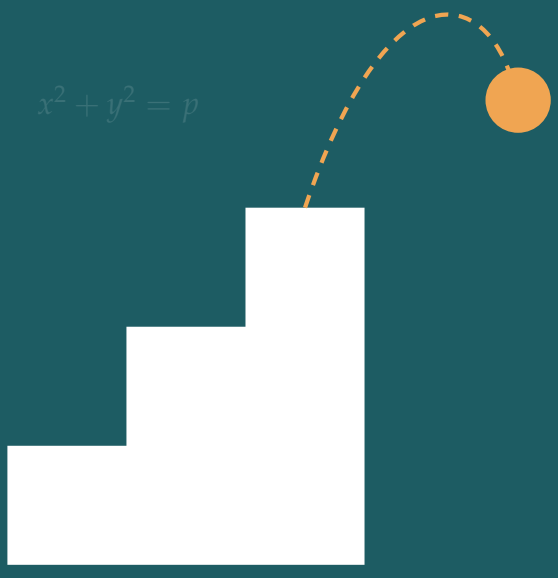
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1992 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1992 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1992 Worked Solutions

Worked solutions to all 48 questions of the 1992 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

Contents

STEP I

Q1 Calendar Arithmetic Modulo 7 Pure . . . 3
Q2 Magic Squares and Counting Pure . . . 6
Q3 Odd and Even Functions Pure . . . 10
Q4 Loci in the Argand Diagram Pure . . . 13
Q5 Repeated Polynomial Factors Pure . . . 17
Q6 Small Angle Approximations Pure . . . 21
Q7 Integer-Valued Polynomials Pure . . . 24
Q8 Fundamental Theorem of Calculus Pure . . . 27
Q9 Differential Equations Pure . . . 30
Q10 Recoil and Momentum Mechanics . . . 33
Q11 Elastic Strings Equilibrium Mechanics . . . 36
Q12 Statics with Friction Mechanics . . . 40
Q13 Escape Speed and Radial Fall Mechanics . . . 44
Q14 Interpreting Statistical Claims Probability and Statistics . . . 47
Q15 Expected Waiting Time Probability and Statistics . . . 51
Q16 Network Reliability Probability and Statistics . . . 55

STEP II

Q1 Limits of Sequences Pure . . . 59
Q2 Clairaut Differential Equation Pure . . . 62
Q3 Hypocycloids and Area Pure . . . 66
Q4 Axioms for Differentiation Pure . . . 69
Q5 Matrix Groups Pure . . . 72
Q6 Tangents and Root Counting Pure . . . 75
Q7 Repeated Roots of Cubics Pure . . . 79
Q8 Techniques of Integration Pure . . . 82
Q9 Scalar Triple Product Pure . . . 85
Q10 Loci in the Argand Diagram Pure . . . 89
Q11 Projectile Motion Mechanics . . . 93

Q12 Moments and Elastic Strings Mechanics . . . 96
Q13 Friction and Connected Particles Mechanics . . . 100
Q14 Connected Particles and Pulleys Mechanics . . . 103
Q15 Random Walks Probability and Statistics . . . 107
Q16 Conditional Probability Probability and Statistics . . . 111

STEP III

Q1 Maclaurin Series Pure . . . 114
Q2 Matrix Powers and Induction Pure . . . 117
Q3 Curves and Polynomial Roots Pure . . . 120
Q4 Orthogonal Trajectories Pure . . . 124
Q5 Vector Geometry in 3D Pure . . . 128
Q6 Trigonometric Reduction Formulae Pure . . . 132
Q7 Vector Geometry Pure . . . 135
Q8 Trigonometric Sums via Roots Pure . . . 138
Q9 Polar Equation of Ellipse Pure . . . 141
Q10 Arc Length and Area Pure . . . 145
Q11 Collisions and Restitution Mechanics . . . 149
Q12 Conservation of Momentum Mechanics . . . 152
Q13 Rotation About a Fixed Axis Mechanics . . . 156
Q14 Unwinding String Dynamics Mechanics . . . 160
Q15 Geometric Probability Probability and Statistics . . . 164
Q16 Poisson Thinning Probability and Statistics . . . 168

WORKED SOLUTION

STEP I 1992, Question 1

Pure • Calendar Arithmetic Modulo 7

STEP I 1992, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q1

The key idea

Days of the week repeat every 7 days, so all we need is the number of days between the two dates, reduced modulo 7. The trick that makes this painless is that $365 = 52 \times 7 + 1$: an ordinary year shifts the day of the week by one, and a leap year by two. So we count whole years and leap days first, then deal with the odd months at the end. "Carefully" means being precise about which February 29ths actually fall inside the interval, and about the one century year that might trip us up.

Solution

We split the interval from 3 April 1905 to 26 June 1992 into two pieces:

$$3 \text{ April } 1905 \longrightarrow 3 \text{ April } 1992 \longrightarrow 26 \text{ June } 1992.$$

Whole years: 3 April 1905 to 3 April 1992

This is $1992 - 1905 = 87$ whole years. Each such year (from 3 April to the next 3 April) contains 365 days, plus one extra day if a 29 February falls inside it.

Which 29 Februaries lie between 3 April 1905 and 3 April 1992? The leap years in this range are those divisible by 4, namely

$$1908,\ 1912,\ \ldots,\ 1988,\ 1992.$$

The 29 February of 1904 came before 3 April 1905, so it is not counted; the 29 February of 1992 comes before 3 April 1992, so it is counted. That gives

$$\frac{1992 - 1908}{4} + 1 = 21 + 1 = 22$$

leap days.

Ask yourself Is every year divisible by 4 in this range really a leap year?

In the Gregorian calendar, century years are leap years only if they are divisible by 400, so 1900 was *not* a leap year (and 2000 was). But neither of these lies in our interval, so the simple "divisible by 4" rule (the one in the rhyme) is exactly right here. Had the question gone back before 1900, we would have had to drop 29 February 1900.

So the number of days from 3 April 1905 to 3 April 1992 is

$$87 \times 365 + 22 = 31\,755 + 22 = 31\,777.$$

Working modulo 7 is quicker: since $365 \equiv 1 \pmod 7$, this is

$$87 + 22 = 109 = 15 \times 7 + 4 \equiv 4 \pmod 7.$$

The odd months: 3 April 1992 to 26 June 1992

Using the rhyme, April has 30 days and May has 31, so

$$\begin{aligned} 3\text{ April} &\to 3\text{ May} : 30\text{ days},\\ 3\text{ May} &\to 3\text{ June} : 31\text{ days},\\ 3\text{ June} &\to 26\text{ June} : 23\text{ days},\end{aligned}$$

a total of $30 + 31 + 23 = 84 = 12 \times 7$ days, which is $\equiv 0 \pmod 7$. Pleasingly, 3 April 1992 and 26 June 1992 fall on the same day of the week.

Putting it together

The total gap is

$$31\,777 + 84 = 31\,861 = 4551 \times 7 + 4 \text{ days},$$

so 3 April 1905 is 4 days of the week *earlier* than 26 June 1992. Counting back four days from Friday: Thursday, Wednesday, Tuesday, Monday. Hence

$$3\text{ April 1905 was a Monday.}$$

Checks

- A computer counted the days between the two dates directly, month by month, and found 31 861 days, which is $4 \pmod 7$; a computer calendar also gives Monday for 3 April 1905 (and Friday for 26 June 1992).
- Sanity check on direction: going *forward* 4 days from Monday gives Friday, as it should.
- Check the leap-day count another way: 87 years contain roughly $87/4 \approx 21.75$ leap days, so 22 is about right.

Common slips

- Including 29 February 1904 (which is before the start date) or excluding 29 February 1992 (which is before the end date): each gives an answer one day out.
- Counting back instead of forward, or vice versa: the remainder 4 means 1905 is four days *before* Friday in the weekly cycle, not after.
- Treating 1900 as a leap year when extending the method to earlier dates.

Beyond STEP

This is arithmetic modulo 7: the key fact $365 \equiv 1 \pmod 7$ turns a large count into a small one. Congruences, and their use in calendar puzzles such as Zeller's congruence for the day of the week of any date, are a standard part of a first course in number theory. Davenport's *The Higher Arithmetic* is a lovely introduction to congruences and much more.

WORKED SOLUTION

STEP I 1992, Question 2

Pure • Magic Squares and Counting

STEP I 1992, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

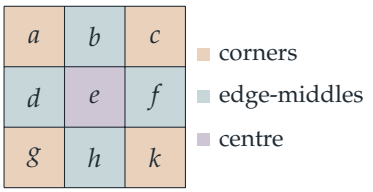
step.maths.org/questions/92-s1-q2

The key idea

Everything comes from adding up lines cleverly. Adding all three rows uses each number exactly once, which gives n. Adding the four lines through the centre uses every number once and the centre four times, which pins down e. After that the question becomes a small, careful case analysis: once e is known, each line through the centre pairs up two numbers with a fixed sum, and that leaves very little room for the large numbers. Symmetry of the square (rotating it) then does the counting for us.

Solution

It helps to name the three kinds of cell: the four *corners* a, c, g, k, the four *edge-middles* b, d, f, h, and the *centre* e.



Part (i)

The three rows together contain each of the nine entries exactly once. So, adding the three row sums,

$$3n = (a+b+c)+(d+e+f)+(g+h+k) = 1+2+\cdots+9 = 45,$$

and hence $n = 15$.

Part (ii)

Four lines pass through the centre: the middle row, the middle column and the two diagonals. Adding them,

$$(d+e+f)+(b+e+h)+(a+e+k)+(c+e+g)=4n=60.$$

On the left, every entry other than e appears exactly once and e appears four times, so the left-hand side is $45+3e$. Therefore $45+3e=60$, giving $e=5$.

A useful consequence: since each line through the centre sums to 15 and $e=5$, the two cells *opposite* each other through the centre always sum to 10:

$$a+k=c+g=b+h=d+f=10.$$

Part (iii)

The entry 9 is not the centre, because $e=5$. So we must show that 9 cannot be in a corner.

Suppose, for contradiction, that 9 is in a corner. The argument is the same for every corner, so take $a=9$. Then the opposite corner is $k=10-9=1$. The top row and the left column both pass through a, so

$$b+c=15-9=6 \qquad \text{and} \qquad d+g=15-9=6.$$

The four numbers b,c,d,g are distinct and are drawn from what is left once $1,5,9$ are used, namely $\{2,3,4,6,7,8\}$. The only way to write 6 as a sum of two *different* numbers from this set is $2+4$ (since $1+5$ uses numbers already placed and $3+3$ repeats). So $\{b,c\}=\{2,4\}$ and $\{d,g\}=\{2,4\}$, which needs the numbers 2 and 4 twice each. That contradicts distinctness.

? **Ask yourself** Why does it suffice to deal with the corner a alone?

For any corner, the same three facts hold: the opposite corner must be 1, and the two lines along the edges of the square through that corner each need their other two entries to sum to 6. So the same contradiction appears. (Equivalently, rotating a magic square through a quarter turn gives another magic square, and rotations carry a to each of the other corners.)

So 9 is neither the centre nor a corner, and it must be one of b,d,f,h.

Part (iv)

Let $b=9$. Then $h=10-9=1$, and the top row gives $a+c=6$. As in (iii), the unused numbers are $\{2,3,4,6,7,8\}$, and the only pair of distinct numbers from them with sum 6 is $\{2,4\}$. So either $a=2,\ c=4$ or $a=4,\ c=2$.

Once a and c are fixed, everything else is forced by opposite pairs and the side columns:

- $a=2,\ c=4$: then $k=10-a=8$, $g=10-c=6$, $d=15-a-g=7$ and $f=15-c-k=3$.

- $a = 4$, $c = 2$: then $k = 6$, $g = 8$, $d = 15 - 4 - 8 = 3$ and $f = 15 - 2 - 6 = 7$.

Forcing values only shows these are the *only* candidates; we must still check that they work. In each case the entries are $1, \ldots, 9$ once each, and every row, column and diagonal sums to 15:

2	9	4
7	5	3
6	1	8

4	9	2
3	5	7
8	1	6

(For example, in the first: rows $15, 15, 15$; columns $2 + 7 + 6$, $9 + 5 + 1$, $4 + 3 + 8$; diagonals $2 + 5 + 8$, $4 + 5 + 6$.) So there are exactly two magic squares with $b = 9$, and the second is the mirror image of the first in the middle column.

Part (v)

Turning a magic square through a quarter turn clockwise gives another magic square: its rows are the old columns, its columns are the old rows (read in reverse), and the two diagonals swap. The quarter turn is reversible (turn back anticlockwise), so it gives a one-to-one correspondence between magic squares.

A quarter turn clockwise moves the cell b to the cell f. So the magic squares with $f = 9$ are exactly the quarter-turns of the magic squares with $b = 9$, and there are exactly 2 of them. Turning again, there are exactly 2 with $h = 9$ and exactly 2 with $d = 9$.

By (iii), every magic square has 9 in exactly one of the positions b, f, h, d, so these four groups are disjoint and cover everything. Hence there are

$$4 \times 2 = 8$$

different 3×3 magic squares. They are the two squares in (iv) together with their rotations through a quarter, half and three-quarter turn; equivalently, they are the eight rotations and reflections of a single square.

Checks

A short computer search over all $9! = 362\,880$ arrangements of $1, \ldots, 9$ finds exactly 8 magic squares. All of them have line sum 15 and centre 5, all have 9 in an edge-middle, and the two with $b = 9$ are the squares found in (iv). By hand, you can check that in every square 1 sits opposite 9 and the even numbers $2, 4, 6, 8$ fill the corners.

Common slips

- In (iv), forgetting to check that the forced squares really are magic. The deduction only shows that nothing else can work.
- In (v), answering "2" or "4" without explaining why rotation produces new, genuinely different squares, or without using (iii) to see that the four cases cover everything without overlap.

Beyond STEP

Part (v) is a first taste of *group actions*: the eight symmetries of the square (four rotations and four reflections) form a group, and the magic squares make up a single orbit under it. Counting objects up to symmetry like this is developed in Burnside's lemma, which you meet in a first course in group theory. Allenby's *Rings, Fields and Groups* is a friendly place to start.

WORKED SOLUTION

STEP I 1992, Question 3

Pure • Odd and Even Functions

STEP I 1992, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q3

The key idea

Every integral here is over the symmetric interval $[-\pi, \pi]$, so the first thing to ask of each integrand is whether it is even or odd. An odd function integrates to zero over a symmetric interval, while an even one gives twice the integral over $[0, \pi]$, where the modulus signs disappear. Spotting the symmetry turns all four parts into at most one short calculation each.

Solution

We use two standard facts. If $f(-x) = f(x)$ (even) then

$$\int_{-a}^{a} f(x)\,\mathrm{d}x = 2\int_{0}^{a} f(x)\,\mathrm{d}x,$$

and if $f(-x) = -f(x)$ (odd) then $\int_{-a}^{a} f(x)\,\mathrm{d}x = 0$. Both follow by substituting $x = -u$ in the integral over $[-a, 0]$, which turns it into $\int_{0}^{a} f(-u)\,\mathrm{d}u$.

We also need $\int_{0}^{\pi} \sin x\,\mathrm{d}x = \left[-\cos x\right]_{0}^{\pi} = 1 - (-1) = 2$.

Part (i)

Since $|\sin(-x)| = |-\sin x| = |\sin x|$, the integrand is even. On $[0, \pi]$ we have $\sin x \geq 0$, so $|\sin x| = \sin x$ there. Hence

$$\int_{-\pi}^{\pi} |\sin x|\,\mathrm{d}x = 2\int_{0}^{\pi} \sin x\,\mathrm{d}x = 2 \times 2 = 4.$$

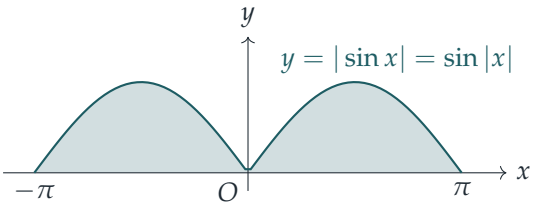
Part (ii)

Since $|-x| = |x|$, the function $\sin|x|$ is even. On $[0, \pi]$, $|x| = x$, so

$$\int_{-\pi}^{\pi} \sin|x|\,\mathrm{d}x = 2\int_{0}^{\pi} \sin x\,\mathrm{d}x = 4.$$

Ask yourself Is it a coincidence that (i) and (ii) give the same answer?

No: on this interval the two integrands are the same function. For $0 \leq x \leq \pi$ both equal $\sin x$. For $-\pi \leq x \leq 0$ we have $\sin x \leq 0$, so $|\sin x| = -\sin x$, while $\sin|x| = \sin(-x) = -\sin x$ as well. The two curves only part company outside $[-\pi, \pi]$: for example, at $x = 3\pi/2$, $|\sin x| = 1$ but $\sin|x| = -1$.



The shaded region consists of two arches of the sine curve, each of area 2.

Part (iii)

Here $(-x)\sin(-x) = (-x)(-\sin x) = x\sin x$, so the integrand is even (a product of two odd functions). Therefore

$$\int_{-\pi}^{\pi} x\sin x\,\mathrm{d}x = 2\int_{0}^{\pi} x\sin x\,\mathrm{d}x.$$

Integrating by parts with $u = x$ and $\mathrm{d}v = \sin x\,\mathrm{d}x$ (so $v = -\cos x$):

$$\begin{aligned}\int_{0}^{\pi} x\sin x\,\mathrm{d}x &= \left[-x\cos x\right]_0^{\pi} + \int_{0}^{\pi}\cos x\,\mathrm{d}x \\ &= \left(-\pi\cos\pi - 0\right) + \left[\sin x\right]_0^{\pi} \\ &= \pi + 0 = \pi.\end{aligned}$$

Hence

$$\int_{-\pi}^{\pi} x\sin x\,\mathrm{d}x = 2\pi.$$

Part (iv)

Now $(-x)^{10}\sin(-x) = x^{10}(-\sin x) = -x^{10}\sin x$, so the integrand is odd (an even function times an odd one). The integral over the symmetric interval is therefore

$$\int_{-\pi}^{\pi} x^{10}\sin x\,\mathrm{d}x = 0.$$

There is no need to integrate by parts ten times! The contributions from $[-\pi, 0]$ and $[0, \pi]$ are equal in size and opposite in sign, and they cancel exactly.

Checks

- Area sense: in (i) the integrand is non-negative and at most 1, so the answer must lie between 0 and $2\pi \approx 6.28$; 4 does.
- In (iii), $x \sin x \geq 0$ on $[-\pi, \pi]$, so the answer must be positive; 2π is. Differentiating $\sin x - x\cos x$ gives $x \sin x$, which confirms the antiderivative.
- Simpson's rule with 20 000 strips gives 4.0000000, 4.0000000, 6.2831853 and 0.0000000 for the four integrals. In (iv) the zero is not just a small number: the integral of $|x^{10} \sin x|$ over the same interval is about 13 275, so the cancellation is exact and complete.

Common slips

- Writing $\int_{-\pi}^{\pi} |\sin x| \, \mathrm{d}x = \big[-\cos x\big]_{-\pi}^{\pi} = 0$, ignoring the modulus. An integrand that is never negative cannot have integral 0 unless it is zero almost everywhere.
- Assuming $\sin|x|$ is odd because $\sin x$ is. Composing with $|x|$ always produces an even function.
- Sign errors in the integration by parts in (iii): remember $\cos\pi = -1$, so $-\pi\cos\pi = +\pi$.

Beyond STEP

Splitting a function into even and odd parts, $f(x) = \frac{1}{2}\big(f(x) + f(-x)\big) + \frac{1}{2}\big(f(x) - f(-x)\big)$, is the starting point of Fourier series. On $[-\pi, \pi]$, the even part is expanded in cosines and the odd part in sines, and integrals like (iii) and (iv) are exactly the ones used to calculate the coefficients. For example, the coefficient of $\sin x$ in the Fourier series of x comes from the integral in (iii). Boyce and DiPrima, *Elementary Differential Equations*, has a readable introduction to Fourier series and their use in solving differential equations.

WORKED SOLUTION

STEP I 1992, Question 4

Pure • Loci in the Argand Diagram

STEP I 1992, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q4

The key idea

Turn each condition into a statement about x and y (with $z = x + \mathrm{i}y$), or about the modulus and argument of z, and the region usually becomes obvious. Parts (i) and (ii) need nothing more than this. Part (iii) is different: it describes a set of *sums*, so we have to show two things. Every sum lies in the region we claim, and every point of that region really is such a sum.

Solution

Throughout, $z = x + \mathrm{i}y$ with x, y real.

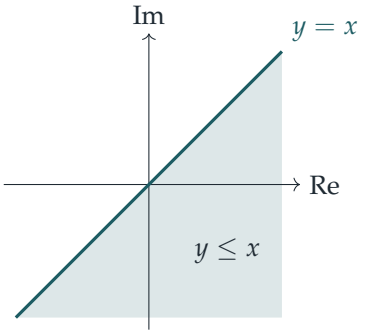
Part (i)

Expanding,

$$(1+\mathrm{i})z = (1+\mathrm{i})(x+\mathrm{i}y) = (x-y) + \mathrm{i}(x+y),$$

so $\mathrm{Re}\big((1+\mathrm{i})z\big) = x - y$. The condition is $x - y \geq 0$, that is $y \leq x$. The set is the closed half-plane on and below the line $y = x$, including the line itself.

Geometrically, multiplying by $1 + \mathrm{i} = \sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$ rotates by $\frac{\pi}{4}$ and scales by $\sqrt{2}$. So the condition says that z, rotated anticlockwise through $\frac{\pi}{4}$, lands in the right half-plane $\mathrm{Re} \geq 0$. That means $-\frac{3\pi}{4} \leq \arg z \leq \frac{\pi}{4}$ (or $z = 0$), which is the same half-plane.



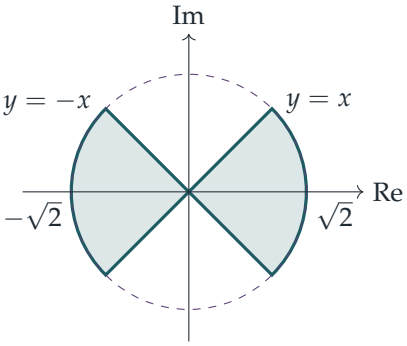
Part (ii)

Since $|z^2| = |z|^2$, the first condition is $|z|^2 \leq 2$, that is $|z| \leq \sqrt{2}$: the closed disc of radius $\sqrt{2}$ centred at the origin.

For the second, $z^2 = (x^2 - y^2) + 2ixy$, so $\mathrm{Re}(z^2) = x^2 - y^2$ and the condition is $x^2 \geq y^2$, that is $|y| \leq |x|$. This is the pair of opposite closed sectors between the lines $y = x$ and $y = -x$ that contain the real axis.

In polar form the same thing is quicker: if $z = re^{i\theta}$ then $z^2 = r^2 e^{2i\theta}$, so we need $r \leq \sqrt{2}$ and $\cos 2\theta \geq 0$, which gives $-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}$ or $\frac{3\pi}{4} \leq \theta \leq \frac{5\pi}{4}$ (or $z = 0$).

The set is the intersection: two closed quarter-discs (each a sector of angle $\frac{\pi}{2}$ and radius $\sqrt{2}$), one opening to the right and one to the left, meeting at the origin. The boundary is included throughout.



Part (iii)

Let S be the set of sums. We claim S is the closed annulus $\{z : 1 \leq |z| \leq 3\}$.

Every sum lies in the annulus. If $|z_1| = 2$ and $|z_2| = 1$, the triangle inequality gives

$$|z_1 + z_2| \leq |z_1| + |z_2| = 3, \qquad |z_1 + z_2| \geq |z_1| - |z_2| = 1.$$

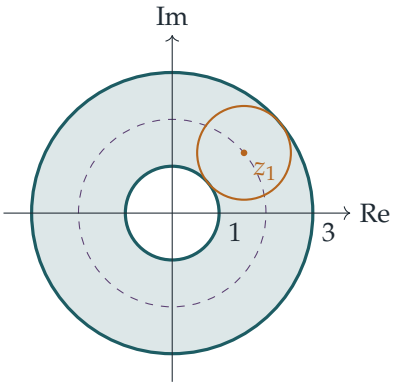
Ask yourself Is that enough? How do we know the whole annulus is filled, and not just some curve inside it?

The inequalities only show that S lies *inside* the annulus, so we still need the reverse inclusion. Take any z with $1 \le |z| \le 3$. We want a unit complex number z_2 with $|z - z_2| = 2$, because then $z_1 = z - z_2$ has modulus 2 and $z = z_1 + z_2$.

Geometrically, z_2 must lie on the unit circle centred at 0 and also on the circle of radius 2 centred at z. Two circles with radii 1 and 2 whose centres are a distance $d = |z|$ apart meet exactly when $2 - 1 \le d \le 2 + 1$, and that is our condition $1 \le |z| \le 3$.

To see this without quoting the circle fact, write $z = re^{i\alpha}$ and consider $f(\varphi) = |z - e^{i\varphi}|$, which is continuous in φ. At $\varphi = \alpha$ the point $e^{i\varphi}$ points the same way as z, so $f(\alpha) = |r - 1| = r - 1 \le 2$. At $\varphi = \alpha + \pi$ it points the opposite way, so $f(\alpha + \pi) = r + 1 \ge 2$. By the intermediate value theorem $f(\varphi) = 2$ for some φ, and $z_2 = e^{i\varphi}$ works.

So S is exactly the closed annulus between the circles $|z| = 1$ and $|z| = 3$, with both boundary circles included. A good way to picture it: fix z_1 on the circle of radius 2, and $z_1 + z_2$ traces a unit circle centred at z_1. As z_1 goes round, these little circles sweep out the annulus.



The amber circle shows the sums for one fixed z_1 on the dashed circle $|z_1| = 2$.

Checks

- In (i), $z = 1$ gives $\mathrm{Re}(1 + i) = 1 \ge 0$, and $z = i$ gives $\mathrm{Re}(-1 + i) = -1 < 0$. This agrees with $y \le x$.
- In (ii), $z = 1$ is in the set and $z = i$ is not ($i^2 = -1$), so the sectors should contain the real axis, as drawn.
- In (iii), $z_1 = 2$, $z_2 = -1$ gives 1 and $z_1 = 2$, $z_2 = 1$ gives 3, so both boundary circles are reached. We tested all three descriptions on 200 000 random points. Random sums had moduli between 1.0000 and 3.0000, and for 1000 random points of the annulus a suitable z_2 was found to within 2×10^{-4}.

Common slips

- In (ii), keeping only the sector $|\arg z| \leq \frac{\pi}{4}$ and forgetting the left-hand one. Both z and $-z$ have the same square.
- In (iii), stopping after the triangle inequality. That shows only that S lies inside the annulus, not that S is the whole annulus.
- Leaving out whether boundaries are included. Every inequality here is non-strict, so every boundary is in.

Beyond STEP

Part (iii) is a small instance of the *Minkowski sum* $A + B = \{a + b : a \in A,\ b \in B\}$ of two sets, an idea that runs through geometry and convex analysis. The two-inclusion argument (show $S \subseteq T$, then $T \subseteq S$) is the standard way to prove that two sets are equal, and you will use it constantly at university. For a geometric view of complex multiplication as rotation and scaling, which makes (i) and (ii) almost instant, see Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP I 1992, Question 5

Pure • Repeated Polynomial Factors

STEP I 1992, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q5

This question has a full guided treatment in *Beyond STEP*, Section 9.7, *Equal Sums of Powers*: https://beyondstep.keithmcnulty.org/ch09/equal-power-sums.html

The key idea

If a polynomial has a repeated factor $(1-x)^m$, then differentiating it (and multiplying by x) leaves a factor $(1-x)^{m-1}$. So 1 is a root of the polynomial and of the next few in the chain. Evaluating each of these polynomials at $x=1$ using its coefficients gives a sum of the form $\sum k^r c_k$. When every coefficient is ± 1, each of these sums says that two sets of integers have equal sums of rth powers.

Solution

The factor $(1-x)^3$

Using the difference of two squares twice,

$$1-x^2=(1-x)(1+x), \qquad 1-x^4=(1-x^2)(1+x^2)=(1-x)(1+x)(1+x^2),$$

so

$$\mathrm{p}_0(x)=(1-x)^3(1+x)^2(1+x^2),$$

and $(1-x)^3$ is a factor of $\mathrm{p}_0(x)$, as required.

$\mathrm{p}_0'(1)=0$ and $\mathrm{p}_1'(1)=0$

We prove one small fact and use it repeatedly.

Lemma. If $\mathrm{p}(x)=(1-x)^m\mathrm{q}(x)$ with $m\geq 1$ and q a polynomial, then $\mathrm{p}'(x)$, and hence $x\mathrm{p}'(x)$, has the factor $(1-x)^{m-1}$.

Proof. By the product rule,

$$\mathrm{p}'(x)=-m(1-x)^{m-1}\mathrm{q}(x)+(1-x)^m\mathrm{q}'(x)=(1-x)^{m-1}\big[(1-x)\mathrm{q}'(x)-m\,\mathrm{q}(x)\big],$$

and the bracket is a polynomial. Multiplying by x doesn't remove the factor. □

Apply this with $m = 3$: $\mathrm{p}_0'(x)$ has the factor $(1-x)^2$, so $\mathrm{p}_0'(1) = 0$. Also $\mathrm{p}_1(x) = x\mathrm{p}_0'(x)$ has the factor $(1-x)^2$. Applying the lemma again with $m = 2$, $\mathrm{p}_1'(x)$ has the factor $1-x$, so $\mathrm{p}_1'(1) = 0$.

Ask yourself Why not just differentiate the product of three brackets and substitute $x = 1$?

We could, but the question asks us to argue from factors, and the factor argument is the one that generalises. A root of multiplicity m survives $m-1$ differentiations, and that is exactly what the final part needs.

The two identities

Multiplying out,

$$(1-x)(1-x^2) = 1 - x - x^2 + x^3,$$

and then

$$\mathrm{p}_0(x) = 1 - x - x^2 + x^3 - x^4 + x^5 + x^6 - x^7.$$

So $c_0 = c_3 = c_5 = c_6 = 1$ and $c_1 = c_2 = c_4 = c_7 = -1$.

Now

$$\mathrm{p}_0'(x) = \sum_{k=1}^{7} k\,c_k x^{k-1}, \qquad \mathrm{p}_1(x) = \sum_{k=0}^{7} k\,c_k x^k, \qquad \mathrm{p}_1'(x) = \sum_{k=1}^{7} k^2 c_k x^{k-1}.$$

Putting $x = 1$ and using $\mathrm{p}_0'(1) = 0$:

$$-1-2+3-4+5+6-7 = 0, \quad \text{that is} \quad 1+2+4+7 = 3+5+6.$$

Using $\mathrm{p}_1'(1) = 0$:

$$-1^2 - 2^2 + 3^2 - 4^2 + 5^2 + 6^2 - 7^2 = 0, \quad \text{that is} \quad 1^2 + 2^2 + 4^2 + 7^2 = 3^2 + 5^2 + 6^2.$$

(Both sides equal 14, and 70 for the squares.) The c_0 term contributes nothing, because it is multiplied by 0 or 0^2.

The final part

Go one step further. Let

$$\mathrm{P}_0(x) = (1-x)(1-x^2)(1-x^4)(1-x^8) = \mathrm{p}_0(x)(1-x^8).$$

Since $1 - x^8 = (1-x)(1+x)(1+x^2)(1+x^4)$, $\mathrm{P}_0(x)$ has the factor $(1-x)^4$. Define $\mathrm{P}_{j+1}(x) = x\mathrm{P}_j'(x)$ for $j = 0, 1, 2$. By the lemma, P_1, P_2 and P_3 have the factors $(1-x)^3$, $(1-x)^2$ and $(1-x)$ respectively, so

$$\mathrm{P}_1(1) = \mathrm{P}_2(1) = \mathrm{P}_3(1) = 0.$$

Write $\mathrm{P}_0(x) = \sum_{k=0}^{15} d_k x^k$. Each application of $x\frac{\mathrm{d}}{\mathrm{d}x}$ multiplies the coefficient of x^k by k, so $\mathrm{P}_r(x) = \sum_{k=0}^{15} k^r d_k x^k$, and

$$\sum_{k=1}^{15} k^r d_k = 0 \qquad \text{for } r = 1, 2, 3.$$

To find the d_k, multiply $\mathrm{p}_0(x)$ by $1 - x^8$. This gives

$$\mathrm{P}_0(x) = \mathrm{p}_0(x) - x^8 \mathrm{p}_0(x).$$

The second term gives $x^8, \ldots, x^{15}$ with the signs of p_0 reversed. There is no overlap, so every d_k is ± 1:

k	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
d_k	+	−	−	+	−	+	+	−	−	+	+	−	+	−	−	+

(In fact $d_k = (-1)^{s(k)}$, where $s(k)$ is the number of 1s in the binary expansion of k, because x^k arises by choosing $-x^{2^i}$ from the brackets matching the binary digits of k.)

Leaving out $k = 0$, the coefficient $d_k = -1$ for $k \in \{1, 2, 4, 7, 8, 11, 13, 14\}$ (eight numbers) and $d_k = +1$ for $k \in \{3, 5, 6, 9, 10, 12, 15\}$ (seven numbers). Together these are exactly $1, 2, \ldots, 15$. So take

$$(a_1, \ldots, a_8) = (1, 2, 4, 7, 8, 11, 13, 14), \qquad (a_9, \ldots, a_{15}) = (3, 5, 6, 9, 10, 12, 15).$$

Then $\sum k^r d_k = 0$ says precisely that

$$a_1^r + \cdots + a_8^r = a_9^r + \cdots + a_{15}^r \qquad (r = 1, 2, 3),$$

which is what we had to show.

> **Ask yourself** Why does the split have eight numbers on one side and seven on the other?

The $+1$ side also contains $k = 0$, which contributes $0^r = 0$ for $r \geq 1$ and so can be dropped. That is why the two sides don't need the same number of terms. (For $r = 0$ the identity would need the 0: then $8 = 8$.)

Checks

Direct computation: the sums are $60 = 60$ for $r = 1$, $620 = 620$ for $r = 2$ and $7200 = 7200$ for $r = 3$. A computer expanded P_0, confirmed its signs and the multiplicity 4 of the root $x = 1$, and verified all three equalities. For $r = 4$ the equality fails (the root has multiplicity only 4), which is a good sign the argument is sharp.

Common slips

- Forgetting the $k = 0$ term and so thinking the two sides must have equal size.
- Concluding $\mathrm{p}_1'(1) = 0$ from $\mathrm{p}_0''(1) = 0$ alone. Since $\mathrm{p}_1' = \mathrm{p}_0' + x\mathrm{p}_0''$, you need both $\mathrm{p}_0'(1) = 0$ and $\mathrm{p}_0''(1) = 0$.

Beyond STEP

This is the Prouhet–Tarry–Escott problem: splitting $\{0, 1, \ldots, 2^n - 1\}$ by the parity of the binary digit sum (the Thue–Morse sequence) gives two sets with equal sums of rth powers for all $r < n$. The tool behind it, that a root of multiplicity m is a root of the first $m - 1$ derivatives, is central to polynomial algebra over any field. For the algebra of polynomials and their roots, see Allenby, *Rings, Fields and Groups*. For number theory more generally, see Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP I 1992, Question 6

Pure • Small Angle Approximations

STEP I 1992, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q6

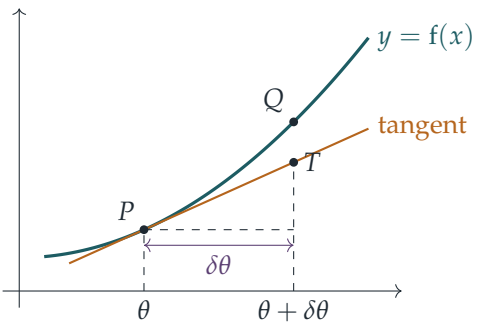
The key idea

The angle at the far point C is exactly ϕ, because $\theta + \phi$ is an exterior angle of triangle ABC. The sine rule then gives AC exactly. Since C is very far away, ϕ must be tiny, and the small-angle approximations (which are the first part's linear approximation in disguise) turn the exact formula into something simple. The error question is that same linear approximation used once more, this time with ϕ as the variable.

Solution

The linear approximation

Near the point $P = (\theta, \mathrm{f}(\theta))$ the graph of f is very close to its tangent line, which has gradient $\mathrm{f}'(\theta)$.



Move a distance $\delta\theta$ to the right. Along the tangent the height rises by (gradient)$\times$(run) $= \mathrm{f}'(\theta)\,\delta\theta$, reaching T. Along the curve it rises to Q, at height $\mathrm{f}(\theta + \delta\theta)$. The gap QT shrinks faster than $\delta\theta$ itself as $\delta\theta \to 0$, because

$$\frac{\mathrm{f}(\theta + \delta\theta) - \mathrm{f}(\theta)}{\delta\theta} \to \mathrm{f}'(\theta)$$

is precisely the definition of the derivative. So for small $\delta\theta$,

$$\mathrm{f}(\theta+\delta\theta) \approx \mathrm{f}(\theta) + \mathrm{f}'(\theta)\,\delta\theta.$$

The distance of C

In triangle ABC the angle at A is θ, and the exterior angle at B is $\theta+\phi$. An exterior angle equals the sum of the two opposite interior angles, so

$$\angle ACB = \phi.$$

The interior angle at B is $\pi-(\theta+\phi)$, and $\sin(\pi-x)=\sin x$, so the sine rule gives the exact results

$$AC = \frac{a\sin(\theta+\phi)}{\sin\phi}, \qquad h = AC\sin\theta = \frac{a\sin\theta\,\sin(\theta+\phi)}{\sin\phi}.$$

Ask yourself Where does "h is very large compared with a" actually enter?

It tells us that ϕ is small. Rearranging the exact formula, $\sin\phi = a\sin\theta\sin(\theta+\phi)/h \le a/h$, so $h \gg a$ forces $\sin\phi$, and hence ϕ, to be tiny. (Seen from a very distant C, the baseline AB subtends almost no angle.) Now use the first part with $\mathrm{f}=\sin$:

$$\sin(\theta+\phi) \approx \sin\theta + \phi\cos\theta \approx \sin\theta, \qquad \sin\phi \approx \sin 0 + \phi\cos 0 = \phi.$$

Therefore

$$h \approx \frac{a\sin^2\theta}{\phi},$$

as required, and

$$AC = \frac{h}{\sin\theta} \approx \frac{a\sin\theta}{\phi}.$$

(Keeping the $\phi\cos\theta$ term gives $AC \approx a\sin\theta/\phi + a\cos\theta$. The extra piece is only of the size of a, which is negligible beside a distance of order h.)

The effect of an error in ϕ

Treat AC as the function $g(\phi) = a\sin\theta\,\phi^{-1}$, with θ fixed because it is measured correctly. By the first part, an error $\delta\phi$ changes the estimate by

$$\delta(AC) \approx g'(\phi)\,\delta\phi = -\frac{a\sin\theta}{\phi^2}\,\delta\phi.$$

So the error in AC has size about $\dfrac{a\sin\theta}{\phi^2}\,|\delta\phi|$; overestimating ϕ underestimates the distance.

Ask yourself If we double a, the formula has a on top. Doesn't the error double?

No, because ϕ is not fixed: it depends on the baseline. The point C (and so h) stays the same, and from $h \approx a\sin^2\theta/\phi$ we get $\phi \approx a\sin^2\theta/h$, so doubling a roughly doubles ϕ. (θ changes only very slightly, since C is so far away.) Writing the error in terms of the fixed quantity h:

$$\frac{a\sin\theta}{\phi^2}\,|\delta\phi| \approx a\sin\theta\cdot\frac{h^2}{a^2\sin^4\theta}\,|\delta\phi| = \frac{h^2}{a\sin^3\theta}\,|\delta\phi|.$$

For the same instrumental error $\delta\phi$, the error in AC is inversely proportional to a. **Doubling the distance between the telescopes roughly halves the error.**

Checks

- Take $a = 1$, $\theta = \pi/6$, $\phi = 10^{-4}$. The exact formulae give $h \approx 2500.43$ and $AC \approx 5000.87$; the approximations give 2500 and 5000. The difference in AC is about $0.87 \approx a\cos\theta$, just as the extra term predicts.
- Computing C directly as the intersection of the two sight lines, for thousands of random a, θ and small ϕ, matches both approximations to within 0.1%, and the change in AC caused by a small $\delta\phi$ matches $-a\sin\theta\,\delta\phi/\phi^2$.
- For a fixed distant C, recomputing ϕ with baselines a and $2a$ and perturbing each by the same $\delta\phi$ gives error ratios of 2.
- Dimensions: $a\sin^2\theta/\phi$ is a length, since angles in radians are dimensionless.

Common slips

- Using $\sin\phi \approx \phi$ with ϕ in degrees. The approximation, like f′ for sin, needs radians.
- Concluding that the error doubles with a by holding ϕ fixed. The target fixes h, not ϕ.
- Missing that $\angle ACB = \phi$ and setting up coordinates instead. That works, but it takes longer.

Beyond STEP

This is triangulation by parallax, the method astronomers use to measure distances to nearby stars, with the Earth's orbit as the baseline. The error analysis is a first taste of error propagation: to first order, an error δx in a measurement becomes an error $g'(x)\,\delta x$ in anything computed from it. That idea, made precise by Taylor's theorem with a remainder term, is a central part of first-year analysis; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1992, Question 7

Pure • Integer-Valued Polynomials

STEP I 1992, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q7

The key idea

Don't try to prove that the coefficients are integers, because for the quadratic they needn't be: $\frac{1}{2}x^2 + \frac{1}{2}x$ takes integer values at every integer. Instead, express the coefficients in terms of the given integer values and substitute back, so that $\mathrm{f}(n)$ appears as a combination of $\mathrm{f}(-1)$, $\mathrm{f}(0)$ and $\mathrm{f}(1)$ with integer multipliers. The final part is then just the middle part again, after a shift of origin.

Solution

The linear function

Since $\mathrm{g}(0) = b$ and $\mathrm{g}(1) = a + b$, we have $b = \mathrm{g}(0)$ and $a = \mathrm{g}(1) - \mathrm{g}(0)$, and both are integers. So for any integer n,

$$\mathrm{g}(n) = an + b = \mathrm{g}(0) + n\big(\mathrm{g}(1) - \mathrm{g}(0)\big),$$

which is a sum of products of integers, so it's an integer.

The quadratic function

Write $p = \mathrm{f}(-1)$, $q = \mathrm{f}(0)$, $r = \mathrm{f}(1)$, all integers. Then

$$p = A - B + C, \qquad q = C, \qquad r = A + B + C,$$

and solving,

$$C = q, \qquad B = \frac{r - p}{2}, \qquad A = \frac{p + r - 2q}{2}.$$

So A and B need not be integers; they could be halves of odd integers. Substituting these into $\mathrm{f}(n) = An^2 + Bn + C$ and collecting the terms in p, q and r:

$$\begin{aligned}\mathrm{f}(n) &= \frac{p + r - 2q}{2}\, n^2 + \frac{r - p}{2}\, n + q \\ &= \frac{n(n-1)}{2}\, p + (1 - n^2)\, q + \frac{n(n+1)}{2}\, r.\end{aligned}$$

Ask yourself The multipliers of p and r have a 2 in the denominator. Why are they integers?

For an integer n, the numbers $n-1$ and n are consecutive integers, so one of them is even, and $n(n-1)$ is even. Likewise $n(n+1)$ is even. So $\frac{1}{2}n(n-1)$ and $\frac{1}{2}n(n+1)$ are integers, as is $1-n^2$. Hence $\mathrm{f}(n)$ is a sum of products of integers, and so it's an integer for every integer n, as required.

(The three multipliers are the quadratics that equal 1 at one of $-1, 0, 1$ and 0 at the other two. This is Lagrange interpolation.)

The final part

Let α be real and define $\mathrm{h}(x) = \mathrm{f}(\alpha + x)$. Expanding,

$$\mathrm{h}(x) = A(\alpha+x)^2 + B(\alpha+x) + C = Ax^2 + (2A\alpha + B)x + (A\alpha^2 + B\alpha + C),$$

so h is also a function of the form $A'x^2 + B'x + C'$ with real coefficients. The middle part used nothing about A, B, C except that they were real numbers. Now

$$\mathrm{h}(-1) = \mathrm{f}(\alpha - 1), \qquad \mathrm{h}(0) = \mathrm{f}(\alpha), \qquad \mathrm{h}(1) = \mathrm{f}(\alpha + 1)$$

are integers by hypothesis. So, by the middle part applied to h, $\mathrm{h}(n) = \mathrm{f}(\alpha + n)$ is an integer for every integer n. Explicitly,

$$\mathrm{f}(\alpha+n) = \frac{n(n-1)}{2}\,\mathrm{f}(\alpha-1) + (1-n^2)\,\mathrm{f}(\alpha) + \frac{n(n+1)}{2}\,\mathrm{f}(\alpha+1).$$

Checks

- Put $n = -1, 0, 1$ into the formula for $\mathrm{f}(n)$: the multipliers become $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$, giving back p, q, r.
- The example $\mathrm{f}(x) = \frac{1}{2}x^2 + \frac{1}{2}x$ has $p = 0$, $q = 0$, $r = 1$, and the formula gives $\mathrm{f}(n) = \frac{1}{2}n(n+1)$, the triangular numbers.
- Numerically, with $\alpha = \sqrt{2}$ and $\mathrm{f}(\alpha - 1) = 3$, $\mathrm{f}(\alpha) = -7$, $\mathrm{f}(\alpha + 1) = 11$, the quadratic through these points gives $\mathrm{f}(\alpha + 5) = 363.000\ldots$, matching $10 \cdot 3 - 24 \cdot (-7) + 15 \cdot 11 = 363$.

Common slips

- Claiming that A, B, C must be integers. They needn't be, and an argument that relies on it is wrong.
- In the final part, trying to show that α must be an integer or rational. It can be any real number; only the shift matters.

Beyond STEP

Polynomials that take integer values at every integer are exactly the integer combinations of the binomial coefficients $\binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}$. For a quadratic these are 1, x and $\frac{1}{2}x(x-1)$, and the result is proved with finite differences: a polynomial of degree d is integer-valued if it takes integer values at $d+1$ consecutive integers. The same idea underlies Newton's forward-difference formula in numerical analysis. For more on integer-valued functions and divisibility, see Davenport, *The Higher Arithmetic*.

WORKED SOLUTION

STEP I 1992, Question 8

Pure • Fundamental Theorem of Calculus

STEP I 1992, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q8

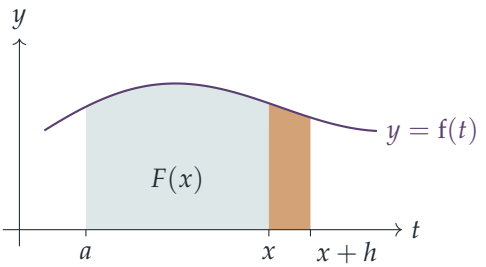
The key idea

An integral with a variable upper limit is an area that grows as the limit moves, and it grows at a rate equal to the height of the curve at that limit. That one fact (the Fundamental Theorem of Calculus) turns each *integral* equation in the question into a *differential* equation. Differentiating loses the information about constants, so we always recover it by substituting a convenient value of x, usually $x = 0$, into the original equation.

Solution

The derivative of an integral

Write $F(x) = \int_a^x \mathrm{f}(t)\,\mathrm{d}t$, the area under $y = \mathrm{f}(t)$ between $t = a$ and $t = x$. Increasing x by a small amount h adds a thin strip of width h:



The strip has area $F(x+h) - F(x)$. If f is continuous, then on $[x, x+h]$ the curve stays within a small amount $\varepsilon(h)$ of $\mathrm{f}(x)$, where $\varepsilon(h) \to 0$ as $h \to 0$. So the strip lies between rectangles of height $\mathrm{f}(x) \pm \varepsilon(h)$:

$$\big(\mathrm{f}(x) - \varepsilon(h)\big)h \le F(x+h) - F(x) \le \big(\mathrm{f}(x) + \varepsilon(h)\big)h.$$

Dividing by $h > 0$ (the case $h < 0$ is the same with the strip on the left) and letting $h \to 0$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_a^x \mathrm{f}(t)\,\mathrm{d}t = \lim_{h\to 0}\frac{F(x+h)-F(x)}{h} = \mathrm{f}(x).$$

Notice that a doesn't appear: changing the lower limit only adds a constant to F.

The equation $\mathrm{f}(x) = \int_0^x \mathrm{f}(t)\,\mathrm{d}t + 1$

The right-hand side is differentiable by the result above, so f is too, and differentiating both sides gives

$$\mathrm{f}'(x) = \mathrm{f}(x).$$

Putting $x = 0$ in the original equation, the integral is over an interval of zero length, so $\mathrm{f}(0) = 1$.

Ask yourself How do we know $\mathrm{f}' = \mathrm{f}$ has no solutions other than multiples of e^x?

Consider $g(x) = \mathrm{e}^{-x}\mathrm{f}(x)$. By the product rule,

$$g'(x) = \mathrm{e}^{-x}\big(\mathrm{f}'(x) - \mathrm{f}(x)\big) = 0,$$

so g is constant, and $g(0) = \mathrm{f}(0) = 1$. Hence $\mathrm{e}^{-x}\mathrm{f}(x) = 1$, that is, $\mathrm{f}(x) = \mathrm{e}^x$, as required. (Conversely, $\int_0^x \mathrm{e}^t\,\mathrm{d}t + 1 = \mathrm{e}^x - 1 + 1 = \mathrm{e}^x$, so it really is a solution.)

The equation $\mathrm{f}(x) = \int_0^x \mathrm{f}(t)\,\mathrm{d}t$

Exactly as before, $\mathrm{f}' = \mathrm{f}$, so $\mathrm{e}^{-x}\mathrm{f}(x)$ is constant. But now $x = 0$ gives $\mathrm{f}(0) = 0$, so the constant is 0, and the only solution is

$$\mathrm{f}(x) = 0 \quad \text{for all } x.$$

The "+1" in the previous equation was what fixed the size of the exponential; without it only the zero function survives.

The final part

Differentiate both sides with respect to x. On the right, x is now the *lower* limit, and

$$\int_x^1 t^2\mathrm{f}(t)\,\mathrm{d}t = -\int_1^x t^2\mathrm{f}(t)\,\mathrm{d}t, \qquad \text{so} \qquad \frac{\mathrm{d}}{\mathrm{d}x}\int_x^1 t^2\mathrm{f}(t)\,\mathrm{d}t = -x^2\mathrm{f}(x).$$

Hence

$$\mathrm{f}(x) = -x^2\mathrm{f}(x) + 1 - x^4,$$
$$(1+x^2)\,\mathrm{f}(x) = 1 - x^4 = (1-x^2)(1+x^2).$$

Since $1 + x^2 > 0$ we may divide by it, giving

$$\mathrm{f}(x) = 1 - x^2.$$

To find C, put $x = 0$ in the given equation. The left-hand side and the terms $x - \frac{x^5}{5}$ vanish, leaving

$$0 = \int_0^1 t^2(1-t^2)\,\mathrm{d}t + C = \left[\tfrac{t^3}{3} - \tfrac{t^5}{5}\right]_0^1 + C = \tfrac{2}{15} + C,$$

so $C = -\frac{2}{15}$, as required.

Ask yourself Differentiating only tells us the two sides have the same derivative. Does $\mathrm{f}(x) = 1 - x^2$ with $C = -\frac{2}{15}$ really satisfy the equation for every x?

Yes. With this f the two sides have equal derivatives for all x, so they differ by a constant; and we chose C precisely so that they agree at $x = 0$. So they agree everywhere. Conversely, any solution must satisfy the differentiated equation, so $1 - x^2$ is the only possible f.

Checks

- At $x = 1$: the left side is $\int_0^1 (1-t^2)\,\mathrm{d}t = \frac{2}{3}$, and the right side is $0 + 1 - \frac{1}{5} - \frac{2}{15} = \frac{10}{15} = \frac{2}{3}$. They match.
- Numerically, Simpson's rule gives $\int_0^1 t^2(1-t^2)\,\mathrm{d}t \approx 0.133333 = \frac{2}{15}$, and both sides of the final equation agree to 10^{-10} at $x = -1.3,\ 0.25,\ 0.5,\ 0.9,\ 2$.

Common slips

- Forgetting the minus sign when the variable is the *lower* limit: $\frac{\mathrm{d}}{\mathrm{d}x}\int_x^1 g(t)\,\mathrm{d}t = -g(x)$.
- Solving $\mathrm{f}' = \mathrm{f}$ and stopping at Ae^x: the original integral equation, at $x = 0$, is what fixes A.
- Writing $\mathrm{f}(x) = \int_0^x \mathrm{f}(x)\,\mathrm{d}x$ style expressions that mix the dummy variable t with the limit x.

Beyond STEP

Equations in which the unknown function appears inside an integral are called *integral equations*; the ones here, with a variable limit, are Volterra equations, and they are equivalent to an initial-value problem for a differential equation. In a first analysis course you prove the Fundamental Theorem of Calculus carefully from the definition of the Riemann integral, and the strip argument above becomes an ε–δ proof. Abbott's *Understanding Analysis* does this very readably.

WORKED SOLUTION

STEP I 1992, Question 9

Pure • Differential Equations

STEP I 1992, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q9

The key idea

Everything follows from one equation: the rate of change of the *volume* of water equals (rate in) minus (rate out). By similar triangles the volume is a simple function of the depth x, so the chain rule turns this into a differential equation for x. In the first part the equation separates neatly; in the second, rather than solving it, we read off the behaviour from the sign of $\mathrm{d}x/\mathrm{d}t$ and from the depth at which it vanishes.

Solution

Emptying the cone

When the water is x cm deep it fills a smaller cone, similar to the whole filter, of height x and radius r. By similar triangles $r/x = a/H$, so $r = ax/H$, and the volume of water is

$$V = \tfrac{1}{3}\pi r^2 x = \frac{\pi a^2 x^3}{3H^2}.$$

By the chain rule,

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\pi a^2 x^2}{H^2}\frac{\mathrm{d}x}{\mathrm{d}t}.$$

Water only leaves, at rate Ax, so $\mathrm{d}V/\mathrm{d}t = -Ax$. Hence

$$\frac{\pi a^2 x^2}{H^2}\frac{\mathrm{d}x}{\mathrm{d}t} = -Ax \quad\Longrightarrow\quad x\frac{\mathrm{d}x}{\mathrm{d}t} = -\frac{AH^2}{\pi a^2},$$

where we divided by x, which is allowed while there is still water in the cone ($x > 0$). The left-hand side is $\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}x^2\right)$, so $\frac{1}{2}x^2$ decreases at the *constant* rate $AH^2/(\pi a^2)$. It starts at $\frac{1}{2}h^2$ and the cone is empty when it reaches 0, so the time taken is

$$T = \frac{\frac{1}{2}h^2}{AH^2/(\pi a^2)} = \frac{\pi a^2 h^2}{2AH^2} \text{ seconds},$$

as required. (Equivalently, separate variables: $\int_h^0 x\,\mathrm{d}x = -\frac{AH^2}{\pi a^2}\int_0^T \mathrm{d}t$.)

Pouring water in as well

Now $\mathrm{d}V/\mathrm{d}t = B - Ax$, so

$$\frac{\pi a^2 x^2}{H^2}\frac{\mathrm{d}x}{\mathrm{d}t} = B - Ax, \qquad \text{that is} \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = \frac{H^2(B - Ax)}{\pi a^2 x^2}.$$

Since $x > 0$, the factor $H^2/(\pi a^2 x^2)$ is positive, so $\mathrm{d}x/\mathrm{d}t$ has the sign of $B - Ax$. Write $c = B/A$. Then

$$\frac{\mathrm{d}x}{\mathrm{d}t} > 0 \text{ when } x < c, \qquad \frac{\mathrm{d}x}{\mathrm{d}t} = 0 \text{ when } x = c, \qquad \frac{\mathrm{d}x}{\mathrm{d}t} < 0 \text{ when } x > c.$$

So $x = c$ is an equilibrium depth, where inflow exactly balances outflow, and the level always moves *towards* c. This is only physically possible inside the filter if $c \le H$, so the cases are decided by comparing c with h and with H.

? **Ask yourself** If the level is moving towards c, does it ever actually get there, or overshoot?

It can't overshoot: to pass c it would first have to equal c, and there $\mathrm{d}x/\mathrm{d}t = 0$. In fact it never reaches c in finite time. Treating t as a function of x, the time to go from h to a depth X between h and c is

$$t = \int_h^X \frac{\pi a^2 x^2}{H^2(B - Ax)}\,\mathrm{d}x = \int_h^X \frac{\pi a^2 x^2}{AH^2(c - x)}\,\mathrm{d}x.$$

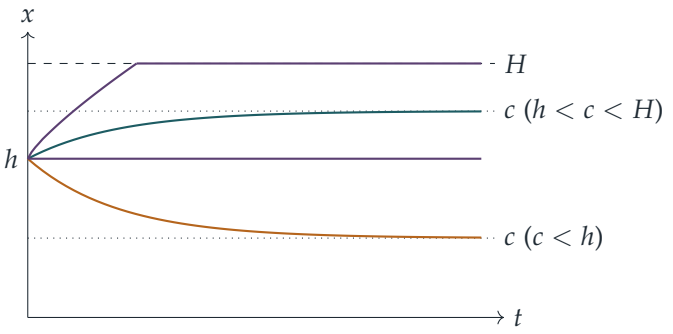
Near $x = c$ the integrand behaves like $\dfrac{\pi a^2 c^2}{AH^2}\cdot\dfrac{1}{c - x}$, whose integral is a multiple of $-\ln(c - x)$; this tends to infinity as $X \to c$. So the level approaches c ever more slowly and only reaches it "after infinite time". The same argument works from either side.

The cases are therefore:

- $B < Ah$ (so $c < h$). Initially $\mathrm{d}x/\mathrm{d}t < 0$: the level falls, slowing down, and tends to $c = B/A$ from above without reaching it. In particular, unlike in the first part, the cone never empties (as $c > 0$).
- $B = Ah$ (so $c = h$). Then $\mathrm{d}x/\mathrm{d}t = 0$ from the start: water flows in as fast as it drains, and the level stays at h.
- $Ah < B < AH$ (so $h < c < H$). The level rises, slowing down, and tends to B/A from below. It stays below H, so the filter never overflows.
- $B = AH$ (so $c = H$). The level rises towards the brim and tends to H, but by the argument above never quite reaches it, so (in this model) it never overflows.
- $B > AH$ (so $c > H$). Now $B - Ax \ge B - AH > 0$ for all $x \le H$, and $x^2 \le H^2$, so

  $$\frac{\mathrm{d}x}{\mathrm{d}t} \ge \frac{H^2(B - AH)}{\pi a^2 H^2} = \frac{B - AH}{\pi a^2} > 0.$$

  The rate of rise is bounded below by a positive constant, so the level reaches the brim in a finite time, at most $\pi a^2(H - h)/(B - AH)$ seconds. After that the filter overflows: the level stays at H and the excess $B - AH$ $\mathrm{cm}^3\,\mathrm{s}^{-1}$ spills over the top.



The sketch shows typical behaviour: falling to c (amber), staying at h or overflowing after a finite time (plum), and rising to c (teal).

Checks

- Putting $B = 0$ gives $c = 0$: the level falls towards 0, matching the first part (there it *does* reach 0, because the outflow Ax is proportional to x while the volume goes like x^3, so $\mathrm{d}x/\mathrm{d}t = -AH^2/(\pi a^2 x)$ does not tend to 0).
- Numerically, with $a = 3$, $H = 8$, $A = 0.7$, $h = 5$, the formula gives $T \approx 7.889\,\mathrm{s}$, and stepping x^2 in time gives the same. Integrating the second equation numerically confirms each case; for instance with $B/A = 9 > H$ the level reaches H after about 41.5 s, while with $B/A = 7$ the time to get within ε of 7 grows by about 71 s each time ε is divided by 10, the logarithmic growth predicted above.

Common slips

- Using the cylinder volume $\pi a^2 x$ instead of the volume of the similar cone of height x.
- Claiming the level reaches B/A in a finite time, or that it overshoots and oscillates.
- Forgetting the brim: the equilibrium B/A is only reached if $B/A \leq H$; otherwise the filter overflows.

Beyond STEP

The second part is a first taste of the *qualitative* theory of differential equations: for $\mathrm{d}x/\mathrm{d}t = f(x)$ you can find the equilibria (zeros of f), decide their stability from the sign of f on either side, and use uniqueness of solutions to show they are approached but never crossed, all without solving anything. Boyce and DiPrima, *Elementary Differential Equations*, treats autonomous equations and stability in this spirit.

WORKED SOLUTION

STEP I 1992, Question 10

Mechanics • Recoil and Momentum

STEP I 1992, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q10

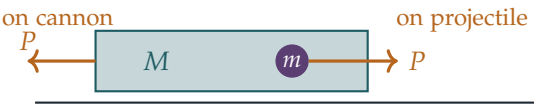
The key idea

The force P pushes the projectile forwards, but by Newton's third law the projectile pushes the cannon backwards with the same force P, and nothing horizontal stops the cannon recoiling. So the projectile does *not* travel a distance l relative to the ground while it is in the barrel: it travels l *relative to the cannon*. Both bodies have constant accelerations, so the cleanest route is to work with the relative motion.

Solution

The model

We treat the projectile and the cannon as particles moving along the same horizontal line, the floor as smooth (the cannon "slides freely"), and the force between projectile and barrel along the barrel as the constant P. Vertically, the weights are balanced by normal reactions (the floor on the cannon, the barrel on the projectile), so they play no part in the horizontal motion. Take the firing direction as positive.



The two accelerations

While the projectile is in the barrel, the only horizontal force on it is P forwards, and the only horizontal force on the cannon is the reaction P backwards (Newton's third law; the floor is smooth). By Newton's second law the accelerations are

$$a_{\text{proj}} = \frac{P}{m}, \qquad a_{\text{cannon}} = -\frac{P}{M}.$$

Both are constant, and both bodies start from rest.

Ask yourself Over what distance does the projectile accelerate: l, or something else?

The barrel has length l, but the barrel itself moves backwards. The projectile leaves when its displacement *relative to the cannon* reaches l. Relative to the cannon, the projectile starts from rest with constant acceleration

$$a_{\text{rel}} = \frac{P}{m} - \left(-\frac{P}{M}\right) = \frac{P}{m} + \frac{P}{M} = \frac{P(M+m)}{mM}.$$

The time in the barrel

Using $s = \frac{1}{2}at^2$ for the relative motion, the projectile emerges at the time T with

$$l = \frac{1}{2} \cdot \frac{P(M+m)}{mM}\, T^2, \qquad \text{so} \qquad T^2 = \frac{2mMl}{P(M+m)}.$$

The exit speed

Relative to the ground the projectile has constant acceleration P/m from rest, so at time T its speed v satisfies

$$\begin{aligned} v^2 = \left(\frac{P}{m}T\right)^2 &= \frac{P^2}{m^2} \cdot \frac{2mMl}{P(M+m)} \\ &= \frac{2PMl}{m(M+m)}. \end{aligned}$$

Since $v > 0$,

$$v = \sqrt{\frac{2PMl}{m(M+m)}},$$

which is the required result.

Another route: energy and momentum

The same answer comes from conservation laws. There is no external horizontal force, so horizontal momentum is conserved: if the cannon has speed V backwards at the moment of exit, then $mv = MV$. If the projectile moves x forwards and the cannon X backwards while the projectile is in the barrel, then $x + X = l$. The force on the projectile does work Px, and the reaction on the cannon, which points the same way as the cannon moves, does work PX; together they do work $P(x + X) = Pl$. Hence

$$Pl = \tfrac{1}{2}mv^2 + \tfrac{1}{2}MV^2 = \tfrac{1}{2}mv^2 + \tfrac{1}{2}M\left(\frac{mv}{M}\right)^2 = \tfrac{1}{2}mv^2\,\frac{M+m}{M},$$

which again gives $v^2 = \dfrac{2PMl}{m(M+m)}$.

Checks

- Heavy cannon: as $M \to \infty$ the factor $M/(M+m) \to 1$ and $v \to \sqrt{2Pl/m}$, the speed from a fixed cannon (work Pl all going to the projectile). A recoiling cannon always gives a slightly smaller speed, as it should, because some energy goes into the recoil.
- Units: Pl is energy and Pl/m has units of $\mathrm{m}^2\,\mathrm{s}^{-2}$, so v is a speed.
- Numbers: with $m = 0.1$, $M = 0.9$, $P = 9$ and $l = 0.5$ (SI units) the formula gives $v = 9\ \mathrm{m\,s^{-1}}$, with the cannon recoiling at $1\ \mathrm{m\,s^{-1}}$. Then momentum $0.1 \times 9 = 0.9 \times 1$ and kinetic energy $4.05 + 0.45 = 4.5 = Pl$. A step-by-step numerical simulation of both bodies, stopped when the separation reaches l, agrees with the formula for many random values of m, M, P and l.

Common slips

- Using $v^2 = 2(P/m)l$, which treats l as the projectile's distance relative to the ground and ignores the recoil.
- Forgetting Newton's third law and leaving the cannon at rest; the cannon slides freely, so it must recoil.
- In the energy method, taking the work done as P times the projectile's ground displacement only; the reaction on the cannon does work too.

Beyond STEP

This is a two-body problem, and the relative acceleration $P(M+m)/(mM)$ is P divided by the *reduced mass* $\mu = mM/(m+M)$. Any two bodies interacting only with each other behave, relative to one another, like a single body of mass μ; the same idea handles planetary orbits, binary stars and colliding particles, and it underlies rocket and recoil problems. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both the reduced mass and the momentum ideas here.

WORKED SOLUTION

STEP I 1992, Question 11

Mechanics • Elastic Strings Equilibrium

STEP I 1992, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q11

The key idea

Each particle is in equilibrium, so the tension just above it exceeds the tension just below it by mg. That links the three tensions, and Hooke's law turns tensions into extensions. The one extra fact we need is geometric: the three stretched lengths must add up to the height of the room. Once the heights are known, "$CD = a$" just says the lowest string carries no tension.

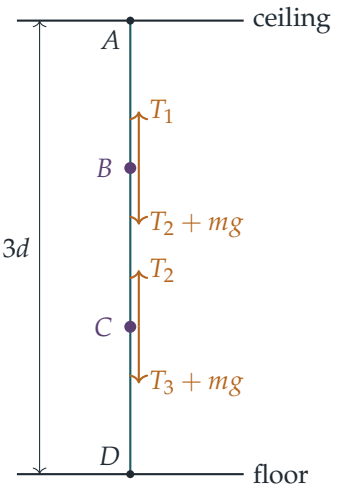
Solution

The model

The strings are light and elastic, obeying Hooke's law (tension $= \lambda \times$ extension$/a$) when stretched; the particles are at rest. Write T_1, T_2, T_3 for the tensions in AB, BC, CD and x_1, x_2, x_3 for their extensions, so $T_i = \lambda x_i / a$. It is convenient to name the extension that a weight mg alone would produce:

$$e = \frac{mga}{\lambda}.$$

We first assume all three strings are taut, and check afterwards when this is true. Measure heights y_B, y_C above the floor.



Heights of B and C

Resolving vertically for each particle (the string above pulls up, the string below and the weight pull down):

$$B: \quad T_1 = T_2 + mg, \qquad C: \quad T_2 = T_3 + mg.$$

Dividing by λ/a turns these into statements about extensions:

$$x_1 = x_2 + e, \qquad x_2 = x_3 + e, \qquad \text{so} \qquad x_2 = x_3 + e, \quad x_1 = x_3 + 2e.$$

The total length of the three strings is the height of the room:

$$3a + x_1 + x_2 + x_3 = 3d \implies 3x_3 + 3e = 3d - 3a \implies x_3 = d - a - e.$$

Then $x_2 = d - a$ and $x_1 = d - a + e$. The heights follow from the lengths of the lower strings:

$$y_C = CD = a + x_3 = d - e = d - \frac{mga}{\lambda},$$
$$y_B = y_C + BC = (d - e) + (a + d - a) = 2d - \frac{mga}{\lambda}.$$

As a check on AB: $3d - y_B = d + e = a + x_1$, as it should be.

? **Ask yourself** When are these answers actually valid?

We assumed every string is taut, so we need every extension to be at least zero. The tightest condition is on the lowest string: $x_3 \geq 0$, that is

$$\frac{mga}{\lambda} \leq d - a.$$

(Then $x_2 = d - a \geq 0$ and $x_1 > 0$ automatically.) It makes sense that CD is the first to go slack: it supports nothing, and only the stretching of the room holds it taut. If

$mga/\lambda > d - a$, the string CD is slack and the system is just two particles hanging from A: then $T_2 = mg$, $T_1 = 2mg$, so $AB = a + 2e$ and $BC = a + e$, giving $y_B = 3d - a - 2e$ and $y_C = 3d - 2a - 3e$, provided this is still above the floor. The question intends the main case with all strings taut.

When can $CD = a$?

The string CD has its natural length exactly when it is just taut with no tension, $T_3 = 0$, i.e. $x_3 = 0$. This is the boundary between the two cases above, and either set of formulas gives

$$y_C = a \iff d - \frac{mga}{\lambda} = a \iff m = \frac{\lambda(d-a)}{ga}.$$

Physically, the particle at C is then held up entirely by BC, with $T_2 = mg$, and $T_1 = 2mg$.

A suitable mass exists exactly when this value of m is positive, which needs $d > a$. Conversely, if $d \leq a$ then no positive m works: if all strings were taut we would need $x_2 = d - a \geq 0$ and $x_3 = d - a - e < 0$, which is impossible, so CD is slack and C hangs at height $3d - 2a - 3e < 3a - 2a = a$ (or rests on the floor), so $CD < a$. Hence the set of values is

$$\{\, d : d > a \,\}.$$

For such d we get $y_C = a$, $y_B = d + a$ and $AB = 2d - a$.

Checks

- Units: mga/λ is a force times a length divided by a force, so a length, as it must be to subtract from d.
- With $m = 0$ the three strings share the stretch equally: each has length d, putting C at d and B at $2d$. Adding mass lowers both particles by the same amount e, since BC keeps extension $d - a$.
- Numerically, minimising the potential energy (gravitational plus elastic, with slack strings allowed) over $0 \leq y_C \leq y_B \leq 3d$ for 250 random choices of a, λ, d, m reproduced the heights in both the taut and the slack-CD cases to within 10^{-14}. With $a = 1$, $\lambda = 10$, $d = 2$ and $m = \lambda(d-a)/(ga) \approx 1.020$ it gave $CD = 1.000000$ and $y_B = 3.000000$; for $d \leq a$ no positive m gave $CD = a$.

Common slips

- Writing the tension as $\lambda\times$ (stretched length) $/a$ instead of $\lambda\times$ extension$/a$.
- Forgetting that the total length is fixed by the room: without $\sum(a + x_i) = 3d$ the equilibrium equations don't determine the extensions.
- Assuming every string must be in tension. Strings can go slack, and $CD = a$ happens precisely on the boundary of slackness.

Beyond STEP

This is a small statically indeterminate structure: equilibrium of the particles alone doesn't fix the tensions, and we needed a compatibility condition (the lengths add up) as well. The same idea, in matrix form, is how engineers analyse frameworks of springs and bars. Another way in is to minimise the total potential energy, which is how the numerical check worked; that variational view is central in later mechanics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers springs, equilibrium and potential energy carefully.

WORKED SOLUTION

STEP I 1992, Question 12

Mechanics • Statics with Friction

STEP I 1992, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q12

The key idea

The two ladders are light, so the only way to get a useful equation for one of them is to take moments about the hinge A: that removes the unknown force from the other ladder and from the load at A. The feet want to slide apart, and the outward push is resisted partly by friction and partly by the rod. At the moment of collapse friction is doing all it can, so the rod has to supply the rest.

Solution

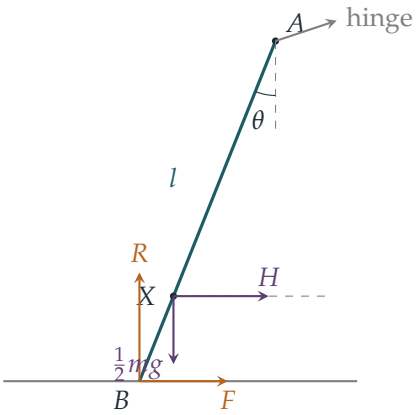
The model

The ladders are light rods, smoothly hinged at A. The wooden rod XY is uniform, of mass m, and pin-jointed at X and Y. The mass M is a particle at A. Write $t = \tan\theta$. By the left–right symmetry of the whole arrangement, the two feet carry equal normal reactions R, and resolving vertically for the whole step-ladder gives

$$2R = (M+m)g \quad \Longrightarrow \quad R = \tfrac{1}{2}(M+m)g.$$

Since the ladders lean outwards, the feet tend to slide apart, so the friction F at each foot acts inwards.

Forces from the rod. The rod is horizontal and has its own weight, so the forces at its ends are *not* along it. By symmetry each end carries half its weight, so the ladder exerts an upward force $\frac{1}{2}mg$ on the rod at X, together with a horizontal force H outwards (the tension). By Newton's third law, the rod pulls ladder AB at X with a horizontal force H *inwards* and a vertical force $\frac{1}{2}mg$ *downwards*.



Moments about A for ladder AB

Measured from A, the foot B is $l\sin\theta$ across and $l\cos\theta$ down, and X is $\frac{3}{4}l\sin\theta$ across and $\frac{3}{4}l\cos\theta$ down. Taking moments about A (all forces at A have no moment):

- R turns the ladder one way with moment $Rl\sin\theta$;
- F and H (both inwards, acting below A) turn it the other way, with moments $Fl\cos\theta$ and $H\cdot\frac{3}{4}l\cos\theta$;
- the downward $\frac{1}{2}mg$ at X acts on the same side as R but in the opposite sense, with moment $\frac{1}{2}mg\cdot\frac{3}{4}l\sin\theta$ opposing R.

So

$$Fl\cos\theta+\tfrac{3}{4}Hl\cos\theta=Rl\sin\theta-\tfrac{3}{8}mgl\sin\theta,$$

and dividing by $l\cos\theta$ and substituting for R,

$$F+\tfrac{3}{4}H=t\left(\tfrac{1}{2}(M+m)g-\tfrac{3}{8}mg\right)=t\left(\tfrac{1}{2}M+\tfrac{1}{8}m\right)g. \qquad (*)$$

Ask yourself Friction and the rod both hold the feet in. Which value of F do we use?

Equation $(*)$ says friction and tension share a fixed amount of work: the more friction provides, the less the rod needs to. So the step-ladder survives as long as some allowed friction ($F\le\mu R$) leaves $H\le T$, and the smallest tension possible comes from the largest friction, $F=\mu R$. This is the limiting state at collapse that the question lets us assume. Putting $F=\mu R=\frac{1}{2}\mu(M+m)g$ into $(*)$:

$$\begin{aligned}\tfrac{3}{4}H&=t\left(\tfrac{1}{2}M+\tfrac{1}{8}m\right)g-\tfrac{1}{2}\mu(M+m)g\\&=\tfrac{1}{2}(t-\mu)Mg+\tfrac{1}{8}(t-4\mu)mg,\end{aligned}$$

so

$$H=\tfrac{2}{3}(t-\mu)Mg+\tfrac{1}{6}(t-4\mu)mg.$$

Since $t > \mu$, H increases with M: without the rod, friction alone could not hold even the load (with $m = 0$ we would need $t \le \mu$).

The largest safe mass

The rod survives provided $H \le T$, that is

$$\tfrac{2}{3}(t-\mu)Mg \le T - \tfrac{1}{6}(t-4\mu)mg,$$

and dividing by the positive quantity $\tfrac{2}{3}(t-\mu)g$,

$$\boxed{M \le \frac{6T - (\tan\theta - 4\mu)mg}{4(\tan\theta - \mu)g}}$$

So the greatest mass that can safely be placed at A is $\dfrac{6T - (\tan\theta - 4\mu)mg}{4(\tan\theta - \mu)g}$.

Failure under its own weight

With nothing at A ($M = 0$), the tension needed is $H = \tfrac{1}{6}(t-4\mu)mg$, so the rod breaks if

$$\tfrac{1}{6}(\tan\theta - 4\mu)mg > T \iff \tan\theta > \frac{6T}{mg} + 4\mu,$$

as required. In that case the bound on M above is negative: no load at all is safe, because the step-ladder cannot even support itself.

Checks

- If T is huge the safe mass is huge, and if $\tan\theta$ only just exceeds μ the denominator is tiny and again almost any load is safe, because friction nearly suffices on its own.
- A numerical solution of the full statics (both ladders, the rod and the pin at A, with twelve unknown force components and limiting friction at both feet) for 2000 random configurations reproduced the tension formula to about 10^{-12}. For example, with $\theta = 45^\circ$, $\mu = 0.2$, $m = 2\,\text{kg}$ and $g = 9.8\,\text{m}\,\text{s}^{-2}$, the rod tension with $M = 0$ is $\tfrac{1}{6}(0.2)(19.6) \approx 0.653\,\text{N}$.

Common slips

- Treating the rod as a light tie, so the force at X is purely along XY. The rod has weight, so each end also carries $\tfrac{1}{2}mg$ vertically, and this is where the $\tfrac{1}{8}m$ and the 4μ come from.
- Putting friction outwards at the feet. Since $\tan\theta > \mu$, the feet tend to splay, so friction points inwards and helps the rod.

Beyond STEP

A frame like this, where the equations of equilibrium alone don't fix every force (here, friction and tension can trade off), is called *statically indeterminate*. Engineers resolve such cases with extra information: a limiting condition, as here, or the elasticity of the members. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to deepen your understanding of forces, torques and equilibrium.

WORKED SOLUTION

STEP I 1992, Question 13

Mechanics • Escape Speed and Radial Fall

STEP I 1992, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

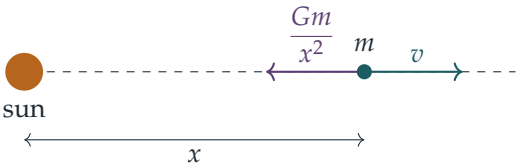
step.maths.org/questions/92-s1-q13

The key idea

The force depends only on the distance x, so the natural tool is the equation of motion written as $v\frac{\mathrm{d}v}{\mathrm{d}x}$, which integrates at once to an energy equation linking speed and distance. In the first part the fact that v^2 can never be negative turns that equation into a ceiling on x. In the second part the energy equation gives $\frac{\mathrm{d}x}{\mathrm{d}t}$ as a function of x, and separating the variables gives the time.

Solution

We model the sun as fixed and the comet as a particle, with no forces other than the given attraction. Since the attraction acts along the line joining the comet to the sun, a comet that is moving directly away from the sun (or is at rest) keeps moving along that line. So x is a coordinate along a straight line, and we let $v = \frac{\mathrm{d}x}{\mathrm{d}t}$, taking positive v to mean away from the sun.



The energy equation

Newton's second law along the line, with the force towards the sun (in the direction of decreasing x), gives

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{Gm}{x^2}.$$

Writing the acceleration as $v\frac{\mathrm{d}v}{\mathrm{d}x}$ and cancelling m,

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{G}{x^2} \qquad \Longrightarrow \qquad \tfrac{1}{2}v^2 = \frac{G}{x} + C$$

for some constant C. This holds throughout the motion, whichever way the comet is moving. (It is conservation of energy per unit mass: kinetic energy $\frac{1}{2}v^2$ plus potential energy $-G/x$.)

The condition for x to become arbitrarily large

At the given moment $x = h$ and $v = V$, so $C = \frac{1}{2}V^2 - \dfrac{G}{h}$ and, at every later time,

$$v^2 = V^2 - \frac{2G}{h} + \frac{2G}{x}.$$

We prove the contrapositive: if $V^2 < 2G/h$ then x is bounded.

Ask yourself The comet could in principle move about in all sorts of ways later on. How can one equation bound x for all time?

Because $v^2 \geq 0$ at every instant, whatever the comet is doing. Suppose $V^2 < 2G/h$, and put $k = \dfrac{2G}{h} - V^2 > 0$. Then the energy equation says

$$\frac{2G}{x} = v^2 + k \;\geq\; k, \qquad \text{so} \qquad x \leq \frac{2G}{k} = \frac{2Gh}{2G - hV^2}$$

at every later time (we may multiply up, since $x > 0$ and $k > 0$). So x can never exceed the fixed value $2Gh/(2G - hV^2)$: the comet rises to that distance, where $v = 0$, and then falls back. Hence x cannot become arbitrarily large unless $V^2 \geq 2G/h$, as required.

(The converse is also true. If $V^2 \geq 2G/h$ then $v^2 \geq 2G/x > 0$ while the comet moves outwards, so it never stops, and since $v^2 \geq 2G/x \geq 2G/X$ whenever $x \leq X$, it cannot linger below any distance X for ever. This is the escape speed.)

The time to fall into the sun

Now the comet starts at rest at a very great distance D. Released from rest, it falls straight towards the sun. By the energy equation with $v = 0$ at $x = D$,

$$\tfrac{1}{2}v^2 = \frac{G}{x} - \frac{G}{D}.$$

Ask yourself What does "a great distance" mean for the constant?

It means we let $D \to \infty$, so that $G/D \to 0$ and the constant is zero: the comet has zero total energy. (A large but finite D only makes the comet a little slower, and the time a little longer, by an amount that tends to zero as $D \to \infty$.) So at distance x its speed is $\sqrt{2G/x}$ and, since x is decreasing,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -\sqrt{\frac{2G}{x}} \qquad \Longrightarrow \qquad \sqrt{x}\,\mathrm{d}x = -\sqrt{2G}\,\mathrm{d}t.$$

Let T be the time to go from $x = h$ to $x = 0$. Integrating,

$$\sqrt{2G}\,T = \int_0^h \sqrt{x}\,\mathrm{d}x = \tfrac{2}{3}h^{3/2},$$

so

$$T = \frac{2h^{3/2}}{3\sqrt{2G}} = \frac{h}{3}\sqrt{\frac{2h}{G}}.$$

(The integrand $\sqrt{x}$ is perfectly well behaved at $x = 0$, so the comet really does reach the sun in finite time, even though its speed grows without limit.)

Checks

- Units: the force Gm/x^2 is a mass times an acceleration, so G has units $\mathrm{m^3\,s^{-2}}$. Then V^2 and $2G/h$ are both in $\mathrm{m^2\,s^{-2}}$, and $h^{3/2}/\sqrt{G}$ is in seconds.
- At $x = h$ the comet's speed is $\sqrt{2G/h}$. If it kept that speed it would take $h/\sqrt{2G/h} = \frac{h}{2}\sqrt{2h/G}$, which is longer than T, as it should be since the comet speeds up. In fact T is two-thirds of this.
- Numerically, with $G = h = 1$: stepping the motion forward from $x = 1$ with $v = \sqrt{1.5}$, the comet turns round at $x = 4.000$, as $2Gh/(2G - hV^2) = 4$ predicts. Starting from $x = 1$ with $v = -\sqrt{2}$, it reaches the sun after 0.4714, matching $T = \frac{1}{3}\sqrt{2} \approx 0.47140$. Starting from rest at $D = 10^6$ instead of infinity changes T only in the seventh decimal place.

Common slips

- Forgetting the minus sign in $\frac{\mathrm{d}x}{\mathrm{d}t} = -\sqrt{2G/x}$, then getting a negative time, or integrating from 0 to h without saying why.
- Showing only that the comet slows down: every comet moving away slows down, even one that escapes. The point is that, when $V^2 < 2G/h$, v^2 would become negative beyond a certain distance.

Beyond STEP

The quantity $\frac{1}{2}v^2 - G/x$ is the energy per unit mass in a Newtonian gravitational field, and its sign decides between bound orbits (ellipses) and escape (parabolas and hyperbolas). The zero-energy fall in the last part is the degenerate, straight-line case of a parabolic orbit, and $T \propto h^{3/2}$ is a relative of Kepler's third law. First-year courses on orbits develop all of this from the same energy and angular-momentum equations; Kleppner and Kolenkow, *An Introduction to Mechanics*, treats central forces and orbits thoroughly.

WORKED SOLUTION

STEP I 1992, Question 14

Probability and Statistics • Interpreting Statistical Claims

STEP I 1992, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q14

This question has a full guided treatment in *Beyond STEP*, Section 13.1, *Reading the Headlines*: https://beyondstep.keithmcnulty.org/ch13/headlines.html

The key idea

Every headline takes one number and leaps to a conclusion. For each one, ask two questions. First, is it comparing like with like: the same thing, measured in the same way, scaled by how much exposure there is to the risk? Second, could the difference just be chance? For the second, a natural yardstick is the Poisson model for counts of rare, independent events, whose standard deviation is the square root of its mean. Knowing when that model fits, and when it doesn't, is the heart of (ii) to (iv).

Solution

Throughout, "on average" means the 1974–1989 annual averages: 1080 pedestrian deaths and 180 air deaths a year.

Part (i)

The ratio $1080/180 = 6$ is where the "six times" comes from, but it compares raw totals, not risks. A risk is deaths per unit of exposure: per journey, per hour, or per kilometre travelled. Almost everyone in Poldavia walks near roads every day, while comparatively few people fly, and then only occasionally. To compare safety we would need something like

$$\frac{\text{pedestrian deaths}}{\text{total hours (or km) walked}} \quad \text{against} \quad \frac{\text{air deaths}}{\text{total hours (or km) flown}},$$

and the headline gives us neither denominator. The answer could even go the other way, depending on the measure: per kilometre, flying looks very safe, but per hour or per journey it may look less so.

There are other problems too. "Walking" here really means being hit by a vehicle while on foot, and the figure says nothing about pedestrians in quiet places. Air deaths include crew as well as passengers. And averages over a period cannot "prove" anything about safety in 1991; at best they are evidence about 1974–1989.

Part (ii)

At first sight 125 is well below the average of 180. Is the drop more than chance?

If deaths happened independently, one at a time, a Poisson model with mean 180 would be reasonable. Its standard deviation is $\sqrt{180} \approx 13.4$, so 125 would be about 4.1 standard deviations below the mean, and very unlikely to happen by chance ($P(X \leq 125) \approx 10^{-5}$).

But air deaths do not happen one at a time. They come in clusters: a single crash can kill tens or hundreds of people. The natural model is that the number of *fatal crashes* is Poisson, and each crash kills a random number of people. This hugely increases the variability. For example, suppose there are on average 6 fatal crashes a year, each killing 30 people. Then the annual death toll is $30N$ with $N \sim \text{Po}(6)$, which has standard deviation $30\sqrt{6} \approx 73$, and

$$P(\text{deaths} \leq 125) = P(N \leq 4) \approx 0.29.$$

On those (illustrative) figures, a year with 125 deaths is entirely ordinary: it needs only one or two fewer crashes than usual. Letting the size of each crash vary as well makes the spread larger still.

So a single year's figure tells us very little. We would also want to know how much flying took place in 1990 (if there was less air traffic, fewer deaths would be expected anyway) and the figures for several years, to see whether there is a genuine trend. The headline's claim is not supported.

Part (iii)

Two separate things are wrong here.

One day cannot show an increase. "Increasing" is a claim about a trend over time, and a single day's figure gives no comparison at all. We would need daily (or annual) figures over a long period.

Seven is not even unusual. The only road figure we have is for pedestrians: 1080 a year, or

$$\lambda = \frac{1080}{365} \approx 2.96 \text{ per day.}$$

Pedestrian deaths are fairly well modelled as Poisson: they are rare, mostly one per accident, and roughly independent. Let $X \sim \text{Po}(2.96)$ be the number of pedestrians killed on a given day. Then

$$P(X \geq 7) = 1 - \sum_{k=0}^{6} \mathrm{e}^{-\lambda} \frac{\lambda^k}{k!} \approx 0.031,$$

so we would expect about $365 \times 0.031 \approx 11$ days a year on which at least 7 pedestrians die. And the headline says "people killed", which includes drivers, passengers and cyclists as well. The total daily road death rate must be higher than 2.96, which makes 7 even less remarkable.

Ask yourself Isn't a day with more than double the average number of deaths alarming?

Not for small counts. When the mean is about 3, the Poisson standard deviation is $\sqrt{3} \approx 1.7$, which is large compared with the mean, so big proportional swings from day to day are normal. Picking out the worst day is also a selection effect: there are 365 days in a year, and a newspaper can always find a bad one. The same day of the week also matters, since traffic and the number of people out on foot vary through the week and the seasons. A Poisson model with a constant rate is only a rough first approximation.

Part (iv)

This is the headline with a real case to answer. First the statistics, then the interpretation.

Is 1350 *more than chance?* Here the Poisson model is a good starting point, for the reasons given in (iii). With mean 1080, the standard deviation is $\sqrt{1080} \approx 32.9$, and

$$\frac{1350 - 1080}{32.9} \approx 8.2.$$

A value 8 standard deviations above the mean has probability of order 10^{-15} under the model. Even if the true year-to-year spread were two or three times the Poisson value (because of bad winters, say), 1350 would still be very unusual compared with a steady rate of 1080. So something has very probably changed. The question is what.

Compared with what? The headline compares 1990 with a 16-year average, not with 1989. If pedestrian deaths had been rising steadily over the period, the average of 1080 would describe the middle of it (around 1981–82), and 1990 should be well above it. For instance, a steady rise of 30 deaths a year, from about 855 in 1974, has average 1080 and gives about 1335 in 1990. In that case 1990 would be no "rise" at all, just the continuation of a long trend. That trend might deserve a headline, but a different one. We need the year-by-year figures to tell the two stories apart.

Has the exposure changed? More people, more vehicles, or more walking would all increase deaths without making walking any more dangerous per journey. The fair comparison is a rate, such as deaths per 100 000 population or per kilometre walked.

Has the measurement changed? A change in how deaths are recorded or classified would cause an apparent jump. Examples are a change in how long after an accident a death still counts as a road death, or in who counts as a pedestrian, or better reporting. Any such change would need to be ruled out.

Is the word right? "Casualties" normally means people killed *or injured*. The figure quoted is for deaths only, so the headline overstates what the number shows. Injuries are far more numerous, and they might show quite a different pattern.

In summary: the rise to 1350 is far too large to be explained by Poisson variation about a constant rate of 1080. But before calling it "alarming" we need the recent annual figures (to separate a sudden rise from a long-term trend) and the population and traffic figures (to turn counts into rates). We also need to check that the definitions have not changed.

Checks

The probabilities were checked by summing the Poisson series exactly and by simulation with 2×10^5 trials. For (iii), $P(X \geq 7) = 0.0315$ when $\lambda = 1080/365$ (simulation: 0.0317). For (ii), with 6 crashes a year of 30 deaths each, $P(N \leq 4) = 0.285$. Letting the crash size vary uniformly from 1 to 59 gave standard deviation 84 and $P(\text{deaths} \leq 125) \approx 0.28$ by simulation. A quick rule of thumb for a Poisson count is that values within about two square roots of the mean are unremarkable. For one day, $3 + 2 \times 1.7 \approx 6.4$, so 7 is only just outside that range, and with 365 days to choose from such days are bound to occur. For a year, $1080 + 2 \times 33 = 1146$, which is nowhere near 1350.

Common slips

- Comparing totals instead of rates in (i): a risk always needs a denominator.
- Using the Poisson standard deviation $\sqrt{180}$ in (ii). Deaths in air crashes are clustered, so the spread is far larger.
- Stopping at "8 standard deviations, so it's a real rise" in (iv), without asking whether the comparison with a 16-year average, the population or the definitions explain it.

Beyond STEP

Part (ii) is about *overdispersion*. A total made up of a Poisson number of clusters, each of random size, is a *compound Poisson* variable, and its variance is $\lambda\,\mathrm{E}[S^2]$, not λ. Part (iv) leads into hypothesis testing, and into the much harder question of what a statistically significant change actually means. Both appear early in a university probability and statistics course. Blitzstein and Hwang, *Introduction to Probability*, is a good place to read about Poisson models and their limits.

WORKED SOLUTION

STEP I 1992, Question 15

Probability and Statistics • Expected Waiting Time

STEP I 1992, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q15

The key idea

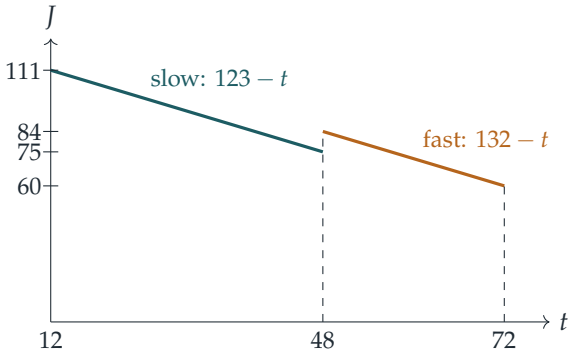
Everything repeats every hour, so it is enough to look at a passenger arriving at a time uniformly distributed over one hour. That hour splits into two gaps: the passengers in the gap before a train all catch that train, and each one's journey is "wait plus travel time". Within a gap the wait is uniform, so its average is half the gap. This turns the whole question into a quadratic in the length of one gap.

Solution

The present timetable

Measure time t in minutes past noon (any hour will do), and let the passenger's arrival time T be uniform on the hour-long interval $(12, 72]$. The trains in this window are the slow train at 48, arriving in London at $48 + 75 = 123$, and the fast train at 72 (that is, 12 minutes past the next hour), arriving at $72 + 60 = 132$.

- If $12 < T \leq 48$, the first available train is the slow one, so the journey time is $J = 123 - T$.
- If $48 < T \leq 72$, the first available train is the fast one, so $J = 132 - T$.



Since T has density $\frac{1}{60}$ on $(12, 72]$,

$$\mathrm{E}(J) = \frac{1}{60}\left[\int_{12}^{48}(123-t)\,\mathrm{d}t + \int_{48}^{72}(132-t)\,\mathrm{d}t\right].$$

Each integrand is linear, so each integral is the length of the interval times the value at its midpoint:

$$\int_{12}^{48}(123-t)\,\mathrm{d}t = 36\times(123-30) = 3348, \qquad \int_{48}^{72}(132-t)\,\mathrm{d}t = 24\times(132-60) = 1728.$$

Hence

$$\mathrm{E}(J) = \frac{3348+1728}{60} = \frac{5076}{60} = 84.6 \text{ minutes},$$

as required.

Retiming the fast train

Now let the fast train leave at x minutes past the hour, the slow one still at 48. Let d be the number of minutes from a fast departure to the next slow departure, so $0 \le d < 60$ and $d \equiv 48 - x \pmod{60}$. (For the original timetable, $d = 36$.) The hour then splits into

- a gap of length d before each slow train: those passengers wait W uniform on $[0, d)$ and then travel for 75 minutes;
- a gap of length $60 - d$ before each fast train: those passengers wait uniformly on $[0, 60-d)$ and then travel for 60 minutes.

A passenger lands in the first gap with probability $\frac{d}{60}$ and in the second with probability $\frac{60-d}{60}$, and a uniform wait on a gap of length g has mean $\frac{g}{2}$. So, by conditioning on which gap the passenger arrives in,

$$\begin{aligned}
60\,\mathrm{E}(J) &= d\left(\frac{d}{2}+75\right) + (60-d)\left(\frac{60-d}{2}+60\right)\\
&= \frac{d^2}{2} + 75d + \frac{3600-120d+d^2}{2} + 3600 - 60d\\
&= d^2 - 45d + 5400.
\end{aligned}$$

As a check, $d = 36$ gives $1296 - 1620 + 5400 = 5076$, matching the first part.

Ask yourself Why is the best gap not $d = 30$, splitting the hour evenly?

Each passenger in the slow train's gap pays a 15-minute penalty for its slower journey. So we want fewer passengers in that gap, which means making it shorter than half an hour, but not so short that the fast train's gap (and hence those passengers' waits) becomes long. The quadratic balances the two effects.

Completing the square,

$$d^2 - 45d + 5400 = \left(d - \tfrac{45}{2}\right)^2 + 5400 - \tfrac{2025}{4},$$

which is least when $d = 22.5$, a value inside $[0, 60)$. Then $60\,\mathrm{E}(J) = \frac{19575}{4}$, so the minimum average journey time is

$$\mathrm{E}(J) = \frac{19575}{240} = \frac{1305}{16} = 81\tfrac{9}{16} \approx 81.56 \text{ minutes}.$$

The fast train must leave 22.5 minutes before the slow one, so

$$x = 48 - 22.5 = 25.5,$$

that is, the fast train should leave at $25\frac{1}{2}$ minutes past the hour. If departures must be on whole minutes, $x = 25$ and $x = 26$ ($d = 23$ and $d = 22$) tie, each giving $\frac{2447}{30} \approx 81.57$ minutes.

Checks

- At $d = 0$ the two trains leave together; everyone takes the fast one and waits on average 30 minutes, so $\mathrm{E}(J) = 90$. The formula gives $\frac{5400}{60} = 90$.
- As $d \to 60$ the slow train leaves just before the fast one and catches nearly everyone: average wait 30, journey 75, total 105. The formula gives $\frac{3600 - 2700 + 5400}{60} = 105$.
- A numerical integration of the journey time over the hour for every x on a grid of step 0.1 gives 84.6 at $x = 12$ and a minimum of 81.5625 at $x = 25.5$.

Common slips

- Taking the passengers who arrive between 48 and 60 past the hour to the fast train "at 12" without adding the hour: their train leaves at 12 past the *next* hour.
- Averaging the two journey times weighted by the number of trains rather than by the lengths of the gaps; it is the length of the gap before a train that decides how many passengers it carries.

Beyond STEP

The quantity computed here is a mean waiting time for a periodic timetable, and the observation that long gaps catch more passengers is a form of the *inspection paradox*: a randomly arriving passenger is more likely to land in a long gap than a short one. This is a central idea in renewal theory and queueing theory, which study random arrivals and services. Grimmett and Welsh, *Probability: An Introduction*, is a good place to meet the underlying tools of conditioning and expectation.

WORKED SOLUTION

STEP I 1992, Question 16

Probability and Statistics • Network Reliability

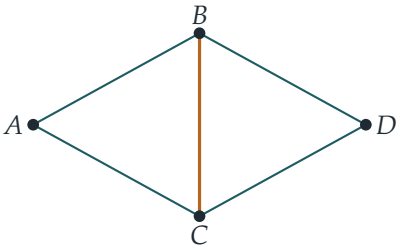
STEP I 1992, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s1-q16

The key idea

The road network is a "bridge": two routes from A to D, one through B and one through C, with the road CB joining them in the middle. The cleanest way to handle the bridge is to condition on whether CB is open, because once we know that, everything reduces to roads in series and in parallel. Snow-proofing a road has the same effect as fusing its two ends into a single town, so we can compare the two proposals by looking at the smaller networks they create.



Solution

Throughout, write $q = 1 - p$ for the probability that a given road is open. The roads are blocked independently, so the probability of any particular pattern of open and blocked roads is a product of ps and qs. There are four routes from A to D that don't revisit a town:

$$ABD, \qquad ACD, \qquad ABCD, \qquad ACBD.$$

The only open route is $ABCD$

For $ABCD$ to be open we need AB, BC and CD all open. For it to be the *only* open route, ABD must be blocked, and since AB is open this forces BD to be blocked. Similarly ACD must be blocked, and since CD is open this forces AC to be blocked. Then $ACBD$

is automatically blocked too (it uses both AC and BD).

So the event is exactly "AB, BC, CD open and AC, BD blocked". By independence its probability is

$$q^3p^2 = p^2(1-p)^3,$$

as required.

The probability of getting through at present

Let E be the event that A and D are joined by unblocked roads, and condition on the road CB.

If CB is open, then B and C behave like a single town. We get from A to that town unless both AB and AC are blocked, and from it to D unless both BD and CD are blocked. These involve different roads, so

$$\mathrm{P}(E \mid CB \text{ open}) = (1-p^2)^2.$$

If CB is blocked, the only possible routes are ABD and ACD, each open with probability q^2, independently since they share no road. So

$$\mathrm{P}(E \mid CB \text{ blocked}) = 1-(1-q^2)^2.$$

It is easier to work with the probability that A is cut off from D. Since $1-(1-p^2)^2 = p^2(2-p^2)$ and $1-q^2 = p(2-p)$,

$$\begin{aligned}\mathrm{P}(\text{cut off}) &= q\,p^2(2-p^2) + p\cdot p^2(2-p)^2\\ &= p^2[(1-p)(2-p^2) + p(4-4p+p^2)]\\ &= p^2\,(2+2p-5p^2+2p^3).\end{aligned}$$

Which road to snow-proof

Snow-proofing CB. Now CB is always open, so we are permanently in the first case above:

$$\mathrm{P}(\text{cut off}) = 1-(1-p^2)^2 = 2p^2-p^4.$$

Snow-proofing AB. Now A and B act as a single town, and we are cut off from D exactly when BD is blocked *and* C offers no way through. The way through C needs CD open and at least one of AC, CB open, so it fails with probability $p+q\,p^2$. Hence

$$\mathrm{P}(\text{cut off}) = p\,(p+qp^2) = p^2+p^3-p^4.$$

Ask yourself Both answers start with a p^2 term. How do we decide which is smaller for every p?

Subtract them:

$$(2p^2-p^4)-(p^2+p^3-p^4) = p^2-p^3 = p^2(1-p) > 0 \quad \text{for } 0<p<1.$$

So snow-proofing AB always leaves the smaller chance of being cut off, whatever the value of p. The government should choose AB; the two choices are never equally good.

This matches intuition. Fixing CB only helps when it is needed as part of a detour, but fixing AB makes A and B a single town, which is joined to D by the direct road BD and also by the path through C.

The final part

With $p = \frac{1}{10}$, the present probability of being cut off is

$$\frac{1}{100}\left(2 + \frac{2}{10} - \frac{5}{100} + \frac{2}{1000}\right) = \frac{1}{100} \times 2.152 = 0.02152.$$

This is greater than $\frac{1}{100}$, so the government will go ahead (and, by the previous part, it will snow-proof AB).

The main term $2p^2 = 0.02$ already settles it: there are two ways of being cut off by just two blocked roads, namely $\{AB, AC\}$ and $\{BD, CD\}$.

Checks

- As $p \to 0$, the cut-off probability is about $2p^2$, which counts the two pairs of roads whose blocking isolates A or D. After snow-proofing AB only the pair $\{BD, CD\}$ remains, giving about p^2, as our formula says. After snow-proofing CB both pairs remain, giving about $2p^2$.
- At $p = 1$ the present formula gives $1 \cdot (2 + 2 - 5 + 2) = 1$: if every road is blocked we are certainly cut off.
- An exact enumeration over all 2^5 patterns of open and blocked roads confirms every formula above for several values of p. At $p = \frac{1}{10}$ it gives 0.02152 now, 0.0109 after snow-proofing AB and 0.0199 after snow-proofing CB. A simulation of 2×10^5 winters gave 0.0215.

Common slips

- Adding the probabilities of the four routes being open. The routes share roads, so these events overlap and the sum overcounts.
- In the first part, forgetting that "the only open route" forces AC and BD to be blocked, and giving just $(1 - p)^3$.
- Comparing the two proposals at $p = \frac{1}{10}$ only. The difference $p^2(1 - p)$ shows that AB is better for every p.

Beyond STEP

This is a small example from *network reliability*, and the trick of conditioning on one edge (or fusing its ends) is the deletion–contraction idea that runs through graph theory. For large random networks the same questions lead to *percolation theory*, where there is a sharp threshold value of p at which long-range connections suddenly disappear. Grimmett and Welsh, *Probability: An Introduction*, is a good place to build the conditioning tools used here.

WORKED SOLUTION

STEP II 1992, Question 1

Pure • Limits of Sequences

STEP II 1992, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q1

The key idea

Each limit needs one standard move, and the skill is picking the right one. For rational functions, divide top and bottom by the highest power of n. For something bounded divided by something growing, squeeze it. For a known limit in disguise, substitute so the disguise falls away. And for a difference of square roots, multiply by the conjugate: that turns a difference of two nearly equal quantities into a quotient you can see clearly.

Solution

Part (i)

Dividing top and bottom by n,

$$\frac{n}{n+1} = \frac{1}{1+\frac{1}{n}} \longrightarrow \frac{1}{1+0} = 1,$$

since $\frac{1}{n} \to 0$. The limit is 1.

Part (ii)

The highest power is n^2, which appears only in the denominator. Dividing top and bottom by n^2,

$$\frac{5n+1}{n^2-3n+4} = \frac{\frac{5}{n}+\frac{1}{n^2}}{1-\frac{3}{n}+\frac{4}{n^2}} \longrightarrow \frac{0}{1} = 0.$$

The limit is 0: the denominator grows like n^2, but the numerator only like n.

Part (iii)

We know nothing useful about $\sin n$ for large integers n except that it is bounded: $|\sin n| \leq 1$. That is enough, because

$$\left|\frac{\sin n}{n}\right| \leq \frac{1}{n} \longrightarrow 0.$$

So $\dfrac{\sin n}{n}$ is squeezed between $-\frac{1}{n}$ and $\frac{1}{n}$, both of which tend to 0, and the limit is 0.

Part (iv)

? **Ask yourself** Isn't this the same as (iii)?

No: here the angle $1/n$ is shrinking to 0, not growing. Put $x = 1/n$. As $n \to \infty$ we have $x \to 0^+$, and the expression is $\dfrac{\sin x}{x}$, whose limit as $x \to 0$ is the standard result

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

(One way to see it: this is the derivative of sin at 0, namely $\cos 0 = 1$. Or use $\sin x = x - \frac{x^3}{6} + \cdots$, so $\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \cdots$.) The limit is 1.

Part (v)

As $n \to \infty$, $\arctan n \to \frac{\pi}{2}$, because $\tan\theta \to +\infty$ as $\theta \to \frac{\pi}{2}$ from below and arctan is continuous and increasing. The limit $\frac{\pi}{2}$ is not zero, so we may take reciprocals:

$$(\arctan n)^{-1} \longrightarrow \frac{1}{\pi/2} = \frac{2}{\pi}.$$

Note that $(\arctan n)^{-1}$ means $1/\arctan n$, not the inverse function of arctan (which would just give tan).

Part (vi)

Top and bottom both tend to 0, so we can't just take limits separately. Multiply each by its conjugate:

$$\sqrt{n+1} - \sqrt{n} = \frac{(n+1) - n}{\sqrt{n+1} + \sqrt{n}} = \frac{1}{\sqrt{n+1} + \sqrt{n}}, \qquad \sqrt{n+2} - \sqrt{n} = \frac{2}{\sqrt{n+2} + \sqrt{n}}.$$

Hence

$$\frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n+2} - \sqrt{n}} = \frac{\sqrt{n+2} + \sqrt{n}}{2\left(\sqrt{n+1} + \sqrt{n}\right)} = \frac{\sqrt{1 + \frac{2}{n}} + 1}{2\left(\sqrt{1 + \frac{1}{n}} + 1\right)},$$

dividing top and bottom by $\sqrt{n}$. As $n \to \infty$ both square roots tend to 1 (since $\sqrt{\ }$ is continuous), so the limit is

$$\frac{1+1}{2(1+1)} = \frac{1}{2}.$$

Ask yourself Why should the answer be $\frac{1}{2}$ rather than, say, 1?

Near a large n the graph of $\sqrt{x}$ is almost a straight line with gradient $\frac{1}{2\sqrt{n}}$. So $\sqrt{n+1} - \sqrt{n} \approx \frac{1}{2\sqrt{n}}$ (a step of 1) and $\sqrt{n+2} - \sqrt{n} \approx \frac{2}{2\sqrt{n}}$ (a step of 2): the ratio is about $\frac{1}{2}$, the ratio of the step lengths.

Checks

Plug in a large n. With $n = 10^6$: (i) gives 0.999999; (ii) gives 5×10^{-6}; (iii) gives about -3.5×10^{-7}; (iv) gives 1.0000000 to seven places; (v) gives 0.636620, against $2/\pi \approx 0.636620$; and (vi), computed from the rationalised form, gives 0.50000012. In (vi) also check the rationalised form against the original at small n: at $n = 1$ both equal $(\sqrt{2}-1)/(\sqrt{3}-1) \approx 0.5658$.

Common slips

- Treating (iii) and (iv) alike: in (iii) the sine stays bounded while the denominator grows; in (iv) both top and bottom shrink to 0, and their ratio tends to 1.
- Reading $(\arctan n)^{-1}$ in (v) as the inverse function.
- Computing (vi) on a calculator straight from the original form at huge n: the subtraction of nearly equal numbers loses accuracy. Rationalise first.

Beyond STEP

At university, "$a_n \to L$" is defined precisely: for every $\varepsilon > 0$ there is an N such that $|a_n - L| < \varepsilon$ for all $n > N$. Every move above, such as the squeeze in (iii), taking reciprocals of a non-zero limit in (v) and pushing a limit through a continuous function in (vi), becomes a theorem proved from that definition in a first analysis course. Abbott's *Understanding Analysis* develops these ideas carefully and readably.

WORKED SOLUTION

STEP II 1992, Question 2

Pure • Clairaut Differential Equation

STEP II 1992, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q2

The key idea

The equation has the form $y = xp - f(p)$ with $p = \mathrm{d}y/\mathrm{d}x$, known as a Clairaut equation. Differentiating once makes the awkward terms p cancel, and what is left factorises into two much simpler equations. The catch is that differentiating loses information (constants disappear), so every candidate solution must be substituted back into $(*)$ to fix its constants.

Solution

Write $p = \dfrac{\mathrm{d}y}{\mathrm{d}x}$, so that $(*)$ reads $y = xp - \cosh p$.

Differentiating

Differentiate both sides with respect to x, using the product rule on xp and the chain rule on $\cosh p$:

$$p = p + x\frac{\mathrm{d}p}{\mathrm{d}x} - \sinh p\,\frac{\mathrm{d}p}{\mathrm{d}x}.$$

The p on each side cancels, leaving

$$\frac{\mathrm{d}p}{\mathrm{d}x}\left(x - \sinh p\right) = 0, \qquad \text{that is,} \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2}\left(x - \sinh\frac{\mathrm{d}y}{\mathrm{d}x}\right) = 0.$$

A product is zero only if one of its factors is, so either $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$ or $x - \sinh\left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right) = 0$, as required.

The two equations

First equation. Integrating $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$ twice gives the general solution

$$y = Ax + B, \qquad A, B \text{ arbitrary constants.}$$

Second equation. Here $\sinh p = x$. Since sinh is a bijection from $\mathbb{R}$ to $\mathbb{R}$, this gives $p = \sinh^{-1} x$ for every x. Integrating by parts with $u = \sinh^{-1} x$, $\mathrm{d}v = \mathrm{d}x$:

$$\begin{aligned} y &= \int \sinh^{-1} x \,\mathrm{d}x = x \sinh^{-1} x - \int \frac{x}{\sqrt{1+x^2}} \,\mathrm{d}x \\ &= x \sinh^{-1} x - \sqrt{1+x^2} + C, \qquad C \text{ an arbitrary constant.} \end{aligned}$$

Solutions of $(*)$

Every solution of $(*)$ satisfies one of the two derived equations, but not conversely, because differentiating threw away a condition. So we substitute each family back into $(*)$.

> **Ask yourself** The derived equations have two and one arbitrary constants. Why can't all of these solve a first-order equation like $(*)$?

A first-order equation should have a one-parameter family of solutions, and each family above has too many constants or an unchecked one. Differentiating $(*)$ only tells us that $y - xp + \cosh p$ is *constant*; we need that constant to be zero.

Lines. With $y = Ax + B$ we have $p = A$, and $(*)$ becomes

$$Ax + B = Ax - \cosh A \iff B = -\cosh A.$$

So we get the one-parameter family of straight lines

$$y = Ax - \cosh A, \qquad A \in \mathbb{R}.$$

The other curve. With $y = x \sinh^{-1} x - \sqrt{1+x^2} + C$ we have $p = \sinh^{-1} x$, so $\sinh p = x$ and $\cosh p = \sqrt{1 + \sinh^2 p} = \sqrt{1+x^2}$ (taking the positive root because $\cosh p > 0$). The right-hand side of $(*)$ is then

$$xp - \cosh p = x \sinh^{-1} x - \sqrt{1+x^2},$$

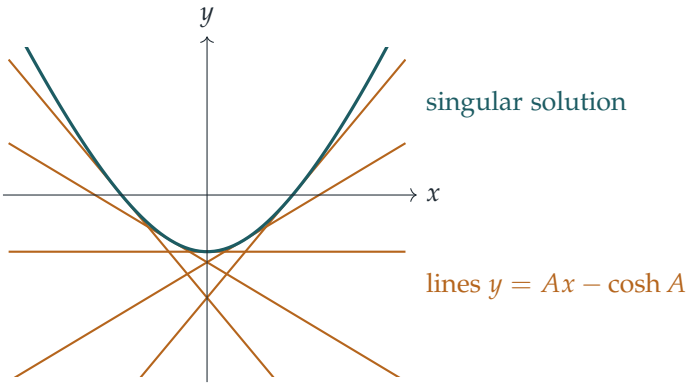
which equals y exactly when $C = 0$. This gives one extra solution, which contains no arbitrary constant:

$$y = x \sinh^{-1} x - \sqrt{1+x^2}.$$

So the solutions of $(*)$ are the lines $y = Ax - \cosh A$ (for each real A) together with the single curve $y = x \sinh^{-1} x - \sqrt{1+x^2}$.

How the two fit together

The line with slope A touches the curve at $x = \sinh A$. There the curve's gradient is $\sinh^{-1}(\sinh A) = A$, and its height is $A\sinh A - \cosh A$, the same as the line's. The curve has $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} = \dfrac{1}{\sqrt{1+x^2}} > 0$, so it is convex and every one of these tangent lines lies below it. The curve is the *envelope* of the family of lines, and is called the *singular solution*.



This also explains a subtlety: since the condition $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}(x - \sinh p) = 0$ holds point by point, a solution may switch between the two types. For example, following the curve up to $x = \sinh A$ and then continuing along its tangent line there gives a function with a continuous derivative that still satisfies $(*)$ everywhere. Pieces like this are built from the solutions found above, and the expected answer is the family of lines plus the singular curve.

Checks

- At $A = 0$ the line is $y = -1$, and the curve has $y(0) = 0 - 1 = -1$ with gradient $\sinh^{-1} 0 = 0$: they touch at $(0, -1)$, as predicted.
- Numerically, with $x = 0.7$: $\sinh^{-1} 0.7 \approx 0.6527$, so the curve gives $y \approx 0.4569 - 1.2207 = -0.7638$, and $xp - \cosh p \approx 0.4569 - 1.2207$ agrees. Adding any $C \neq 0$ to y leaves a mismatch of exactly C.
- Checking $(*)$ by finite differences at random points gives residuals at rounding-error level for every line $y = Ax - \cosh A$ and for the singular curve.

Common slips

- Stopping at $y = Ax + B$ or at $y = x\sinh^{-1} x - \sqrt{1+x^2} + C$: these solve the *differentiated* equation, and $(*)$ itself forces $B = -\cosh A$ and $C = 0$.

- Missing the singular solution altogether by "cancelling" the factor $\dfrac{\mathrm{d}p}{\mathrm{d}x}$.

Beyond STEP

Equations of the form $y = xp - f(p)$ are Clairaut equations. Their general solution is always the family of lines $y = cx - f(c)$, and the singular solution is the envelope of that family. Here $f = \cosh$, and the envelope $y = x\sinh^{-1}x - \sqrt{1+x^2}$ is the Legendre transform of cosh, a construction that appears in mechanics (passing from the Lagrangian to the Hamiltonian) and in thermodynamics. The example also shows that solutions of first-order equations need not be unique when the conditions of the existence and uniqueness theorem fail. Boyce and DiPrima, *Elementary Differential Equations*, covers first-order equations and that uniqueness theorem.

WORKED SOLUTION

STEP II 1992, Question 3

Pure • Hypocycloids and Area

STEP II 1992, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q3

The key idea

Everything follows from *rolling without slipping*: the length of arc the small circle has rolled along the big circle equals the length of arc of the small circle that has been in contact. That fixes how far the small circle has turned, and then B is just "centre P plus a unit vector in the right direction". For the sketch, it pays to notice that the coordinates simplify very neatly with the triple-angle formulae.

Solution

Finding the coordinates of B

Since the small circle (radius 1) stays in contact with the inside of the big one (radius 4), its centre is always at distance $4-1=3$ from O. With ϕ the angle AOP,

$$P = (3\cos\phi,\ 3\sin\phi).$$

Let T be the current point of contact. It lies on the ray OP produced, so $T = (4\cos\phi, 4\sin\phi)$ and the direction from P to T makes angle ϕ with the x-axis.

Initially B was at A, the point of contact. Since there is no slipping, the arc AT of the big circle has the same length as the arc BT of the small circle:

$$4\phi = 1 \times \angle BPT \quad \Longrightarrow \quad \angle BPT = 4\phi.$$

Ask yourself Is B at angle $\phi + 4\phi$ or $\phi - 4\phi$, measured at P?

As P moves anticlockwise around O, the small circle rolls along the inside of the big one and so turns *clockwise* (the arrow in the figure). The point B was the contact point earlier, so it has been left behind the new contact point T: going from T to B around the small circle is a clockwise turn of 4ϕ. Hence the direction $\overrightarrow{PB}$ makes angle $\phi - 4\phi = -3\phi$

with the x-axis, and since $PB = 1$,

$$\begin{aligned} B &= P + (\cos(-3\phi),\ \sin(-3\phi)) \\ &= (3\cos\phi + \cos 3\phi,\ 3\sin\phi - \sin 3\phi), \end{aligned}$$

as required. (Check: $\phi = 0$ gives $(4, 0) = A$.)

Sketching C

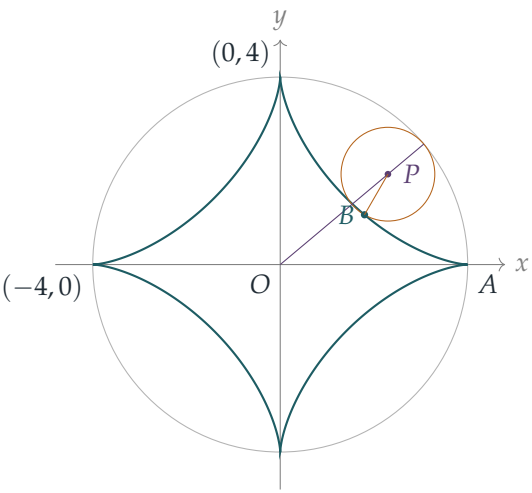
Using $\cos 3\phi = 4\cos^3\phi - 3\cos\phi$ and $\sin 3\phi = 3\sin\phi - 4\sin^3\phi$,

$$x = 4\cos^3\phi, \qquad y = 4\sin^3\phi.$$

So C is the curve $x^{2/3} + y^{2/3} = 4^{2/3}$, an *astroid*. Some features:

- B is on the big circle exactly when $\cos^6\phi + \sin^6\phi = 1$, i.e. $\phi = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, giving the points $(\pm 4, 0)$, $(0, \pm 4)$. Here B is the contact point, so it is momentarily at rest and the curve has a cusp.
- The curve is symmetric in both axes (replace ϕ by $-\phi$ or $\pi - \phi$).
- In the first quadrant, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{12\sin^2\phi\cos\phi}{-12\cos^2\phi\sin\phi} = -\tan\phi$, so the curve leaves $(4, 0)$ along the x-axis and arrives at $(0, 4)$ along the y-axis. It bows *inwards*: at $\phi = \frac{\pi}{4}$ it passes through $(\sqrt{2}, \sqrt{2})$, well inside the line $x + y = 4$.

After one circuit of P (ϕ from 0 to 2π) the small circle has turned 3 times relative to OP and B is back at A, so C is a closed curve with four cusps.



The area

We have $x = 3\cos\phi + \cos 3\phi$ and

$$\frac{\mathrm{d}y}{\mathrm{d}\phi} = 3\cos\phi - 3\cos 3\phi,$$

so

$$x\frac{\mathrm{d}y}{\mathrm{d}\phi} = 9\cos^2\phi - 6\cos\phi\cos 3\phi - 3\cos^2 3\phi.$$

Over a whole period, $\int_0^{2\pi}\cos^2 k\phi\,\mathrm{d}\phi = \pi$ for any integer $k \neq 0$ (write it as $\frac{1}{2}(1+\cos 2k\phi)$), and

$$\int_0^{2\pi}\cos\phi\cos 3\phi\,\mathrm{d}\phi = \frac{1}{2}\int_0^{2\pi}(\cos 4\phi + \cos 2\phi)\,\mathrm{d}\phi = 0.$$

Therefore

$$\text{Area} = \int_0^{2\pi} x\frac{\mathrm{d}y}{\mathrm{d}\phi}\,\mathrm{d}\phi = 9\pi - 0 - 3\pi = 6\pi.$$

The answer is positive, as it should be: B goes round C anticlockwise as ϕ increases.

Checks

- Since C bows inwards, the region lies inside the square with vertices at the four cusps, of area 32; and indeed $6\pi \approx 18.85 < 32$.
- Numerically, Simpson's rule on the integral and the shoelace area of a 200 000-gon inscribed in C both give $18.8496\ldots = 6\pi$.
- The standard astroid formula, area $\frac{3}{8}\pi a^2$ with $a = 4$, also gives 6π.
- At each cusp the numerical speed of B is zero, confirming that B is at rest when it touches the big circle, exactly as rolling demands.

Common slips

- Taking the angle of $\overrightarrow{PB}$ as $+5\phi$ or $+3\phi$: the small circle turns the *opposite* way to P's motion round O.
- Using the radius 4 instead of 3 for the path of P.
- Drawing rounded ends instead of sharp cusps, or arcs that bulge outwards.

Beyond STEP

Curves traced by a point on a circle rolling inside another are *hypocycloids*; rolling outside gives *epicycloids*. When the radii are in ratio 4 : 1 we get the astroid, and ratio 3 : 1 gives the three-cusped deltoid. The area formula $\int x\,\mathrm{d}y$ is a special case of Green's theorem, which relates an integral round a closed curve to a double integral over the region inside; it is a highlight of first-year vector calculus. Needham's *Visual Complex Analysis* treats these rolling curves elegantly using $z = 3e^{i\phi} + e^{-3i\phi}$.

WORKED SOLUTION

STEP II 1992, Question 4

Pure • Axioms for Differentiation

STEP II 1992, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q4

This question has a full guided treatment in *Beyond STEP*, Section 6.5, *Differentiation from Axioms*: https://beyondstep.keithmcnulty.org/ch06/derivations.html

The key idea

The four rules are exactly the familiar properties of differentiation: the product rule, linearity and $\frac{\mathrm{d}}{\mathrm{d}x}x = 1$. The question asks us to show that these properties alone pin the operation down completely on polynomials. The trick for constants is to feed the product rule a product whose answer we already know, such as $1 \times 1 = 1$. After that, the product rule builds up every power of x one step at a time, and linearity handles the rest.

Throughout, we must use *only* the four given rules. We may not assume anything about $\lozenge$ just because it looks like differentiation; that is what we are trying to prove.

Solution

Constants

? **Ask yourself** We know nothing about $\lozenge 1$. How can the rules tell us what it is?

Apply the product rule (i) to the polynomials $\mathrm{f}(x) = \mathrm{g}(x) = 1$. Their product is 1 again, so

$$\lozenge 1 = \lozenge(1 \cdot 1) = 1 \cdot \lozenge 1 + 1 \cdot \lozenge 1 = 2\lozenge 1.$$

Subtracting $\lozenge 1$ from both sides gives $\lozenge 1 = 0$, the zero polynomial.

Now let $\mathrm{f}(x) = c$ be any constant polynomial. Then $\mathrm{f}(x) = c \cdot 1$, and c is a constant, so rule (iv) gives

$$\lozenge c = \lozenge(c \cdot 1) = c\lozenge 1 = c \cdot 0 = 0.$$

This covers $c = 0$ too, so $\lozenge \mathrm{f}(x) = 0$ for every constant polynomial, as required.

(Another route to $\Diamond 1 = 0$: write $x = x \cdot 1$, so rule (i) and rule (iii) give $1 = 1 \cdot 1 + x\,\Diamond 1$, hence $x\,\Diamond 1 = 0$, and a polynomial that gives 0 when multiplied by x must be the zero polynomial.)

The powers x^2 and x^3

Using the product rule with $\mathrm{f}(x) = \mathrm{g}(x) = x$, and then rule (iii),

$$\Diamond x^2 = \Diamond(x \cdot x) = x\,\Diamond x + x\,\Diamond x = x + x = 2x.$$

Next, $x^3 = x^2 \cdot x$, so by the product rule, rule (iii) and the result we have just found,

$$\Diamond x^3 = x\,\Diamond x^2 + x^2\,\Diamond x = x \cdot 2x + x^2 \cdot 1 = 3x^2.$$

Any polynomial

Step 1: powers of x. We prove by induction that

$$\Diamond x^n = n x^{n-1} \qquad \text{for every integer } n \geq 1.$$

For $n = 1$ this is rule (iii). Suppose it holds for some $n = k \geq 1$. Writing $x^{k+1} = x^k \cdot x$ and using the product rule, then rule (iii) and the induction hypothesis,

$$\Diamond x^{k+1} = x\,\Diamond x^k + x^k\,\Diamond x = x \cdot k x^{k-1} + x^k = (k+1)x^k,$$

so the result holds for $n = k + 1$. By induction it holds for all $n \geq 1$. For $n = 0$, $x^0 = 1$ is a constant, so $\Diamond x^0 = 0$ by the first part. In every case $\Diamond x^n = \frac{\mathrm{d}}{\mathrm{d}x} x^n$.

Step 2: sums. Rule (ii) is stated for two polynomials, but it extends to any finite sum by induction on the number of terms: if $\Diamond(\mathrm{f}_1 + \cdots + \mathrm{f}_m) = \Diamond \mathrm{f}_1 + \cdots + \Diamond \mathrm{f}_m$, then applying rule (ii) to $(\mathrm{f}_1 + \cdots + \mathrm{f}_m) + \mathrm{f}_{m+1}$ gives the same statement for $m + 1$ terms.

Step 3: putting it together. Any polynomial can be written $\mathrm{h}(x) = a_0 + a_1 x + \cdots + a_n x^n$ with constant coefficients a_k. By Step 2, then rule (iv) on each term, then Step 1,

$$\begin{aligned} \Diamond \mathrm{h}(x) &= \Diamond(a_0) + \Diamond(a_1 x) + \cdots + \Diamond(a_n x^n) \\ &= a_0\,\Diamond 1 + a_1\,\Diamond x + \cdots + a_n\,\Diamond x^n \\ &= 0 + a_1 + 2a_2 x + \cdots + n a_n x^{n-1} \\ &= \frac{\mathrm{d}}{\mathrm{d}x}\big(\mathrm{h}(x)\big), \end{aligned}$$

which is what we had to prove.

Checks

It's worth confirming that differentiation really does obey all four rules, otherwise the result would be vacuous; it does (the product rule, linearity and $\frac{\mathrm{d}}{\mathrm{d}x} x = 1$). We also built $\Diamond$ by computer using only the four rules, as in the solution, and compared it with the ordinary derivative on 500 random polynomials of degree up to 7 with fractional coefficients. They always agreed, and agreed with a numerical difference quotient; for example the rules give $\Diamond x^3 = 3x^2$.

Common slips

- Saying "$\Diamond c = 0$ because the derivative of a constant is zero". That assumes what we are trying to prove; the zero has to come from the rules.
- Forgetting the constant term in the final part. The induction only starts at $n = 1$, so a_0 needs the result for constants.
- Using rule (ii) for a sum of many terms without noting that it extends from two terms by induction.

Beyond STEP

An operation obeying rules (i), (ii) and (iv) is called a *derivation*, and derivations are studied throughout algebra and geometry. This question shows that on polynomials a derivation is fixed entirely by its value on x. More generally, $\Diamond \mathrm{h} = \mathrm{p}(x)\,\mathrm{h}'(x)$ where $\mathrm{p} = \Diamond x$, and in differential geometry this idea becomes the definition of a vector field. For the algebraic setting of polynomial rings, see Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP II 1992, Question 5

Pure • Matrix Groups

STEP II 1992, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q5

The key idea

Everything here rests on the condition $wz - xy = 1$, which makes the inverse of a matrix in S especially simple. The order-3 part is best handled by noticing that every 2×2 matrix satisfies a quadratic equation built from its trace and determinant. For a matrix in S this lets us write $\mathbf{A}^2$, and then $\mathbf{A}^3$, in terms of $\mathbf{A}$ and $\mathbf{I}$, so the condition $\mathbf{A}^3 = \mathbf{I}$ becomes a condition on the trace $w + z$.

(The paper prints the bottom-right entry of $\mathbf{A}$ as x; the final part refers to z, so the intended matrix clearly has z there, and we use that throughout.)

Solution

The order of an element

The order of $g \in G$ is the smallest positive integer n with $g^n = e$, where e is the identity of G. If no positive power of g equals e, then g has infinite order. The identity is the only element of order 1.

The inverse

Let

$$\mathbf{A} = \begin{pmatrix} w & x \\ y & z \end{pmatrix}, \qquad wz - xy = 1.$$

The usual formula for a 2×2 inverse divides by the determinant, which here is 1, so

$$\mathbf{A}^{-1} = \begin{pmatrix} z & -x \\ -y & w \end{pmatrix}.$$

Multiplying out confirms this: $\mathbf{A}\mathbf{A}^{-1} = \begin{pmatrix} wz - xy & 0 \\ 0 & wz - xy \end{pmatrix} = \mathbf{I}$, and similarly $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$.

S is a group

We check the four group axioms.

- *Closure.* If $\mathbf{A}, \mathbf{B} \in S$, then $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B} = 1 \times 1 = 1$, so $\mathbf{AB} \in S$.
- *Associativity.* We may assume this for matrix multiplication.
- *Identity.* $\det \mathbf{I} = 1$, so $\mathbf{I} \in S$, and $\mathbf{IA} = \mathbf{AI} = \mathbf{A}$ for every $\mathbf{A}$.
- *Inverses.* The matrix $\mathbf{A}^{-1}$ found above has determinant $wz - (-x)(-y) = wz - xy = 1$, so it lies in S.

Hence S is a group under matrix multiplication.

Self-inverse elements and elements of order 2

Comparing $\mathbf{A}^{-1}$ with $\mathbf{A}$ entry by entry, $\mathbf{A}^{-1} = \mathbf{A}$ exactly when

$$z = w, \qquad -x = x, \qquad -y = y,$$

that is, $x = y = 0$ and $z = w$. The determinant condition then reads $w^2 = 1$, so $w = \pm 1$. The self-inverse elements are therefore

$$\mathbf{A} = \mathbf{I} \quad \text{and} \quad \mathbf{A} = -\mathbf{I}.$$

An element has order 2 when $\mathbf{A}^2 = \mathbf{I}$ (equivalently $\mathbf{A}^{-1} = \mathbf{A}$) and $\mathbf{A} \neq \mathbf{I}$. So the only element of order 2 is $-\mathbf{I}$.

Elements of order 3

Write $t = w + z$ for the trace of $\mathbf{A}$. Squaring,

$$\mathbf{A}^2 = \begin{pmatrix} w^2 + xy & x(w+z) \\ y(w+z) & xy + z^2 \end{pmatrix}.$$

Since $xy = wz - 1$, the top-left entry is $w^2 + wz - 1 = tw - 1$, and the bottom-right is $tz - 1$. Hence

$$\mathbf{A}^2 = t\mathbf{A} - \mathbf{I}. \qquad (*)$$

Ask yourself How does $(*)$ help with $\mathbf{A}^3$?

Multiply $(*)$ by $\mathbf{A}$ and use $(*)$ again to remove the $\mathbf{A}^2$:

$$\mathbf{A}^3 = t\mathbf{A}^2 - \mathbf{A} = t(t\mathbf{A} - \mathbf{I}) - \mathbf{A} = (t^2 - 1)\mathbf{A} - t\mathbf{I}.$$

So any power of $\mathbf{A}$ is a combination of $\mathbf{A}$ and $\mathbf{I}$, and we can compare coefficients.

If $w + z + 1 = 0$: then $t = -1$, so $t^2 - 1 = 0$ and $\mathbf{A}^3 = \mathbf{I}$. Also $\mathbf{A} \neq \mathbf{I}$, because $\mathbf{I}$ has trace $2 \neq -1$. The order of $\mathbf{A}$ divides 3 and isn't 1, so it is 3.

Conversely, suppose $\mathbf{A}$ has order 3. Then $\mathbf{A}^3 = \mathbf{I}$ and $\mathbf{A} \neq \mathbf{I}$, so

$$(t^2 - 1)\mathbf{A} = (1 + t)\mathbf{I}.$$

- If $t^2 \neq 1$, then $\mathbf{A} = \dfrac{1+t}{t^2-1}\mathbf{I} = \dfrac{1}{t-1}\mathbf{I}$, a multiple $\lambda\mathbf{I}$ of the identity. Its determinant is $\lambda^2 = 1$, so $\lambda = \pm 1$. But $\mathbf{A} = \mathbf{I}$ is excluded, and $\mathbf{A} = -\mathbf{I}$ has order 2, not 3. So this case can't happen.
- If $t = 1$, the equation becomes $\mathbf{0} = 2\mathbf{I}$, which is false.

The only remaining possibility is $t = -1$, that is, $w + z + 1 = 0$. This proves the "if and only if".

An example. Take $w = 0$ and $z = -1$, and choose $x = -1$, $y = 1$ so that $wz - xy = 0 + 1 = 1$:

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad \mathbf{A}^2 = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \mathbf{A}^3 = \mathbf{I}.$$

Checks

- Multiply out the example above to see $\mathbf{A}^3 = \mathbf{I}$ directly.
- The rotation through 120°, $\begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$, has determinant 1 and trace -1, and of course has order 3.
- We checked by computer that $\mathbf{A}^2 = t\mathbf{A} - \mathbf{I}$ for 20 000 random rational matrices in S. We also searched every matrix in S whose entries are multiples of $\frac{1}{2}$ between -4 and 4: exactly the 48 with trace -1 have order 3, and $-\mathbf{I}$ is the only one of order 2.

Common slips

- Including $\mathbf{I}$ among the elements of order 2. It is self-inverse, but its order is 1.
- Proving only one direction of the order-3 result. The converse needs the cases $t^2 \neq 1$ and $t = 1$ ruled out.
- Forgetting to check that the inverse itself has determinant 1 when verifying the group axioms.

Beyond STEP

The group S is the *special linear group* SL_2, one of the most important groups in mathematics. The identity $\mathbf{A}^2 - (\operatorname{tr}\mathbf{A})\mathbf{A} + (\det\mathbf{A})\mathbf{I} = \mathbf{0}$ is the 2×2 case of the Cayley–Hamilton theorem: every square matrix satisfies its own characteristic equation. It also shows how much the trace alone tells us about an element of SL_2: the argument above generalises to show that an element of finite order n has trace $2\cos(2\pi k/n)$ for some integer k. For more on groups of matrices, see Allenby, *Rings, Fields and Groups*; for Cayley–Hamilton and eigenvalues, see Strang, *Introduction to Linear Algebra*.

WORKED SOLUTION

STEP II 1992, Question 6

Pure • Tangents and Root Counting

STEP II 1992, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q6

The key idea

The solutions of $kx = \ln(2\sec x)$ are the points where the straight line $y = kx$ through the origin meets the curve $y = \ln(2\sec x)$. That curve starts above the origin, is convex and shoots off to infinity, so a line through the origin misses it, touches it, or cuts it twice. The special value k_0 is the gradient of the *tangent from the origin*. Tangency means the two curves have the same height and the same gradient, and those two equations give everything else.

Solution

The two graphs

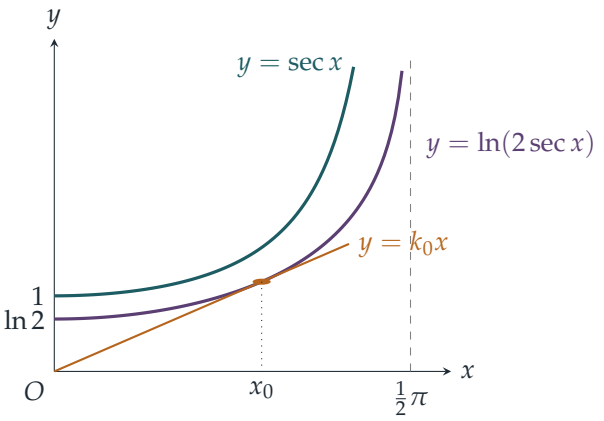
Write $f(x) = \ln(2\sec x) = \ln 2 - \ln\cos x$. Then

$$f'(x) = \frac{\sin x}{\cos x} = \tan x, \qquad f''(x) = \sec^2 x > 0.$$

So $f(0) = \ln 2 \approx 0.69$, $f'(0) = 0$, f is increasing on $(0, \frac{1}{2}\pi)$, it is convex (concave up), and $f(x) \to \infty$ as $x \to \frac{1}{2}\pi^-$ because $\cos x \to 0^+$.

Similarly $y = \sec x$ has $\sec 0 = 1$, gradient $\sec x \tan x$, which is 0 at $x = 0$ and positive afterwards, and $\sec x \to \infty$ as $x \to \frac{1}{2}\pi^-$. Both curves have the vertical asymptote $x = \frac{1}{2}\pi$, and neither is defined there.

The log curve always lies below the secant curve. Putting $u = \sec x \geq 1$, we have $\ln(2u) < u$, because $\ln(2u) - u$ has derivative $\frac{1}{u} - 1 \leq 0$ and so is at most its value at $u = 1$, which is $\ln 2 - 1 < 0$. The log curve also grows far more slowly: as $x \to \frac{1}{2}\pi$ it behaves like ln of the secant.



No solutions for small k, two for large k

Since f is increasing, $f(x) \geq \ln 2$ on $[0, \frac{1}{2}\pi)$, while $kx < \frac{1}{2}\pi k$ there. If $k \leq 2\ln 2/\pi \approx 0.44$, then $kx < \ln 2 \leq f(x)$ throughout, so the line passes underneath the curve and there is no solution.

For large k the line starts below the curve (at $x = 0$ it has height $0 < \ln 2$), but it is steep enough to get above the curve somewhere: at $x = \frac{1}{4}\pi$ the curve has height $\ln(2\sqrt{2}) = \frac{3}{2}\ln 2 \approx 1.04$, and the line is higher there as soon as $k > \frac{6}{\pi}\ln 2 \approx 1.32$. Near $\frac{1}{2}\pi$ the curve escapes to infinity while the line stays at most $\frac{1}{2}\pi k$, so the curve ends up above the line again. The line crosses the curve going up and then again coming down: two solutions.

There cannot be more than two. The function $f(x) - kx$ has second derivative $\sec^2 x > 0$, so it is convex, and a convex function has at most two zeros.

Why k_0 exists

Every solution has $x > 0$, since $f(0) = \ln 2 \neq 0$, so we can divide by x and rewrite the equation as

$$k = g(x), \qquad \text{where } g(x) = \frac{\ln(2\sec x)}{x}, \quad 0 < x < \tfrac{1}{2}\pi.$$

So the number of solutions is the number of times the graph of g reaches height k. Now

$$g'(x) = \frac{xf'(x) - f(x)}{x^2} = \frac{h(x)}{x^2}, \qquad h(x) = x\tan x - \ln(2\sec x),$$

and $h'(x) = \tan x + x\sec^2 x - \tan x = x\sec^2 x > 0$. Since $h(0) = -\ln 2 < 0$ and $h(x) \to \infty$ as $x \to \frac{1}{2}\pi$, h has exactly one zero, x_0. So g decreases on $(0, x_0)$ and increases on $(x_0, \frac{1}{2}\pi)$, and $g \to \infty$ at both ends (the top tends to $\ln 2$ as $x \to 0$, and the top tends to ∞ as $x \to \frac{1}{2}\pi$). Hence, with $k_0 = g(x_0)$, the minimum value of g:

- $k < k_0$: no solutions;

- $k = k_0$: exactly one solution, $x = x_0$;
- $k > k_0$: two solutions.

Graphically, $y = k_0 x$ is the tangent to $y = \ln(2\sec x)$ that passes through the origin. This confirms the sketch argument and pins down the threshold exactly.

Ask yourself Why does "exactly one solution" mean that the line is a tangent?

If the line cut the curve transversally at x_0, then rotating it slightly about O (making k a little smaller or larger) would still give a crossing nearby, and the convexity argument above would give a second one. A single, isolated meeting point that turns into no meeting point when k drops is exactly a point of tangency. Algebraically, this is $h(x_0) = 0$, i.e. $g'(x_0) = 0$.

The equation for x_0

At the point of tangency the line and the curve have the same height and the same gradient:

$$k_0 x_0 = \ln(2\sec x_0), \qquad k_0 = f'(x_0) = \tan x_0.$$

(The second equation is also $h(x_0) = 0$ from above.) Substituting the second into the first gives $x_0 \tan x_0 = \ln(2\sec x_0)$, and since $0 < x_0 < \frac{1}{2}\pi$, $\tan x_0 \neq 0$, so

$$x_0 = \cot x_0 \, \ln(2\sec x_0),$$

as required.

Finding x_0 and k_0

It is tempting to iterate $x_{n+1} = \cot x_n \ln(2\sec x_n)$ directly. That works only slowly: the derivative of the right-hand side at x_0 is $1 - 2x_0/\sin 2x_0 \approx -0.89$, so the iterates oscillate and close in by only about 11% per step. Newton's method on $h(x) = x\tan x - \ln(2\sec x)$ is much better, and $h'(x) = x\sec^2 x$ is very simple:

$$x_{n+1} = x_n - \frac{x_n \tan x_n - \ln(2\sec x_n)}{x_n \sec^2 x_n}.$$

Starting from $x_1 = 1$ (in radians throughout):

$$x_2 \approx 0.9274, \qquad x_3 \approx 0.9147, \qquad x_4 \approx 0.91440.$$

To confirm two decimal places we check a sign change: $h(0.905) \approx -0.023 < 0$ and $h(0.915) \approx +0.0015 > 0$, so $0.905 < x_0 < 0.915$ and

$$x_0 \approx 0.91.$$

Then $k_0 = \tan x_0 \approx \tan 0.9144 \approx 1.30$. Equivalently, $k_0 = \ln(2\sec x_0)/x_0$, which gives the same value.

Checks

- Both expressions for k_0 agree: $\tan(0.91440) \approx 1.2981$ and $\ln(2\sec 0.91440)/0.91440 \approx 1.2981$.
- A numerical scan of $g(x) = \ln(2\sec x)/x$ over $(0, \frac{1}{2}\pi)$ has minimum 1.2981, at $x \approx 0.914$.
- Counting sign changes of $kx - \ln(2\sec x)$: no roots for $k = 1.29$ and two roots for $k = 1.31$.
- The answer fits the rough bounds found earlier: $0.44 < k_0 < 1.32$.

Common slips

- Using degrees. Every value of x here, including the Newton iterates and $\tan x_0$, must be in radians.
- Only one equation for tangency. Meeting the curve is not enough: the gradients must match too, and it is this second condition that pins down k_0.
- Claiming two decimal places without a check. Show a sign change of h across $[0.905, 0.915]$, or iterate until the second decimal place is stable.

Beyond STEP

Counting the roots of $kx = f(x)$ by looking at the range of $f(x)/x$ is a small piece of a big idea. The tangent lines through the origin, and the convexity of f, are what the *Legendre transform* and convex duality are built on, in analysis and in mechanics. Newton's method converges so quickly here because it is quadratically convergent near a simple root, while the fixed-point iteration converges only linearly, with ratio $|\varphi'(x_0)|$. Both facts are proved in a first course in analysis or numerical methods. Bartle and Sherbert's *Introduction to Real Analysis* covers convex functions and the error analysis of Newton's method.

WORKED SOLUTION

STEP II 1992, Question 7

Pure • Repeated Roots of Cubics

STEP II 1992, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q7

The key idea

Comparing coefficients of $(x-a)(x-b)(x-c)$ with the cubic gives p, q and r as symmetric functions of the roots. In (i), a repeated root together with "roots sum to zero" pins the roots down to the shape $\alpha, \alpha, -2\alpha$, and everything follows from that. For (ii), the expression in the question looks like the answer to (i) with q and r replaced by more complicated things. That suggests shifting x to kill the x^2 term and then applying (i) to the new cubic.

Solution

Since a, b, c are the roots and the leading coefficient is 1,

$$x^3 - px^2 + qx - r = (x-a)(x-b)(x-c) = x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc$$

for all x. Comparing coefficients,

$$p = a+b+c, \qquad q = ab+bc+ca, \qquad r = abc.$$

Part (i)

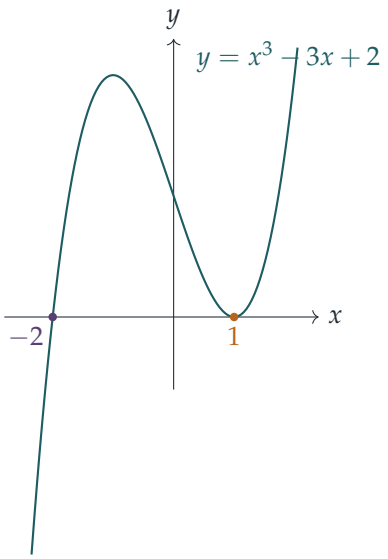
Label the roots so that the repeated one is $a = b = \alpha$. Then $p = 0$ gives $c = -(a+b) = -2\alpha$, so the roots are $\alpha, \alpha, -2\alpha$. Hence

$$q = \alpha^2 + \alpha(-2\alpha) + (-2\alpha)\alpha = -3\alpha^2,$$
$$r = \alpha \cdot \alpha \cdot (-2\alpha) = -2\alpha^3.$$

Therefore

$$4q^3 + 27r^2 = 4(-27\alpha^6) + 27(4\alpha^6) = -108\alpha^6 + 108\alpha^6 = 0,$$

as required. Nothing here needs α to be real, and the case of a triple root ($\alpha = 0$, all coefficients zero) is included.



For example, $x^3 - 3x + 2 = (x-1)^2(x+2)$ has $\alpha = 1$, $q = -3$, $r = -2$, and $4(-27) + 27 \cdot 4 = 0$. The double root shows up as the graph touching the axis.

Part (ii)

Now p need not be zero. The roots of the original cubic sum to p, so their mean is $p/3$. Shifting by the mean, put

$$x = y + \tfrac{p}{3}.$$

Then $y = x - \frac{p}{3}$, so the roots of the new cubic in y are $a - \frac{p}{3}$, $b - \frac{p}{3}$, $c - \frac{p}{3}$. Their sum is $p - p = 0$, and if two of a, b, c are equal then two of these are equal too.

Ask yourself What exactly is the cubic in y?

Expanding each term,

$$\begin{aligned} \left(y + \tfrac{p}{3}\right)^3 &= y^3 + py^2 + \tfrac{p^2}{3}y + \tfrac{p^3}{27}, \\ -p\left(y + \tfrac{p}{3}\right)^2 &= -py^2 - \tfrac{2p^2}{3}y - \tfrac{p^3}{9}, \\ q\left(y + \tfrac{p}{3}\right) &= qy + \tfrac{pq}{3}, \end{aligned}$$

and adding these together with $-r$, the y^2 terms cancel (that is the point of the shift) and

$$x^3 - px^2 + qx - r = y^3 + \left(q - \frac{p^2}{3}\right)y - \left(\frac{2p^3}{27} - \frac{pq}{3} + r\right),$$

using $\frac{p^3}{27} - \frac{p^3}{9} = -\frac{2p^3}{27}$. This cubic in y has the form $y^3 - Py^2 + Qy - R$ with

$$P = 0, \qquad Q = q - \frac{p^2}{3}, \qquad R = \frac{2p^3}{27} - \frac{pq}{3} + r.$$

It has zero y^2 coefficient and two equal roots, so part (i) applies to it and gives $4Q^3 + 27R^2 = 0$, that is,

$$4\left(q - \frac{p^2}{3}\right)^3 + 27\left(\frac{2p^3}{27} - \frac{pq}{3} + r\right)^2 = 0,$$

which is the required result.

Checks

- With $p = 0$ the result of (ii) reduces to $4q^3 + 27r^2 = 0$, which is part (i).
- For $(x-1)^2(x-2) = x^3 - 4x^2 + 5x - 2$ we have $p = 4$, $q = 5$, $r = 2$, so $Q = -\frac{1}{3}$ and $R = \frac{128}{27} - \frac{20}{3} + 2 = \frac{2}{27}$. Then $4Q^3 + 27R^2 = -\frac{4}{27} + \frac{4}{27} = 0$.
- We also tested both identities in exact rational arithmetic on hundreds of random cubics with a repeated root, and checked the shift identity at random points.

Common slips

- Sign errors in Vieta's formulae: with the signs $-p$, $+q$, $-r$ chosen in this question, all three of p, q, r come out positive in terms of the roots.
- Shifting by $p/3$ in the wrong direction. The roots of the new cubic are $a - \frac{p}{3}$ and so on, which is what makes their sum zero.
- Expanding everything in (ii) in terms of a, a, c by brute force. It works, but it is long and error-prone; the shift reuses (i) for free.

Beyond STEP

For a general cubic the left-hand side of (ii) equals $-(a-b)^2(b-c)^2(c-a)^2$, the negative of the *discriminant*. It vanishes exactly when two roots coincide, and its sign tells you whether a real cubic has three real roots or only one. The shift $x = y + \frac{p}{3}$ is also the first step of Cardano's formula for solving cubics. Symmetric functions of roots, discriminants and the question of which equations can be solved by radicals lead into Galois theory; Allenby's *Rings, Fields and Groups* is a good place to start.

WORKED SOLUTION

STEP II 1992, Question 8

Pure • Techniques of Integration

STEP II 1992, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q8

The key idea

Three standard techniques, one per part. In (i) we factorise the denominator completely and notice a *repeated* factor, so the partial fractions need a squared term. In (ii) the substitution $t = \tan\frac{x}{2}$ turns any rational function of $\sin x$ and $\cos x$ into a rational function of t, and here the resulting quadratic factorises nicely. In (iii) we write $\sinh x$ in exponentials and substitute $u = \mathrm{e}^x$, which again leaves partial fractions.

Solution

Part (i)

Since $x^2 - 1 = (x-1)(x+1)$, the denominator is $(x-1)^2(x+1)$. The factor $x-1$ is repeated, so we look for

$$\frac{x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1},$$

that is,

$$x = A(x-1)(x+1) + B(x+1) + C(x-1)^2.$$

Putting $x = 1$ gives $1 = 2B$, so $B = \frac{1}{2}$. Putting $x = -1$ gives $-1 = 4C$, so $C = -\frac{1}{4}$. Comparing coefficients of x^2 gives $0 = A + C$, so $A = \frac{1}{4}$. Hence

$$\begin{aligned}\int \frac{x}{(x-1)(x^2-1)}\,\mathrm{d}x &= \int \left(\frac{1}{4(x-1)} + \frac{1}{2(x-1)^2} - \frac{1}{4(x+1)}\right)\mathrm{d}x \\ &= \tfrac{1}{4}\ln|x-1| - \frac{1}{2(x-1)} - \tfrac{1}{4}\ln|x+1| + c \\ &= \frac{1}{4}\ln\left|\frac{x-1}{x+1}\right| - \frac{1}{2(x-1)} + c.\end{aligned}$$

Part (ii)

Let $t = \tan\frac{x}{2}$. Then

$$\cos x = \frac{1-t^2}{1+t^2}, \qquad \sin x = \frac{2t}{1+t^2}, \qquad \mathrm{d}x = \frac{2\,\mathrm{d}t}{1+t^2},$$

the last because $\mathrm{d}t/\mathrm{d}x = \frac{1}{2}\sec^2\frac{x}{2} = \frac{1}{2}(1+t^2)$. So

$$3\cos x + 4\sin x = \frac{3-3t^2+8t}{1+t^2} = \frac{(1+3t)(3-t)}{1+t^2},$$

and the factors $1+t^2$ cancel:

$$\int \frac{\mathrm{d}x}{3\cos x + 4\sin x} = \int \frac{2\,\mathrm{d}t}{(1+3t)(3-t)}.$$

? **Ask yourself** How do we know the quadratic $3+8t-3t^2$ will factorise?

Its discriminant is $64+36 = 100 = 10^2$, a perfect square, so its roots $t = 3$ and $t = -\frac{1}{3}$ are rational. (That is no accident: $3^2+4^2 = 5^2$, which is also why the harmonic form below has a whole-number amplitude.)

Now write $\dfrac{2}{(1+3t)(3-t)} = \dfrac{a}{1+3t} + \dfrac{b}{3-t}$, so $2 = a(3-t) + b(1+3t)$. Putting $t = 3$ gives $b = \frac{1}{5}$; putting $t = -\frac{1}{3}$ gives $\frac{10}{3}a = 2$, so $a = \frac{3}{5}$. Therefore

$$\begin{aligned}\int \frac{\mathrm{d}x}{3\cos x + 4\sin x} &= \int \left(\frac{3}{5(1+3t)} + \frac{1}{5(3-t)}\right)\mathrm{d}t = \tfrac{1}{5}\ln|1+3t| - \tfrac{1}{5}\ln|3-t| + c \\ &= \frac{1}{5}\ln\left|\frac{1+3\tan\frac{x}{2}}{3-\tan\frac{x}{2}}\right| + c.\end{aligned}$$

Alternatively, write $3\cos x + 4\sin x = 5\sin(x+\beta)$, where $\cos\beta = \frac{4}{5}$ and $\sin\beta = \frac{3}{5}$. Then the integral is $\frac{1}{5}\int \operatorname{cosec}(x+\beta)\,\mathrm{d}x = \frac{1}{5}\ln\left|\tan\frac{1}{2}(x+\beta)\right| + c$. The two answers agree: with $\tan\frac{\beta}{2} = \frac{1}{3}$ (since $\tan\frac{\beta}{2} = \frac{\sin\beta}{1+\cos\beta} = \frac{3/5}{9/5}$), the addition formula gives

$$\tan\tfrac{1}{2}(x+\beta) = \frac{t+\frac{1}{3}}{1-\frac{1}{3}t} = \frac{1+3t}{3-t}.$$

Part (iii)

Since $\sinh x = \frac{1}{2}(\mathrm{e}^x - \mathrm{e}^{-x})$, multiplying top and bottom by e^x gives

$$\frac{1}{\sinh x} = \frac{2\mathrm{e}^x}{\mathrm{e}^{2x}-1}.$$

Substitute $u = \mathrm{e}^x$, so $\mathrm{d}u = \mathrm{e}^x\,\mathrm{d}x$:

$$\begin{aligned}\int \frac{\mathrm{d}x}{\sinh x} &= \int \frac{2\,\mathrm{d}u}{u^2-1} = \int \left(\frac{1}{u-1} - \frac{1}{u+1}\right)\mathrm{d}u = \ln\left|\frac{u-1}{u+1}\right| + c \\ &= \ln\left|\frac{\mathrm{e}^x-1}{\mathrm{e}^x+1}\right| + c = \ln\left|\tanh\frac{x}{2}\right| + c,\end{aligned}$$

where the last step divides top and bottom by $e^{x/2}$: $\dfrac{e^{x/2}-e^{-x/2}}{e^{x/2}+e^{-x/2}} = \tanh\frac{x}{2}$. Note the neat parallel with $\int \operatorname{cosec} x\,\mathrm{d}x = \ln|\tan\frac{x}{2}| + c$ used in the alternative to (ii).

Checks

Differentiate each answer. In (iii), for example, $\frac{\mathrm{d}}{\mathrm{d}x}\ln\tanh\frac{x}{2} = \frac{\frac{1}{2}\operatorname{sech}^2\frac{x}{2}}{\tanh\frac{x}{2}} = \frac{1}{2\sinh\frac{x}{2}\cosh\frac{x}{2}} = \frac{1}{\sinh x}$. Numerically, Simpson's rule gives $\int_2^5$ of the integrand in (i) as 0.548287, $\int_0^1$ of the integrand in (ii) as 0.234276 and $\int_{1/2}^2 \operatorname{cosech} x\,\mathrm{d}x$ as 1.134488, each matching the antiderivative to at least 12 decimal places.

Common slips

- In (i), missing the repeated factor and trying $\frac{A}{x-1} + \frac{C}{x+1}$ alone, or forgetting that $x^2 - 1$ factorises at all.
- Dropping the modulus signs inside the logarithms: the integrands are defined for negative values of $x - 1$, $1 + 3t$ and $\tanh\frac{x}{2}$ too.
- In (ii), forgetting the factor $\frac{2}{1+t^2}$ from $\mathrm{d}x$.

Beyond STEP

The substitution $t = \tan\frac{x}{2}$ (sometimes called the Weierstrass substitution) shows that *every* rational function of $\sin x$ and $\cos x$ has an elementary antiderivative, because partial fractions handle every rational function of t. Behind it is a parametrisation of the unit circle by rational functions, the same one that generates all Pythagorean triples. Questions of which functions have elementary antiderivatives at all lead to Liouville's theorem and differential algebra; for the analytic foundations of integration, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1992, Question 9

Pure • Scalar Triple Product

STEP II 1992, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q9

The key idea

Rearranging the expression for $\mathbf{r}$ as $\mathbf{a}$ plus combinations of $\mathbf{b}-\mathbf{a}$ and $\mathbf{c}-\mathbf{a}$ shows it is a plane through A, B and C. A normal to that plane is the cross product of two vectors lying in it, and expanding that cross product produces exactly the three triple products in the question. After that, the identity holds for *every* λ and μ, so we may choose convenient values.

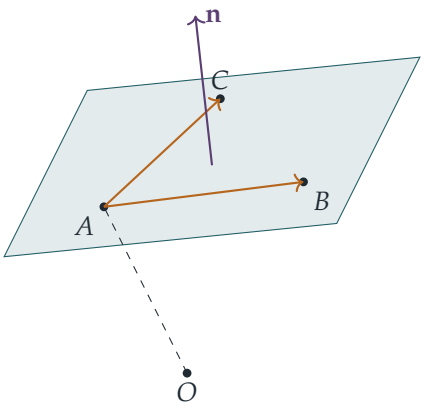
Solution

The locus

Rewrite the equation as

$$\mathbf{r} = \mathbf{a} + \lambda(\mathbf{b}-\mathbf{a}) + \mu(\mathbf{c}-\mathbf{a}).$$

This is the point A plus any combination of the vectors $\overrightarrow{AB} = \mathbf{b}-\mathbf{a}$ and $\overrightarrow{AC} = \mathbf{c}-\mathbf{a}$. These two vectors aren't parallel: if they were, A, B, C would lie on a line, and a line together with O always lies in some plane, which is ruled out. So as λ and μ range over all real numbers, $\mathbf{r}$ sweeps out the **plane through** A**,** B **and** C. Taking $(\lambda,\mu) = (0,0)$, $(1,0)$, $(0,1)$ gives A, B, C themselves. Since O is not in the plane ABC, this plane does not pass through the origin.



The equation $\mathbf{r}\cdot\mathbf{n}=p$

A normal to the plane is perpendicular to both $\overrightarrow{AB}$ and $\overrightarrow{AC}$, so we take

$$\mathbf{n}=(\mathbf{b}-\mathbf{a})\times(\mathbf{c}-\mathbf{a}),$$

which is nonzero because the two vectors are not parallel. Expanding, and using $\mathbf{a}\times\mathbf{a}=\mathbf{0}$ and $\mathbf{u}\times\mathbf{v}=-\mathbf{v}\times\mathbf{u}$,

$$\begin{aligned}\mathbf{n}&=\mathbf{b}\times\mathbf{c}-\mathbf{b}\times\mathbf{a}-\mathbf{a}\times\mathbf{c}+\mathbf{a}\times\mathbf{a}\\&=\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}+\mathbf{a}\times\mathbf{b}.\end{aligned}$$

Every point of the plane differs from A by a vector perpendicular to $\mathbf{n}$, so $\mathbf{r}\cdot\mathbf{n}=\mathbf{a}\cdot\mathbf{n}$. Since $\mathbf{a}$ is perpendicular to $\mathbf{c}\times\mathbf{a}$ and to $\mathbf{a}\times\mathbf{b}$,

$$p=\mathbf{a}\cdot\mathbf{n}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}).$$

So the plane is $\mathbf{r}\cdot(\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}+\mathbf{a}\times\mathbf{b})=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$.

? **Ask yourself** What is $\mathbf{r}\cdot\mathbf{n}$ when we substitute the parametric form of $\mathbf{r}$?

We need $\mathbf{a}\cdot\mathbf{n}$, $\mathbf{b}\cdot\mathbf{n}$ and $\mathbf{c}\cdot\mathbf{n}$. In each, two of the three terms vanish, because a vector is perpendicular to any cross product that contains it:

$$\mathbf{a}\cdot\mathbf{n}=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}),\qquad\mathbf{b}\cdot\mathbf{n}=\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a}),\qquad\mathbf{c}\cdot\mathbf{n}=\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b}).$$

The point $\mathbf{r}=(1-\lambda-\mu)\mathbf{a}+\lambda\mathbf{b}+\mu\mathbf{c}$ lies in the plane for every λ, μ, so $\mathbf{r}\cdot\mathbf{n}=p$ gives

$$(1-\lambda-\mu)\,\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})+\lambda\,\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a})+\mu\,\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})=\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c}).$$

Subtracting $\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})$ from both sides,

$$-(\lambda+\mu)\,\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})+\lambda\,\mathbf{b}\cdot(\mathbf{c}\times\mathbf{a})+\mu\,\mathbf{c}\cdot(\mathbf{a}\times\mathbf{b})=0$$

for all scalars λ, μ, as required.

The deduction

Since the identity holds for all λ and μ, choose values that isolate terms.

- $\lambda = 1$, $\mu = 0$: $-\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = 0$.
- $\lambda = 0$, $\mu = 1$: $-\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) + \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = 0$.

Together these give

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \mathbf{b} \cdot (\mathbf{c} \times \mathbf{a}) = \mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}).$$

(Equivalently, the left-hand side is linear in λ and μ and identically zero, so the coefficients of λ and of μ must both vanish.)

When A, B, C and O are coplanar

Then $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ all lie in one plane through O. Now $\mathbf{b} \times \mathbf{c}$ is perpendicular to that plane (or zero), so $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, and likewise the other two triple products vanish. Geometrically, the parallelepiped on $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ is flat and has zero volume. The result is still true, but trivially: $0 = 0 = 0$. If A, B, C are not collinear the locus is still a plane, but it passes through the origin, so $p = 0$ and its equation is $\mathbf{r} \cdot \mathbf{n} = 0$. If A, B, C are collinear, $\mathbf{n} = \mathbf{0}$ and the locus is only a line (or a single point if all three coincide), so the argument gives no information.

Checks

- With $\mathbf{a} = \mathbf{i}$, $\mathbf{b} = \mathbf{j}$, $\mathbf{c} = \mathbf{k}$, all three triple products equal 1, and $\mathbf{n} = \mathbf{i} + \mathbf{j} + \mathbf{k}$, $p = 1$: the plane $x + y + z = 1$, which indeed contains the three points.
- We tested 2000 random integer triples $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ with exact fractions: the expanded form of $\mathbf{n}$, the value of p, the equation $\mathbf{r} \cdot \mathbf{n} = p$ for random λ, μ, and the equality of the three triple products all held exactly. For coplanar triples every triple product was 0.

Common slips

- Sign errors when expanding $(\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a})$: the term $-\mathbf{b} \times \mathbf{a}$ becomes $+\mathbf{a} \times \mathbf{b}$, and $-\mathbf{a} \times \mathbf{c}$ becomes $+\mathbf{c} \times \mathbf{a}$.
- Forgetting to say why $\mathbf{n} \neq \mathbf{0}$, which is exactly where the non-coplanarity condition (via non-collinearity of A, B, C) is used.

Beyond STEP

The scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ is the 3×3 determinant with rows $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, and the cyclic symmetry proved here is the fact that a cyclic permutation of rows (two swaps) leaves a determinant unchanged. Its value is the signed volume of the parallelepiped on

the three vectors, and it vanishes precisely when they are linearly dependent, which is the coplanar case. In a first-year course this becomes the general theory of determinants as volume scale factors; Strang's *Introduction to Linear Algebra* develops it clearly.

WORKED SOLUTION

STEP II 1992, Question 10

Pure • Loci in the Argand Diagram

STEP II 1992, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q10

The key idea

Write $\dfrac{z-1}{z} = \dfrac{1-z}{0-z}$: its argument is the angle, turned anticlockwise, from the direction $z \to 0$ to the direction $z \to 1$. So fixing the argument fixes the angle that the segment from 0 to 1 subtends at z, and circle theorems (angles in the same segment, opposite angles of a cyclic quadrilateral) take over. The final part is the same idea with the chord z_4z_2 in place of 0 to 1.

Solution

(The statement of the question has a misprint: the condition is meant to be $0 < \alpha \leq \frac{1}{2}\pi$.)

Two facts do all the work. First, since $\arg\dfrac{z-1}{z} = \arg(1-z) - \arg(0-z)$, the argument measures the angle at z from the line $z0$ round to the line $z1$, anticlockwise. Second, it tells us which side of the real axis z is on:

$$\operatorname{Im}\frac{z-1}{z} = \operatorname{Im}\left(1 - \frac{1}{z}\right) = \operatorname{Im}\left(-\frac{\bar{z}}{|z|^2}\right) = \frac{\operatorname{Im} z}{|z|^2}.$$

So $\arg\frac{z-1}{z} \in (0, \pi)$ exactly when $\operatorname{Im} z > 0$, and lies in $(-\pi, 0)$ exactly when $\operatorname{Im} z < 0$. Points on the real axis give a real quotient (argument 0 or π), and $z = 0, 1$ give no argument at all.

Part (i)

Here $\operatorname{Im} z > 0$, and the segment from 0 to 1 subtends the angle α at z. The points above the real axis at which a fixed chord subtends a fixed angle form an arc of a circle through the chord's ends (angles in the same segment are equal). Conversely, every point of that arc sees the chord at angle α, turned anticlockwise because the point lies above the chord, so the whole arc belongs to the locus.

To pin the circle down: the angle at the centre is 2α, so a chord of length 1 gives radius

$\dfrac{1}{2\sin\alpha}$, and the centre lies on $\operatorname{Re} z = \frac{1}{2}$ at height $\frac{1}{2}\cot\alpha \geq 0$. The locus is the arc of

$$\left|z - \tfrac{1}{2} - \tfrac{\mathrm{i}}{2}\cot\alpha\right| = \frac{1}{2\sin\alpha}$$

lying above the real axis, with the end points 0 and 1 excluded. Since $\alpha \leq \frac{1}{2}\pi$ this is the major arc. When $\alpha = \frac{1}{2}\pi$ it is the upper semicircle on the diameter from 0 to 1 (the angle in a semicircle).

Part (ii)

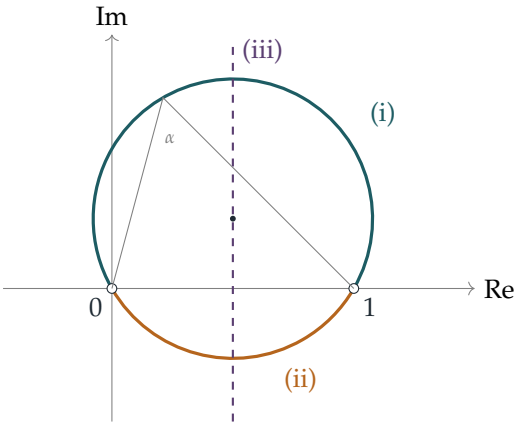
Now the argument $\alpha - \pi$ lies in $(-\pi, -\frac{1}{2}\pi]$, so $\operatorname{Im} z < 0$, and the chord subtends an angle of size $\pi - \alpha$ at z, turned clockwise.

Ask yourself Which circle has the chord subtending $\pi - \alpha$ below the axis?

The same circle as in (i). If P is on the arc from (i) and Q is on the other arc of that circle, then $0PQ1$ (taken in a suitable order) is a cyclic quadrilateral with P and Q opposite, so the angles at P and Q add up to π. Hence the minor arc below the axis is exactly where the angle is $\pi - \alpha$. The locus is the arc of the circle from (i) below the real axis, again without 0 and 1. Parts (i) and (ii) together make up the whole circle apart from 0 and 1. (If $\alpha = \frac{1}{2}\pi$, (ii) is the lower semicircle.)

Part (iii)

$\left|\frac{z-1}{z}\right| = 1$ means $|z-1| = |z|$ with $z \neq 0$: z is equidistant from 0 and 1. The locus is the perpendicular bisector, the line $\operatorname{Re} z = \frac{1}{2}$.



The sketch shows $\alpha = \frac{1}{3}\pi$: the teal major arc is (i), the amber minor arc is (ii), the dashed plum line is (iii), and the open circles mark the excluded points 0 and 1.

The final part

Let $f(z) = \dfrac{z - z_2}{z - z_4}$. Since $(z_1 - z_4)(z_3 - z_2) = (z_4 - z_1)(z_2 - z_3)$,

$$\frac{f(z_1)}{f(z_3)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_4 - z_1)(z_2 - z_3)} = w,$$

so $\arg w = \arg f(z_1) - \arg f(z_3) \pmod{2\pi}$.

The function f is the function of parts (i) and (ii) in new coordinates. Put $z = z_4 + (z_2 - z_4)u$, which is a rotation and enlargement followed by a translation, taking $u = 0, 1$ to z_4, z_2. Then

$$f(z) = \frac{(z_2 - z_4)(u - 1)}{(z_2 - z_4)u} = \frac{u - 1}{u}.$$

The circle through the four points corresponds to a circle in the u-plane through 0 and 1.

? **Ask yourself** Why are z_1 and z_3 on different arcs?

The points come in the order z_1, z_2, z_3, z_4 round the circle, so the chord z_2z_4 cuts the circle into two arcs, one containing z_1 and the other containing z_3. In the u-plane, one of them lies above the real axis and the other below it.

From (i) and (ii), the argument of $\frac{u-1}{u}$ takes a constant value $\beta \in (0, \pi)$ on the upper arc and the value $\beta - \pi$ on the lower arc. Here we are not assuming $\beta \leq \frac{1}{2}\pi$: the circle-theorem argument above never used that restriction. Therefore

$$\arg w = \arg f(z_1) - \arg f(z_3) = \pm\big(\beta - (\beta - \pi)\big) = \pm\pi \pmod{2\pi},$$

so w is real, and in fact negative. ■

Checks

For the square $z_1 = 1$, $z_2 = \mathrm{i}$, $z_3 = -1$, $z_4 = -\mathrm{i}$ we get $w = \dfrac{(1-\mathrm{i})(-1+\mathrm{i})}{(-1-\mathrm{i})(1+\mathrm{i})} = \dfrac{-(1-\mathrm{i})^2}{-(1+\mathrm{i})^2} = \dfrac{-2\mathrm{i}}{2\mathrm{i}} = -1$, which is real and negative. For (i), take $\alpha = \frac{1}{2}\pi$ and $z = \frac{1}{2} + \frac{1}{2}\mathrm{i}$. Then $\frac{z-1}{z} = \frac{-1+\mathrm{i}}{1+\mathrm{i}} = \mathrm{i}$, whose argument is $\frac{1}{2}\pi$. We also checked numerically that the arcs carry the stated arguments for several values of α, and that w was real and negative for 10^4 random sets of four points in order on random circles.

Common slips

- Giving the whole circle for (i). Only one arc has the right argument; the other arc gives $\alpha - \pi$, which is exactly part (ii).
- Forgetting to exclude 0 and 1, where the argument is undefined.

- Getting the sign of $\arg w$ wrong because the minus signs in $(z_4 - z_1)(z_2 - z_3)$ are mishandled. Pairing them as above avoids this.

Beyond STEP

The quantity w is (up to sign and ordering) the *cross-ratio* of four points. It is unchanged by every Möbius transformation $z \mapsto \frac{az+b}{cz+d}$, and four points lie on a circle or line exactly when their cross-ratio is real. Since Möbius maps send circles and lines to circles and lines, this gives a neat way to prove concyclicity. Needham's *Visual Complex Analysis* develops this geometric view beautifully.

WORKED SOLUTION

STEP II 1992, Question 11

Mechanics • Projectile Motion

STEP II 1992, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q11

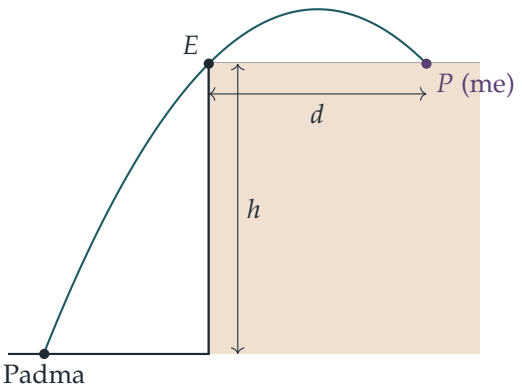
The key idea

The awkward part of the throw is not getting the wallet up to height h but getting it across the top of the cliff: it must be at height at least h all the way from the cliff edge to me, a horizontal distance of at least d. So look at the motion only from the moment the wallet first reaches height h. From then on it is an ordinary projectile on level ground (the level of the cliff top), whose speed we know from energy conservation, and the familiar maximum range u^2/g does the rest.

Solution

The model

Treat the wallet as a particle moving freely under gravity, with no air resistance, thrown from ground level (Padma's height is negligible). Let E be the top edge of the cliff and P the point where I stand, so P is at height h and a horizontal distance d beyond E. The cliff face is vertical, and the beach lies on the far side of it from me, so wherever Padma stands, the wallet has to pass over E (or at worst just graze it) and then come down to P.



We use one fact throughout. If a particle at some level has speed u and moves at angle α above the horizontal, it returns to that level after time $2u\sin\alpha/g$, having travelled horizontally

$$\frac{2u^2\sin\alpha\cos\alpha}{g} = \frac{u^2\sin 2\alpha}{g} \leq \frac{u^2}{g},$$

with equality exactly when $\alpha = 45^\circ$.

The condition is necessary

Suppose some throw gets the wallet to P. It must reach height h, so let Q be the first point of the path at height h. Taking the cliff foot as the zero of potential energy, conservation of energy gives the speed u at Q:

$$u^2 = V^2 - 2gh.$$

Ask yourself Why must Q be at least a horizontal distance d from P?

When the wallet's horizontal position is that of the cliff face, it must be at height at least h, or it would hit the face (or pass below the edge). Its height rises continuously from 0, so it has already reached height h by then: Q is on the beach side of E, or at E. As P is a horizontal distance d beyond E, the horizontal distance from Q to P is at least d.

Between Q and P the wallet goes from height h back to height h, starting with speed u. By the range fact, the horizontal distance it covers is at most u^2/g. Therefore

$$d \leq \frac{u^2}{g} = \frac{V^2 - 2gh}{g}, \qquad \text{that is,} \qquad V^2 \geq g(2h + d).$$

The condition is sufficient

Now suppose $V^2 \geq g(2h + d)$. Then $u^2 = V^2 - 2gh \geq gd > 0$, so $gd/u^2 \leq 1$ and we can choose an angle α with $0 < \alpha \leq 45^\circ$ and

$$\sin 2\alpha = \frac{gd}{u^2}.$$

Consider the parabolic path that passes through E with speed u, moving away from the beach at angle α above the horizontal. By the range fact, it comes back down to height h a horizontal distance $u^2\sin 2\alpha/g = d$ beyond E, which is exactly P. Between E and P it is above height h, so it clears the cliff top.

Run this path backwards in time from E: the wallet is lower and further out over the beach at earlier times (its vertical velocity was larger and upward), so the path meets the ground at some point of the beach, a positive horizontal distance from the cliff foot. Before reaching E the path is entirely on the beach side of the cliff face, so nothing is in the way, and since the beach is broad Padma can stand at that point. By energy conservation the speed there is $\sqrt{u^2 + 2gh} = V$. So if Padma throws from that point at speed V, in the direction of the path, the wallet follows the path and reaches me.

(When $V^2 = g(2h + d)$ exactly, $\alpha = 45°$ and the wallet just grazes the edge E; for larger V there are two angles in $(0°, 90°)$ that work, and the wallet passes strictly above the edge for both.)

Combining the two halves, Padma can throw the wallet to me if and only if $V^2 \geq g(2h + d)$, as required.

Checks

- If $d = 0$ the condition becomes $V^2 \geq 2gh$, which is just what is needed for a vertical throw to reach height h.
- If $h = 0$ it becomes $V^2 \geq gd$, the usual maximum range on level ground.
- Ignoring the cliff edge, the best Padma could do is stand at the cliff foot, and the bounding parabola of all trajectories gives only $V^2 \geq g\left(h + \sqrt{h^2 + d^2}\right)$, which is weaker since $\sqrt{h^2 + d^2} \leq h + d$. So the edge really is the binding constraint.
- Numerically, with $g = 9.8$, $h = 20$ and $d = 15$, the critical speed is $\sqrt{539} \approx 23.2\ \mathrm{m\,s^{-1}}$; Padma stands about 11.4 m from the cliff foot, and a step-by-step simulation of the throw lands it 15.0 m beyond the edge, grazing E. A scan over all launch angles for random h, V confirmed that the largest horizontal gap between the two height-h points of the path is $(V^2 - 2gh)/g$.

Common slips

- Using only the bounding parabola for the point P from the cliff foot. This ignores the cliff edge and gives a condition that is necessary but not sufficient.
- Proving only one direction. The "if" needs an explicit throw, including a check that it starts on the beach and doesn't hit the cliff.

Beyond STEP

The trick of restarting the motion at height h works because the equations of motion don't change with time or under horizontal shifts, so any point of a trajectory can be treated as a fresh launch. Problems of the form "which targets can be reached?" are about the envelope of a family of curves, and with an obstacle in the way the answer is decided by whichever constraint binds first, an idea that reappears in optimisation and the calculus of variations. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for projectile motion and energy methods.

WORKED SOLUTION

STEP II 1992, Question 12

Mechanics • Moments and Elastic Strings

STEP II 1992, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q12

The key idea

Each wheel can turn freely, so the only condition for it to be in equilibrium is that the moments about its centre balance: the hanging particle tries to turn the wheel one way and the elastic string pulls it back the other way. The geometry gives the extension of the elastic string in terms of θ, so Hooke's law gives the tension, and taking moments gives one equation in θ. Counting equilibria then means counting solutions of that equation for the angles that actually make sense.

Solution

Setting up the model

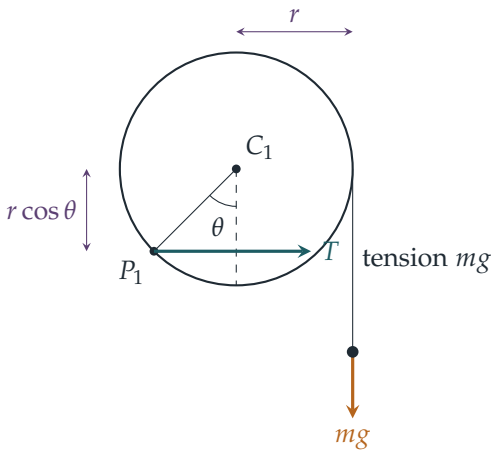
We model the particles as particles, the strings carrying them as light and inextensible, and the elastic string as light and obeying Hooke's law. The axles are smooth, so the reaction at each centre has no moment about that centre.

Take C_1C_2 as horizontal. In the figure, P_1 lies a distance $r\sin\theta$ outside C_1 (to the left) and $r\cos\theta$ below it; P_2 is the mirror image. So P_1 and P_2 are at the same height and the elastic string is horizontal, with length

$$P_1P_2 = d + 2r\sin\theta.$$

Its natural length is d, so the extension is $2r\sin\theta$ and, by Hooke's law, the tension is

$$T = \frac{\lambda\,(2r\sin\theta)}{d}.$$



The particle on W_1 hangs in equilibrium, so its string has tension mg; that string leaves the wheel tangentially on the side nearer W_2, at horizontal distance r from C_1. The elastic string pulls P_1 horizontally towards W_2 with force T, and its line of action is a distance $r\cos\theta$ below C_1.

The equilibrium condition

Taking moments about C_1: the weight turns W_1 clockwise (as drawn) with moment mgr, and the tension turns it anticlockwise with moment $Tr\cos\theta$. So

$$mgr = Tr\cos\theta = \frac{2\lambda r\sin\theta}{d}\, r\cos\theta = \frac{\lambda r^2 \sin 2\theta}{d},$$

using $2\sin\theta\cos\theta = \sin 2\theta$. Dividing by λr^2 gives

$$\sin 2\theta = \frac{mgd}{\lambda r},$$

as required. By symmetry the same equation comes from taking moments about C_2.

Which angles are allowed?

> **Ask yourself** Why can't θ be any angle at all, so that $\sin 2\theta = k$ has infinitely many solutions?

The moment of the tension must oppose the weight. If $\sin\theta \le 0$, the points P_1 and P_2 are no more than d apart, so the elastic string is slack, $T = 0$, and nothing balances mgr. If $\cos\theta \le 0$, the string runs level with or above C_1, so its moment about C_1 is zero or turns the wheel the same way as the weight. Either way there is no equilibrium. Hence an equilibrium needs

$$0 < \theta < \tfrac{\pi}{2}, \qquad \text{that is,} \qquad 0 < 2\theta < \pi.$$

(Could the two wheels sit at *different* angles? The moment of the tension about each centre is T times the perpendicular distance from that centre to the line P_1P_2, and both moments must equal mgr. So C_1 and C_2 are the same distance from the line and on the same side of it, below the centres. Hence the line is parallel to C_1C_2, so P_1 and P_2 are at the same depth and the two angles agree, because if they had opposite signs the string would be exactly length d and slack. So every equilibrium is of the symmetric kind shown.)

Write $k = \dfrac{mgd}{\lambda r} > 0$. On $0 < 2\theta < \pi$, $\sin 2\theta$ rises from 0 to 1 at $2\theta = \frac{\pi}{2}$ and falls back to 0, taking every value in $(0,1)$ exactly twice and the value 1 exactly once. So the equation $\sin 2\theta = k$ has

- no solution if $k > 1$;
- exactly one solution, $\theta = \frac{\pi}{4}$, if $k = 1$;
- exactly two solutions, $\theta = \alpha$ and $\theta = \frac{\pi}{2} - \alpha$ with $\alpha = \frac{1}{2}\arcsin k \in (0, \frac{\pi}{4})$, if $0 < k < 1$.

Since $k > 1 \iff \lambda < \dfrac{mgd}{r}$, the answers are:

Part (i)

No equilibrium positions when $\lambda < \dfrac{mgd}{r}$: the elastic string is too weak to hold the particles at any angle.

Part (ii)

Just one equilibrium position when $\lambda = \dfrac{mgd}{r}$, with $\theta = \frac{\pi}{4}$.

Part (iii)

Exactly two equilibrium positions when $\lambda > \dfrac{mgd}{r}$, at $\theta = \alpha$ and $\theta = \frac{\pi}{2} - \alpha$.

Part (iv)

There is no value of λ that gives more than two equilibrium positions, because θ is confined to $(0, \frac{\pi}{2})$.

Checks

- Dimensions: λ is a force and so is mg, and d/r is a ratio of lengths, so $mgd/(\lambda r)$ is dimensionless, as a sine must be.
- Limiting case: as $\lambda \to \infty$ the string becomes inextensible, and the two solutions tend to $\theta \to 0$ (tiny extension, large tension) and $\theta \to \frac{\pi}{2}$ (string almost level with the

centres, so a tiny moment arm). Both make sense.

- Numerically, with $m = 1$, $d = 4$, $r = 1$, $g = 9.8$ and $\lambda = 3mgd/r$, so $k = \frac{1}{3}$, a full model of both wheels (allowing the two angles to differ) has exactly two equilibria, at $\theta \approx 0.1699$ and $\theta \approx 1.4009$; these are $\frac{1}{2}\arcsin\frac{1}{3}$ and $\frac{\pi}{2}$ minus it.

Common slips

- Taking the extension as $r\sin\theta$ rather than $2r\sin\theta$: both ends of the string move out.
- Using the moment arm $r\sin\theta$ for the tension. The tension is horizontal, so its arm is the *vertical* distance $r\cos\theta$.
- Answering (iv) with "infinitely many" by forgetting that only $0 < \theta < \frac{\pi}{2}$ is physically possible.

Beyond STEP

Counting equilibria as a parameter changes is the start of *bifurcation theory*. Here the two equilibria for $\lambda > mgd/r$ merge at $\lambda = mgd/r$ and disappear, which is called a saddle-node (or fold) bifurcation. Looking at the potential energy, one of the pair turns out to be stable and the other unstable, which is typical of a fold. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers moments, rigid-body equilibrium and stability from potential energy.

WORKED SOLUTION

STEP II 1992, Question 13

Mechanics • Friction and Connected Particles

STEP II 1992, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

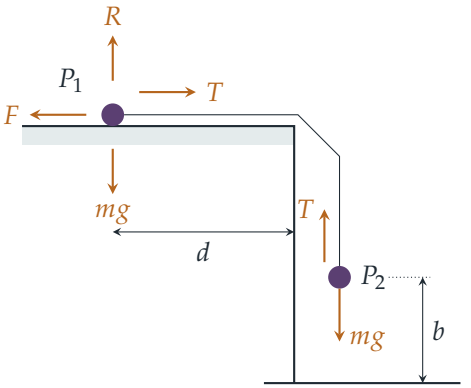
step.maths.org/questions/92-s2-q13

The key idea

The motion has two stages. While P_2 is falling the string is taut and the two particles move together with one common acceleration. Once P_2 hits the floor the string goes slack, and P_1 simply skids to a halt under friction. Everything follows from the speed at the moment of impact: the first part compares a stopping *distance* with the distance left to the edge, and the second compares two *times*.

Solution

We model P_1 and P_2 as particles and the string as light and inextensible, passing over a smooth edge, so the tension T is the same throughout. The table is rough for P_1. We assume $b < d$, so that P_2 lands before P_1 can reach the edge (the string's length plays no further role).



The first part

Stage 1: string taut. Resolving vertically for P_1 gives $R = mg$. It is sliding, so friction is limiting: $F = \mu R = \mu mg$, opposing the motion. With common acceleration a,

$$\begin{aligned} P_2 \text{ (down)}: &\quad mg - T = ma, \\ P_1 \text{ (towards edge)}: &\quad T - \mu mg = ma. \end{aligned}$$

Adding, $2ma = mg(1-\mu)$, so $a = \frac{1}{2}g(1-\mu)$, which is positive because $\mu < 1$; the system really does start to move. P_2 falls a distance b from rest, so its speed V on reaching the floor satisfies

$$V^2 = 2ab = gb(1-\mu).$$

At this moment P_1 has the same speed V, and it has moved a distance b, so it is $d - b$ from the edge.

Stage 2: string slack. After the impact P_2 is on the floor or bouncing up, while P_1 carries on towards the edge, so the string is slack and $T = 0$.

> **Ask yourself** Why can't the string tighten again while P_2 is bouncing?

The string was exactly taut at the moment of impact. After that, P_1 moves closer to the edge, which lets out more string over the edge, and P_2 is no lower than the floor, so it needs no more string than it did at impact. The string stays slack from then on.

So P_1 decelerates at μg alone, and it comes to rest after a further distance

$$s = \frac{V^2}{2\mu g} = \frac{b(1-\mu)}{2\mu},$$

provided that the edge is not reached first. Hence P_1 falls off if and only if $s > d - b$:

$$\begin{aligned} \frac{b(1-\mu)}{2\mu} > d - b &\iff b - b\mu > 2\mu d - 2\mu b \qquad \text{(multiplying by } 2\mu > 0\text{)} \\ &\iff b > \mu(2d - b) \\ &\iff \mu < \frac{b}{2d-b}, \end{aligned}$$

where the last step divides by $2d - b$, which is positive because $d > b$. Every step is reversible, so this is the required "if and only if". (If μ equals $b/(2d-b)$, P_1 stops exactly at the edge and does not fall.)

The second part

Now $\mu > b/(2d-b)$, so P_1 stops on the table. Measure time from the moment P_2 first hits the floor. At that moment both particles have speed V.

P_1: decelerating at μg from speed V, it stops after time

$$t_1 = \frac{V}{\mu g}.$$

P_2: by Newton's law of restitution it rebounds with speed eV. With the string slack it moves freely under gravity, so it returns to the floor when $eVt - \frac{1}{2}gt^2 = 0$, that is, after time

$$t_2 = \frac{2eV}{g}.$$

(The impact is instantaneous, so no time is lost at the first bounce.)

P_1 comes to rest before the second bounce exactly when $t_1 < t_2$:

$$\frac{V}{\mu g} < \frac{2eV}{g} \iff \frac{1}{\mu} < 2e \iff e > \frac{1}{2\mu},$$

dividing by $V/g > 0$. So if $e > 1/(2\mu)$, P_1 comes to rest before P_2 bounces a second time, as required. In fact the converse holds too.

Checks

- Limiting cases: as $\mu \to 0$ the bound $b/(2d - b)$ is beaten trivially, and P_1 certainly slides off a smooth table. If $b = d$, the bound becomes 1, so with $\mu < 1$ it always falls off, which is right because P_1 is already at the edge, still moving, when P_2 lands.
- Since $\mu < 1$, the second condition needs $e > \frac{1}{2}$. A nearly dead floor gives a short hop, so P_2 lands again first, as we'd expect.
- Numerically, with $d = 2$, $b = 1$, $\mu = 0.5$ and $e = 0.8$ (taking $g = 9.8$), stepping the motion gives $t_1 \approx 0.452\,\text{s}$ and $t_2 \approx 0.361\,\text{s}$, matching $V/(\mu g)$ and $2eV/g$ with $V^2 = 4.9$. Here $e = 0.8 < 1/(2\mu) = 1$, so P_2 bounces again first. Random tests of both criteria in a step-by-step simulation agreed in every case.

Common slips

- Keeping the tension in the second stage: once P_2 lands, the string is slack and P_1 is slowed by friction alone.
- Forgetting that P_1 has already moved a distance b, and comparing s with d rather than $d - b$.
- Writing the system's acceleration as $g(1 - \mu)$, forgetting that the force $mg(1 - \mu)$ has to accelerate *both* masses.

Beyond STEP

The string here is a *unilateral* constraint: it can pull but not push, so the equations of motion switch when it goes slack. Systems whose governing equations change at impacts or when contact is lost are called piecewise-smooth or hybrid systems, and they are studied seriously in engineering dynamics and robotics. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats strings, friction and collisions carefully and is a good next step.

WORKED SOLUTION

STEP II 1992, Question 14

Mechanics • Connected Particles and Pulleys

STEP II 1992, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q14

The key idea

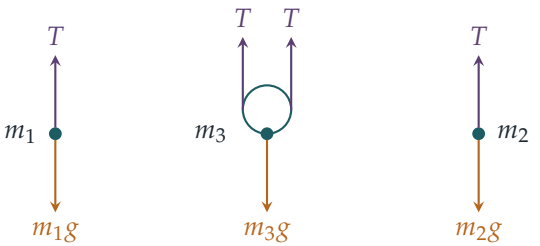
There are three moving bodies but only one string, so there is one tension T throughout and one constraint linking the three accelerations. Write Newton's second law for each particle (the pulley P_3 and its particle move together), write the string-length condition, and you have four linear equations in four unknowns. The rest condition then just says that the acceleration of P_3 is zero.

Solution

The model

The pulleys are smooth and light and the string is light and inextensible, so the tension has the same value T all along the string. Since P_3 is light, it and the particle of mass m_3 form a single body of mass m_3, held up by *two* vertical strands of string, each pulling up with force T.

Measure all displacements *downwards*: let x_1, x_2, x_3 be the depths below the fixed pulleys of the particle m_1, the particle m_2 and the pulley P_3, with accelerations a_1, a_2, a_3 (positive downwards).



Equations of motion

Resolving vertically (downwards positive) for each body:

$$\begin{aligned} m_1 a_1 &= m_1 g - T, \\ m_2 a_2 &= m_2 g - T, \\ m_3 a_3 &= m_3 g - 2T. \end{aligned}$$

The string constraint

Ignoring the fixed lengths wrapped round the pulleys, the string consists of the strand from P_1 down to m_1 (length x_1), the two strands from the fixed pulleys down to P_3 (length x_3 each) and the strand from P_2 down to m_2 (length x_2). Since the string is inextensible,

$$x_1 + x_2 + 2x_3 = \text{constant}, \qquad \text{so} \qquad a_1 + a_2 + 2a_3 = 0$$

after differentiating twice.

Ask yourself Why is the coefficient of x_3 equal to 2 and not 1?

When P_3 drops by a distance d, *both* strands supporting it get longer by d, so $2d$ of string is used up. That has to come from the outer strands, so $x_1 + x_2$ falls by $2d$. Missing this factor is the commonest way to get this question wrong.

The tension

Dividing each equation of motion by its mass,

$$a_1 = g - \frac{T}{m_1}, \qquad a_2 = g - \frac{T}{m_2}, \qquad a_3 = g - \frac{2T}{m_3}.$$

Substituting into $a_1 + a_2 + 2a_3 = 0$:

$$4g - T\left(\frac{1}{m_1} + \frac{1}{m_2} + \frac{4}{m_3}\right) = 0.$$

Multiplying through by $m_1 m_2 m_3$ and solving for T,

$$\boxed{T = \frac{4m_1 m_2 m_3\, g}{m_3(m_1 + m_2) + 4m_1 m_2}}$$

This holds throughout the motion (until a particle reaches a pulley), since nothing in the equations depends on position.

When P_3 stays at rest

The system starts from rest, so P_3 remains at rest exactly when $a_3 = 0$, that is, when $2T = m_3 g$. Using our value of T:

$$\begin{aligned} \frac{8m_1m_2m_3\,g}{m_3(m_1+m_2)+4m_1m_2} &= m_3 g \\ \iff \quad 8m_1m_2 &= m_3(m_1+m_2)+4m_1m_2 \\ \iff \quad 4m_1m_2 &= m_3(m_1+m_2), \end{aligned}$$

where we divided by $m_3 g > 0$ and multiplied by the positive denominator. So if $4m_1m_2 = m_3(m_1+m_2)$ then $a_3 = 0$, and P_3 remains at rest, as required. (Each step is reversible, so the condition is necessary as well as sufficient.)

Note that the outer particles need not be at rest: then $T = \frac{1}{2}m_3 g$ and $a_1 = g - T/m_1 = -a_2$, which is non-zero unless $m_1 = m_2$. One particle rises and the other falls at the same rate, and the string slides under P_3 without moving it.

Checks

- Symmetric case $m_1 = m_2 = m$: the condition becomes $4m^2 = 2mm_3$, i.e. $m_3 = 2m$. That is just static balance, $2T = 2mg = m_3 g$: obvious, and reassuring.
- If $m_3 \to \infty$, then $T \to \dfrac{4m_1m_2g}{m_1+m_2}$, which is twice the familiar Atwood-machine tension $\dfrac{2m_1m_2g}{m_1+m_2}$. That is right: a heavy P_3 falls freely with $a_3 \to g$, so the strands accelerate quickly. If $m_3 \to 0$ then $T \to 0$, as a massless pulley can't support any net force.
- Numerically, with $m_1 = 2$, $m_2 = 3$, $m_3 = 4$ and $g = 9.8$ (SI units), the formula gives $T \approx 21.38\,\text{N}$. Solving the problem independently by an energy (Lagrangian) method gives the same accelerations $a_1 \approx -0.891$, $a_2 \approx 2.673$, $a_3 \approx -0.891\,\text{m s}^{-2}$. With $m_1 = 1$, $m_2 = 3$, $m_3 = 3$ the rest condition holds, and indeed $a_3 = 0$ while $a_1 = -a_2 = -4.9\,\text{m s}^{-2}$.

Common slips

- Writing the constraint as $a_1 + a_2 + a_3 = 0$, forgetting that P_3 hangs on two strands.
- Forgetting that the force on P_3 is $2T$, not T.
- Assuming that P_3 at rest forces the whole system to be at rest. It doesn't unless $m_1 = m_2$.

Beyond STEP

The constraint $x_1 + x_2 + 2x_3 = \text{constant}$ cuts the system to two degrees of freedom, and the cleanest way to handle such systems is Lagrangian mechanics: you write the energy

in terms of independent coordinates and never need the tension at all (it is a "constraint force"). This is how our numerical check worked. You'll meet it in first-year dynamics. A good place to see Newton's laws applied carefully to constrained systems such as pulleys is Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1992, Question 15

Probability and Statistics • Random Walks

STEP II 1992, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q15

The key idea

If the point makes R positive steps out of s, it ends at $x = R - (s - R) = 2R - s$, and R has a binomial distribution. So every probability here is a sum of binomial coefficients. Parity does a lot of the work: after an even number of steps only $x = 2$ and $x = 4$ are possible, and after an odd number only $x = 3$ and $x = 5$, so each $\mathrm{P}(s)$ is a sum of just two neighbouring binomial coefficients. Pascal's rule then collapses that sum to a single coefficient. For the maximum, look at the ratio of consecutive terms, separately for even and odd s.

Solution

Let R be the number of steps in the positive direction among the first s. The steps are independent, each positive with probability $\frac{1}{2}$, so $R \sim \mathrm{B}(s, \frac{1}{2})$ and

$$\mathrm{P}(R = r) = \binom{s}{r}\left(\frac{1}{2}\right)^s.$$

The position after s steps is $X = 2R - s$, so $X = x$ exactly when $R = \frac{1}{2}(s + x)$. In particular X has the same parity as s.

The values $\mathrm{P}(5)$ and $\mathrm{P}(6)$

With $s = 5$, only $x = 3$ and $x = 5$ can happen, needing $R = 4$ and $R = 5$:

$$\mathrm{P}(5) = \left[\binom{5}{4} + \binom{5}{5}\right]\frac{1}{2^5} = \frac{5+1}{32} = \frac{3}{16}.$$

With $s = 6$, only $x = 2$ and $x = 4$ can happen, needing $R = 4$ and $R = 5$:

$$\mathrm{P}(6) = \left[\binom{6}{4} + \binom{6}{5}\right]\frac{1}{2^6} = \frac{15+6}{64} = \frac{21}{64}.$$

The general even case

For $s = 2k$, the possible target points are $x = 2$ and $x = 4$, which need $R = k+1$ and $R = k+2$. By Pascal's rule $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$,

$$\mathrm{P}(2k) = \left[\binom{2k}{k+1} + \binom{2k}{k+2}\right]\left(\frac{1}{2}\right)^{2k} = \binom{2k+1}{k+2}\left(\frac{1}{2}\right)^{2k} = \binom{2k+1}{k-1}\left(\frac{1}{2}\right)^{2k},$$

using $\binom{n}{r} = \binom{n}{n-r}$ with $(2k+1) - (k+2) = k-1$ in the last step. This is the required result. It holds for $k = 1$ too: there $\binom{2}{3} = 0$, so the $x = 4$ term is simply absent, and the formula gives $\frac{1}{4}$, the chance of two positive steps.

The odd case

For $s = 2k+1$ the target points are $x = 3$ and $x = 5$, needing $R = k+2$ and $R = k+3$. In the same way,

$$\mathrm{P}(2k+1) = \left[\binom{2k+1}{k+2} + \binom{2k+1}{k+3}\right]\left(\frac{1}{2}\right)^{2k+1} = \binom{2k+2}{k-1}\left(\frac{1}{2}\right)^{2k+1}.$$

For $k = 2$ this gives $\binom{6}{1}/32 = \frac{3}{16}$, agreeing with $\mathrm{P}(5)$.

The greatest value

Write $a_k = \mathrm{P}(2k)$ for $k \geq 1$ and $b_k = \mathrm{P}(2k+1)$ for $k \geq 1$ (note $b_0 = \mathrm{P}(1) = 0$). All of these are positive, so we can compare consecutive terms by their ratio.

Ask yourself How do we simplify a ratio of two binomial coefficients?

Write both in factorials and cancel. For the even terms,

$$\frac{\binom{2k+3}{k}}{\binom{2k+1}{k-1}} = \frac{(2k+3)!}{k!\,(k+3)!} \cdot \frac{(k-1)!\,(k+2)!}{(2k+1)!} = \frac{(2k+3)(2k+2)}{k(k+3)},$$

so

$$\frac{a_{k+1}}{a_k} = \frac{(2k+3)(2k+2)}{4k(k+3)} = \frac{4k^2+10k+6}{4k^2+12k}.$$

Since the denominator is positive, this ratio is greater than 1 exactly when $10k+6 > 12k$, that is $k < 3$; it equals 1 when $k = 3$; and it is less than 1 when $k > 3$. Hence

$$a_1 < a_2 < a_3 = a_4 > a_5 > a_6 > \cdots,$$

and the largest even-step value is $a_3 = a_4 = \mathrm{P}(6) = \mathrm{P}(8) = \frac{21}{64}$. (Check: $\mathrm{P}(8) = \binom{9}{3}/2^8 = 84/256 = 21/64$.)

For the odd terms the same cancellation gives

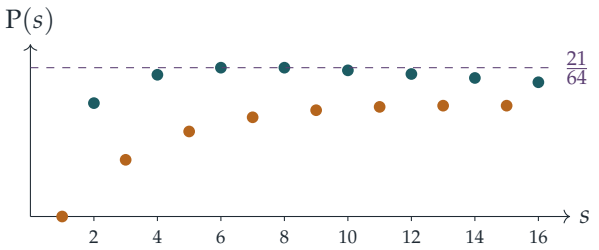
$$\frac{b_{k+1}}{b_k} = \frac{(2k+4)(2k+3)}{4k(k+4)} = \frac{4k^2+14k+12}{4k^2+16k},$$

which is greater than 1 for $k < 6$, equal to 1 at $k = 6$, and less than 1 for $k > 6$. So the largest odd-step value is

$$b_6 = b_7 = \mathrm{P}(13) = \mathrm{P}(15) = \binom{14}{5}\frac{1}{2^{13}} = \frac{2002}{8192} = \frac{1001}{4096}.$$

Finally, $\frac{21}{64} = \frac{1344}{4096} > \frac{1001}{4096}$, so no odd value beats the best even one. Therefore $\mathrm{P}(s)$ takes its greatest value, $\frac{21}{64}$, when

$$s = 6 \quad \text{and} \quad s = 8.$$



The teal points are even s and the amber points odd s: two separate sequences, each rising then falling, with the even one always on top.

Checks

- Small cases by hand: $\mathrm{P}(2) = \frac{1}{4}$ (two steps right), $\mathrm{P}(3) = \frac{1}{8}$ (three steps right), and with $k = 1$ the formulas give $\binom{3}{0}/2^2 = \frac{1}{4}$ and $\binom{4}{0}/2^3 = \frac{1}{8}$, as they should.
- An exact computer enumeration of the walk (fractions, $s \leq 60$) confirms both formulas, $\mathrm{P}(6) = \mathrm{P}(8) = \frac{21}{64} \approx 0.328$ as the overall maximum, and $\mathrm{P}(13) = \mathrm{P}(15) = \frac{1001}{4096} \approx 0.244$ as the odd maximum.

Common slips

- Adding all four target points for every s: half of them are impossible on parity grounds.
- Finding the even maximum and stopping. The odd values form a separate sequence, and you must check that its peak is lower.
- Stopping at "ratio > 1 for $k < 3$" and giving only $s = 6$: the ratio equals 1 at $k = 3$, so $s = 8$ ties.

Beyond STEP

This is the simple symmetric random walk, one of the central objects of probability. Its position after s steps spreads out like $\sqrt{s}$, so the chance of being in any fixed finite

window first rises and eventually falls to zero; the peak near $s = 7$ is where the spread is comparable to the window. The same binomial counting leads to the reflection principle and to the fact that a one-dimensional walk returns to the origin with probability 1. Grimmett and Welsh, *Probability: An Introduction*, has a good chapter on random walks.

WORKED SOLUTION

STEP II 1992, Question 16

Probability and Statistics • Conditional Probability

STEP II 1992, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.17, *Banach's Matchboxes*: https://beyondstep.keithmcnulty.org/ch12/matchboxes.html

The key idea

This is a version of Banach's matchbox problem. Since we're told *when* the shortage was discovered (on reach N), this is a conditional probability. So we compute two joint probabilities: that the first failure happens on reach N and is a toffee, and that it happens on reach N and is a mint. Each one is a "first $N-1$ reaches contain exactly so many of each kind, then one particular kind" calculation, which is binomial.

Solution

Assume the reaches are independent, each a toffee with probability p and a mint with probability q, where $0 < p < 1$. Let T be the event that on the Nth reach she reaches for a toffee and finds the toffee packet empty, and that she has found no packet empty on an earlier reach. Let M be the same event with mints in place of toffees. We want

$$\mathrm{P}(\text{toffee} \mid \text{first discovery on reach } N) = \frac{\mathrm{P}(T)}{\mathrm{P}(T) + \mathrm{P}(M)}.$$

The probability of T

For T to happen, the toffee packet must be empty after $N-1$ reaches. No reach so far has failed, so every one of those reaches took a sweet. So exactly n of the first $N-1$ reaches were for toffees (one more would have been a failure), and the other $N-1-n$ were for mints. Then reach N is for a toffee.

Conversely, suppose the first $N-1$ reaches contain exactly n toffee-reaches and $N-1-n \leq m$ mint-reaches, in any order. Then no earlier reach can have failed: at no stage had she reached for more than n toffees or more than m mints. Then if reach N is a

toffee, T happens.

Ask yourself Doesn't the order of the first $N-1$ reaches matter, so that she doesn't run out early?

No. She only *discovers* that a packet is empty when she reaches into it while it is empty. Emptying the toffee packet early (on the nth toffee) is fine as long as she doesn't reach for a toffee again before reach N. That is automatic, because the first $N-1$ reaches contain only n toffee-reaches. So all $\binom{N-1}{n}$ orders are allowed, and

$$\mathrm{P}(T) = \binom{N-1}{n} p^n q^{N-1-n} \cdot p = \binom{N-1}{n} p^{n+1} q^{N-1-n}, \qquad n \le N-1 \le n+m,$$

and $\mathrm{P}(T) = 0$ otherwise. If $N-1 < n$, she can't have used up the toffees. If $N-1 > n+m$, she would have needed more than m mints.

The probability of M

In exactly the same way, with the roles swapped,

$$\mathrm{P}(M) = \binom{N-1}{m} q^{m+1} p^{N-1-m}, \qquad m \le N-1 \le n+m,$$

and $\mathrm{P}(M) = 0$ otherwise.

The answer

The discovery must happen somewhere in reaches $\min(n,m)+1$ to $n+m+1$. Reach $n+m+2$ is impossible, because by then both packets would be empty and something would already have failed.

- If $N \le m$ (so $\mathrm{P}(M) = 0$ but $\mathrm{P}(T) > 0$), she was certainly reaching for a toffee: the probability is 1.
- If $N \le n$ (so $\mathrm{P}(T) = 0$), the probability is 0.
- If $\max(n,m)+1 \le N \le n+m+1$, both are possible, and

$$\mathrm{P}(\text{toffee}) = \frac{\binom{N-1}{n} p^{n+1} q^{N-1-n}}{\binom{N-1}{n} p^{n+1} q^{N-1-n} + \binom{N-1}{m} q^{m+1} p^{N-1-m}}.$$

In the last case we can tidy this up by dividing the top and bottom by $\mathrm{P}(T)$. The common factor $(N-1)!$ cancels from the binomial coefficients, and the powers combine to $(q/p)^{n+m+2-N}$. This gives

$$\mathrm{P}(\text{toffee}) = \left[1 + \frac{n!\,(N-1-n)!}{m!\,(N-1-m)!} \left(\frac{q}{p}\right)^{n+m+2-N}\right]^{-1}.$$

Checks

- Symmetry: if $n = m$ and $p = q = \frac{1}{2}$, the two expressions are equal and the answer is $\frac{1}{2}$, as it should be.
- Take $n = 2$, $m = 3$, $p = \frac{1}{2}$, $N = 5$. Then $\mathrm{P}(T) = \binom{4}{2}/2^5 = \frac{3}{16}$ and $\mathrm{P}(M) = \binom{4}{3}/2^5 = \frac{1}{8}$, so the answer is $\frac{3/16}{5/16} = \frac{3}{5}$. With fewer toffees, running out of toffees is more likely.
- $\sum_N \big(\mathrm{P}(T) + \mathrm{P}(M)\big) = 1$ over $N = 1, \ldots, n + m + 1$, because some reach must eventually fail. We checked this, and the formula, against an exact step-by-step enumeration for all $n, m \leq 6$ and several values of p. A 2×10^5-trial simulation with $n = 4$, $m = 3$, $N = 7$, $p = 0.55$ gave 0.527, against the exact 0.528.

Common slips

- Forgetting the final factor p (or q) for the Nth reach itself.
- Answering $\mathrm{P}(T)$ rather than the conditional probability $\mathrm{P}(T)/(\mathrm{P}(T) + \mathrm{P}(M))$. We are told that the discovery happened on reach N.
- Thinking the event is "the toffees run out on reach N". Taking the last toffee is not a discovery. The discovery is the next attempt.

Beyond STEP

The time of the first failure is a mixture of two *negative binomial* distributions. The negative binomial distribution counts the trials needed to get a given number of successes. Banach's original matchbox problem ($n = m$, $p = \frac{1}{2}$) asks how many matches are left in the other box at the moment of discovery. Its answer can be approximated with Stirling's formula, giving a mean of about $2\sqrt{n/\pi}$. For the negative binomial distribution, see Grimmett and Welsh, *Probability: An Introduction*. Feller's classic *An Introduction to Probability Theory and Its Applications*, Volume 1, has a famous discussion of the matchbox problem.

WORKED SOLUTION

STEP III 1992, Question 1

Pure • Maclaurin Series

STEP III 1992, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q1

The key idea

In part (i) the two identities are a machine for producing derivatives: once f'' is written in terms of f', every higher derivative follows by differentiating again, and at $x = 0$ everything is built from $f'(0) = \frac{1}{2}$. In part (ii) the trouble is the factor $\sinh x$, which vanishes at the origin; multiplying by x cancels that zero, and the expansion is then the product of two series we can find by hand. Symmetry (odd and even functions) tells us in advance which powers can appear.

Solution

Part (i)

Differentiating $f(x) = \ln(1 + e^x)$ by the chain rule,

$$f'(x) = \frac{e^x}{1 + e^x} > 0,$$

so its logarithm is defined, and

$$\ln[f'(x)] = \ln e^x - \ln(1 + e^x) = x - f(x),$$

which is the first result. Differentiating this identity, the left-hand side gives $f''(x)/f'(x)$, so

$$\frac{f''(x)}{f'(x)} = 1 - f'(x) \quad \Longrightarrow \quad f''(x) = f'(x) - [f'(x)]^2,$$

as required. It is convenient to write this as $f'' = f'(1 - f')$.

? **Ask yourself** How do we get f''' and f'''' without differentiating $\ln(1 + e^x)$ four times?

Differentiate the identity $\mathrm{f}'' = \mathrm{f}' - (\mathrm{f}')^2$ itself, using the product rule on $(\mathrm{f}')^2$:

$$\mathrm{f}''' = \mathrm{f}'' - 2\mathrm{f}'\mathrm{f}'' = \mathrm{f}''(1 - 2\mathrm{f}'),$$
$$\mathrm{f}'''' = \mathrm{f}'''(1 - 2\mathrm{f}') - 2(\mathrm{f}'')^2.$$

Now evaluate at $x = 0$, where $\mathrm{f}(0) = \ln 2$ and $\mathrm{f}'(0) = \frac{1}{1+1} = \frac{1}{2}$:

$$\mathrm{f}''(0) = \tfrac{1}{2} - \tfrac{1}{4} = \tfrac{1}{4},$$
$$\mathrm{f}'''(0) = \tfrac{1}{4}\left(1 - 2 \cdot \tfrac{1}{2}\right) = 0,$$
$$\mathrm{f}''''(0) = 0 - 2\left(\tfrac{1}{4}\right)^2 = -\tfrac{1}{8}.$$

By Maclaurin's theorem, $\mathrm{f}(x) = \sum \mathrm{f}^{(n)}(0)\, x^n / n!$, so

$$\mathrm{f}(x) = \ln 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} + \cdots,$$

the x^3 term being zero (and $-\frac{1}{8} \div 24 = -\frac{1}{192}$).

Part (ii)

Why g has no such expansion. A series $a_0 + a_1 x + a_2 x^2 + \cdots$ in non-negative powers takes the finite value a_0 at $x = 0$, and is continuous there. But $\sinh 0 = 0$ and $\cosh 0 = 1$, so the denominator $\sinh x \cosh 2x$ tends to 0 as $x \to 0$; indeed, since $\sinh x \approx x$ for small x, we have $\mathrm{g}(x) \approx 1/x$, which is unbounded near 0. So $\mathrm{g}(x)$ cannot equal such a series near the origin.

Why $x\mathrm{g}(x)$ does. Write

$$x\,\mathrm{g}(x) = \frac{x}{\sinh x} \cdot \frac{1}{\cosh 2x}.$$

Since $\sinh x = x\left(1 + \frac{x^2}{6} + \frac{x^4}{120} + \cdots\right)$, the factor x cancels and

$$\frac{x}{\sinh x} = \frac{1}{1 + \frac{x^2}{6} + \frac{x^4}{120} + \cdots},$$

which is 1 over a series whose constant term is 1; such a reciprocal can be expanded binomially in non-negative powers. Likewise $\cosh 2x \geq 1$ never vanishes and $1/\cosh 2x$ expands in the same way. The product of two such series is again one.

Why only even powers. sinh is odd and cosh is even, so $\mathrm{g}(-x) = -\mathrm{g}(x)$: g is odd. Hence $h(x) = x\,\mathrm{g}(x)$ satisfies $h(-x) = (-x)(-\mathrm{g}(x)) = h(x)$, so h is even. If $h(x) = \sum a_n x^n$, then $h(x) - h(-x) = 2\sum_{n \text{ odd}} a_n x^n$ is identically zero, and since a power series that vanishes identically has all coefficients zero, $a_n = 0$ for every odd n. (You can also see it directly: both factors above are series in x^2.)

The expansion. Using $(1+u)^{-1} = 1 - u + u^2 - \cdots$ with $u = \frac{x^2}{6} + \frac{x^4}{120}$, and keeping terms up to x^4,

$$\frac{x}{\sinh x} = 1 - \frac{x^2}{6} - \frac{x^4}{120} + \frac{x^4}{36} + \cdots = 1 - \frac{x^2}{6} + \frac{7x^4}{360} + \cdots,$$

since $\frac{1}{36} - \frac{1}{120} = \frac{10-3}{360}$. Next, $\cosh 2x = 1 + 2x^2 + \frac{2}{3}x^4 + \cdots$ (from $\frac{(2x)^2}{2}$ and $\frac{(2x)^4}{24}$), so with $v = 2x^2 + \frac{2}{3}x^4$,

$$\frac{1}{\cosh 2x} = 1 - v + v^2 - \cdots = 1 - 2x^2 - \tfrac{2}{3}x^4 + 4x^4 + \cdots = 1 - 2x^2 + \tfrac{10}{3}x^4 + \cdots.$$

Multiplying the two series and collecting powers:

$$\begin{aligned} x\,\mathrm{g}(x) &= \left(1 - \tfrac{1}{6}x^2 + \tfrac{7}{360}x^4\right)\left(1 - 2x^2 + \tfrac{10}{3}x^4\right) + \cdots \\ &= 1 - \left(2 + \tfrac{1}{6}\right)x^2 + \left(\tfrac{10}{3} + \tfrac{2}{6} + \tfrac{7}{360}\right)x^4 + \cdots \\ &= 1 - \frac{13}{6}x^2 + \frac{1327}{360}x^4 + \cdots, \end{aligned}$$

since $\frac{10}{3} + \frac{1}{3} = \frac{11}{3} = \frac{1320}{360}$.

Checks

- The missing x^3 term in (i) is no accident: $\mathrm{f}(x) - \frac{x}{2} = \ln(\mathrm{e}^{-x/2} + \mathrm{e}^{x/2}) = \ln(2\cosh\frac{x}{2})$ is even, so after $\frac{x}{2}$ only even powers appear.
- Numerically, at $x = 0.1$: $\ln(1+\mathrm{e}^x) \approx 0.7443966600$ and the quartic $\ln 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192}$ agree to about 3.5×10^{-10}.
- At $x = 0.05$, $x\,\mathrm{g}(x)$ and $1 - \frac{13}{6}x^2 + \frac{1327}{360}x^4$ differ by about 9×10^{-8}, roughly $6x^6$, as expected when the next term is in x^6.

Common slips

- Expanding $1/(1+u)$ as $1 - u$ only: the u^2 term contributes to x^4 (the $\frac{1}{36}$ and the $4x^4$ above).
- Writing $\cosh 2x = 1 + 2x^2 + \frac{x^4}{24}\cdots$, forgetting that the 4th power of $2x$ is $16x^4$.

Beyond STEP

Part (ii) is about functions with a *pole*: g blows up like $1/x$, and multiplying by x removes the singularity. In complex analysis such a function has a *Laurent series* $\frac{1}{x}\left(1 - \frac{13}{6}x^2 + \cdots\right)$, and the coefficient of $1/x$ (the *residue*, here 1) is the key to evaluating many real integrals. The radius of convergence of the series is set by the nearest other zero of the denominator, where $\cosh 2x = 0$ at $x = \pm\frac{\pi}{4}\mathrm{i}$. Needham's *Visual Complex Analysis* develops these ideas beautifully.

WORKED SOLUTION

STEP III 1992, Question 2

Pure • Matrix Powers and Induction

STEP III 1992, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q2

The key idea

Everything rests on one fact: $\mathbf{J}^2 = 2\mathbf{J}$. That means any product of matrices of the form $\mathbf{I} + p\mathbf{J}$ is again of that form, so we only ever have to track a single number. The awkward-looking expressions $\frac{1}{2}[(1+2a)^k - 1]$ become transparent once we notice that it is $1 + 2p$, not p, that behaves nicely under multiplication.

Solution

A multiplication rule

Since

$$\mathbf{J}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = 2\mathbf{J},$$

and $\mathbf{I}$ commutes with everything, for any real p and q

$$(\mathbf{I} + p\mathbf{J})(\mathbf{I} + q\mathbf{J}) = \mathbf{I} + (p+q)\mathbf{J} + pq\mathbf{J}^2 = \mathbf{I} + (p + q + 2pq)\mathbf{J}. \qquad (*)$$

Notice that

$$1 + 2(p + q + 2pq) = (1 + 2p)(1 + 2q),$$

so if we attach the number $1 + 2p$ to the matrix $\mathbf{I} + p\mathbf{J}$, multiplying matrices corresponds to multiplying these numbers. For $\mathbf{A} = \mathbf{I} + a\mathbf{J}$ the attached number is $1 + 2a$.

The squares and cubes

By $(*)$ with $p = q = a$,

$$\mathbf{A}^2 = \mathbf{I} + (2a + 2a^2)\mathbf{J}, \qquad \text{and} \qquad \tfrac{1}{2}\left[(1+2a)^2 - 1\right] = \tfrac{1}{2}(4a + 4a^2) = 2a + 2a^2,$$

so $\mathbf{A}^2 = \mathbf{I} + \frac{1}{2}[(1+2a)^2 - 1]\mathbf{J}$, as required.

For the cube, write $\mathbf{A}^3 = \mathbf{A}^2\mathbf{A}$ and use $(*)$ with $p = 2a + 2a^2$, $q = a$:

$$\begin{aligned} p + q + 2pq &= 2a + 2a^2 + a + 2a(2a + 2a^2) \\ &= 3a + 6a^2 + 4a^3 \\ &= \tfrac{1}{2}(6a + 12a^2 + 8a^3) = \tfrac{1}{2}\left[(1 + 2a)^3 - 1\right], \end{aligned}$$

using $(1 + 2a)^3 = 1 + 6a + 12a^2 + 8a^3$. So $\mathbf{A}^3 = \mathbf{I} + \frac{1}{2}[(1 + 2a)^3 - 1]\mathbf{J}$, as required.

For the fourth power, the quickest route is $\mathbf{A}^4 = \mathbf{A}^2\mathbf{A}^2$. With $p = q = \frac{1}{2}[(1 + 2a)^2 - 1]$ we have $1 + 2p = (1 + 2a)^2$, so the number attached to $\mathbf{A}^4$ is $(1 + 2a)^2(1 + 2a)^2 = (1 + 2a)^4$. Hence

$$\mathbf{A}^4 = \mathbf{I} + \tfrac{1}{2}\left[(1 + 2a)^4 - 1\right]\mathbf{J} = \mathbf{I} + (4a + 12a^2 + 16a^3 + 8a^4)\mathbf{J}.$$

The general power

The pattern suggests

$$p_k = \tfrac{1}{2}\left[(1 + 2a)^k - 1\right], \qquad \text{equivalently} \qquad 1 + 2p_k = (1 + 2a)^k.$$

We prove $\mathbf{A}^k = \mathbf{I} + p_k\mathbf{J}$ for all integers $k \geq 1$ by induction.

Base case. For $k = 1$, $p_1 = \frac{1}{2}(1 + 2a - 1) = a$, and $\mathbf{A}^1 = \mathbf{I} + a\mathbf{J}$. (The formula also gives $p_0 = 0$, matching $\mathbf{A}^0 = \mathbf{I}$.)

Inductive step. Suppose $\mathbf{A}^k = \mathbf{I} + p_k\mathbf{J}$ with p_k as above. Then by $(*)$,

$$\mathbf{A}^{k+1} = \mathbf{A}^k\mathbf{A} = \mathbf{I} + (p_k + a + 2ap_k)\mathbf{J},$$

and

$$\begin{aligned} p_k + a + 2ap_k &= (1 + 2a)p_k + a \\ &= \tfrac{1}{2}(1 + 2a)\left[(1 + 2a)^k - 1\right] + a \\ &= \tfrac{1}{2}(1 + 2a)^{k+1} - \tfrac{1}{2}(1 + 2a) + a \\ &= \tfrac{1}{2}\left[(1 + 2a)^{k+1} - 1\right] = p_{k+1}. \end{aligned}$$

So $\mathbf{A}^{k+1} = \mathbf{I} + p_{k+1}\mathbf{J}$, and by induction

$$\mathbf{A}^k = \mathbf{I} + \tfrac{1}{2}\left[(1 + 2a)^k - 1\right]\mathbf{J} \qquad \text{for all integers } k \geq 1.$$

Ask yourself Where did the guess for p_k come from, if I hadn't been handed $\mathbf{A}^2$ and $\mathbf{A}^3$?

From the recurrence itself: the inductive step shows $p_{k+1} = (1 + 2a)p_k + a$, a first-order linear recurrence. Adding $\frac{1}{2}$ to both sides gives $p_{k+1} + \frac{1}{2} = (1 + 2a)(p_k + \frac{1}{2})$, so $p_k + \frac{1}{2}$ is geometric with ratio $1 + 2a$ and first term $a + \frac{1}{2}$. That is exactly $p_k + \frac{1}{2} = \frac{1}{2}(1 + 2a)^k$.

Negative powers (a bonus). If $a \neq -\frac{1}{2}$ the formula works for $k = -1$ too: $p_{-1} = -\dfrac{a}{1+2a}$, and $(*)$ gives $a + p_{-1} + 2ap_{-1} = a - \dfrac{a(1+2a)}{1+2a} = 0$, so $(\mathbf{I}+a\mathbf{J})(\mathbf{I}+p_{-1}\mathbf{J}) = \mathbf{I}$. If $a = -\frac{1}{2}$ then $\mathbf{A} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$ is singular, and the formula gives $p_k = -\frac{1}{2}$ for every $k \geq 1$: indeed $\mathbf{A}^2 = \mathbf{A}$.

Checks

- Take $a = 1$: $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, and multiplying out gives $\mathbf{A}^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$, $\mathbf{A}^4 = \begin{pmatrix} 41 & 40 \\ 40 & 41 \end{pmatrix}$. The formula gives $p_2 = \frac{1}{2}(9-1) = 4$ and $p_4 = \frac{1}{2}(81-1) = 40$.
- $\mathbf{A}$ has eigenvector $(1,1)$ with eigenvalue $1+2a$, and $(1,-1)$ with eigenvalue 1. So $\mathbf{A}^k$ has eigenvalues $(1+2a)^k$ and 1; the matrix $\mathbf{I}+p_k\mathbf{J}$ has eigenvalues $1+2p_k$ and 1, which agree.
- We checked $\mathbf{A}^k = \mathbf{I} + p_k\mathbf{J}$ in exact rational arithmetic for $0 \leq k \leq 12$ and forty random values of a, including $a = -\frac{1}{2}$.

Common slips

- Expanding $(\mathbf{I}+a\mathbf{J})^k$ by the binomial theorem is legitimate (since $\mathbf{I}$ and $\mathbf{J}$ commute) but forgetting $\mathbf{J}^r = 2^{r-1}\mathbf{J}$, not $\mathbf{J}$, gives the wrong coefficients.
- In the inductive step, writing the conclusion for p_{k+1} first and working backwards is not a proof; start from $\mathbf{A}^k\mathbf{A}$.

Beyond STEP

The number $1+2p$ we attached to $\mathbf{I}+p\mathbf{J}$ is the eigenvalue of that matrix on the vector $(1,1)$; every such matrix shares the eigenvectors $(1,1)$ and $(1,-1)$, so they are all diagonalised by the same change of basis, and powers are computed by simply raising eigenvalues to the kth power. This is the idea behind diagonalisation, which turns matrix powers, matrix exponentials and linear recurrences into scalar calculations. Strang's *Introduction to Linear Algebra* develops it with many worked examples.

WORKED SOLUTION

STEP III 1992, Question 3

Pure • Curves and Polynomial Roots

STEP III 1992, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q3

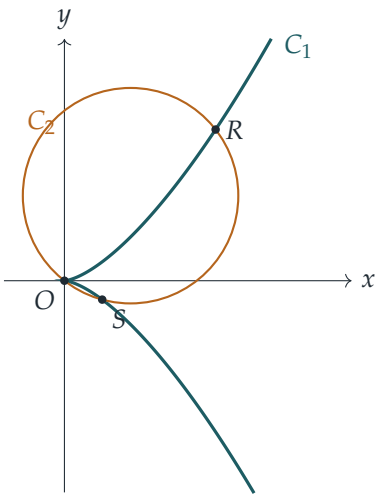
The key idea

Substitute the parametrisation straight into the general equation of a circle through the origin. This gives a polynomial equation in t whose roots are the parameters of the intersection points, and the origin shows up as a factor t^2. What is left is a quartic with no t^3 term, so its roots have a fixed sum and a fixed sum of products in pairs. Knowing two of the roots, r and s, then pins down the other two completely.

Solution

The sketch

Since $x = t^2 \geq 0$, the curve lies in the half-plane $x \geq 0$. Replacing t by $-t$ sends (x, y) to $(x, -y)$, so the curve is symmetric about the x-axis. Eliminating t gives $y^2 = x^3$, so $y = \pm x^{3/2}$. At $t = 0$ we have $\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}y}{\mathrm{d}t} = 0$, and near the origin $|y| = x^{3/2}$ is much smaller than x. So both branches leave O tangent to the x-axis, and the curve has a *cusp* at O. For large t the branches grow steeply, since $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3t}{2}$.



The curve C_1 is in teal. The amber circle is one example of C_2, the circle through O, R and S with $r = 1$ and $s = -\frac{1}{2}$.

The quartic

A circle through the origin has an equation with no constant term:

$$x^2 + y^2 + 2gx + 2fy = 0,$$

for real constants g and f, not both zero. The point with parameter t lies on the circle exactly when

$$t^4 + t^6 + 2gt^2 + 2ft^3 = 0, \qquad \text{that is,} \qquad t^2\left(t^4 + t^2 + 2ft + 2g\right) = 0.$$

The factor t^2 gives $t = 0$, which is the origin. Every intersection other than O has $t \neq 0$, so its parameter satisfies

$$t^4 + t^2 + at + b = 0, \qquad a = 2f, \quad b = 2g.$$

In particular r and s are roots of this equation, and a, b are real constants, as required.

Each point of C_1 has exactly one parameter, since $t = y/x$ when $x \neq 0$ (and $t = 0$ when $x = 0$). So distinct points R and S have distinct parameters, $r \neq s$, and the intersections of C_1 and C_2 correspond exactly to the real roots of the equation $t^2(t^4 + t^2 + at + b) = 0$.

The other two roots

Let the four roots of the quartic (real or complex) be r, s, p and q. Comparing coefficients of

$$(t - r)(t - s)(t - p)(t - q) = t^4 + 0 \cdot t^3 + 1 \cdot t^2 + at + b$$

gives

$$\begin{aligned} r + s + p + q &= 0, \\ rs + (r + s)(p + q) + pq &= 1. \end{aligned}$$

(Here we've grouped the six products in pairs as rs, pq and the four cross terms $rp + rq + sp + sq = (r+s)(p+q)$.)

> **Ask yourself** We only know two of the four roots. How can we get at the other two without knowing a and b?

The two missing coefficients, t^3 and t^2, don't depend on the circle at all. The first equation gives $p + q = -(r+s)$. Substituting into the second,

$$pq = 1 - rs + (r+s)^2 = r^2 + rs + s^2 + 1.$$

So p and q are the roots of the quadratic

$$t^2 + (r+s)\,t + \left(r^2 + rs + s^2 + 1\right) = 0,$$

whose coefficients depend only on r and s. Any further intersection of C_1 and C_2 would have to have a parameter that is a real root of this quadratic.

(As a by-product, $a = -(r+s)(r^2+s^2+1)$ and $b = rspq = rs(r^2+rs+s^2+1)$, so the circle through O, R and S is determined by r and s.)

No further real intersections

The discriminant of the quadratic is

$$\begin{aligned}\Delta &= (r+s)^2 - 4\left(r^2 + rs + s^2 + 1\right)\\ &= -3r^2 - 2rs - 3s^2 - 4\\ &= -\left[2(r^2+s^2) + (r+s)^2 + 4\right].\end{aligned}$$

The bracket is at least 4, because each of $2(r^2+s^2)$ and $(r+s)^2$ is non-negative. So $\Delta \le -4 < 0$, and the quadratic has no real roots: p and q are a pair of non-real complex conjugates.

So the only real roots of $t^2(t^4 + t^2 + at + b) = 0$ are 0, r and s. Since each real root corresponds to exactly one point of C_1, the only real intersections of C_1 and C_2 are O, R and S. This is what we had to show.

Checks

Take $r = 1$, $s = -\frac{1}{2}$. The formulas give $a = -\frac{9}{8}$ and $b = -\frac{7}{8}$, so $f = -\frac{9}{16}$ and $g = -\frac{7}{16}$. The circle has centre $(\frac{7}{16}, \frac{9}{16})$, and both $R = (1,1)$ and $S = (\frac{1}{4}, -\frac{1}{8})$ are at squared distance $\frac{130}{256}$ from it, as is O. The leftover quadratic is $t^2 + \frac{1}{2}t + \frac{7}{4}$, with discriminant $\frac{1}{4} - 7 < 0$. This is the circle in the figure, and you can see it meets C_1 only at O, R and S. We also checked 400 random pairs (r,s) numerically. For each pair we found the circle through O, R, S, confirmed the factorisation, and scanned $-30 \le t \le 30$: the circle equation changed sign only at $t = r$ and $t = s$.

Common slips

- Forgetting that a circle through O has no constant term. The general circle has three constants, and losing the constant term is exactly what makes the factor t^2 appear.
- Thinking the double root $t = 0$ means the circle is tangent to C_1 at O. It comes from the cusp: every circle through O gives t^2 as a factor.
- Saying the discriminant is "clearly negative" without writing it as minus a sum of squares plus a positive constant.

Beyond STEP

The curve $y^2 = x^3$ is the standard example of a singular cubic curve. Counting intersections with multiplicity, as here, is the start of Bézout's theorem: curves of degrees m and n meet in mn points, counted properly. A cubic and a circle give 6, which matches our two at O, plus R, S and the two complex points. The same Vieta-type bookkeeping drives the group law on elliptic curves. Reid's *Undergraduate Algebraic Geometry* and Silverman and Tate's *Rational Points on Elliptic Curves* both take these ideas further.

WORKED SOLUTION

STEP III 1992, Question 4

Pure • Orthogonal Trajectories

STEP III 1992, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q4

The key idea

Write each curve of S_1 as $y = 1 + \dfrac{a}{x-a}$: it is a rectangular hyperbola with asymptotes $x = a$ and $y = 1$, and every one passes through the origin. To find a differential equation for the whole family, differentiate and then use the curve's own equation to get rid of a. Two curves cross at right angles exactly when the product of their gradients is -1, so replacing $\mathrm{d}y/\mathrm{d}x$ by $-1\big/\frac{\mathrm{d}y}{\mathrm{d}x}$ turns the equation for S_1 into the one for S_2, which separates.

Solution

The curves of S_1

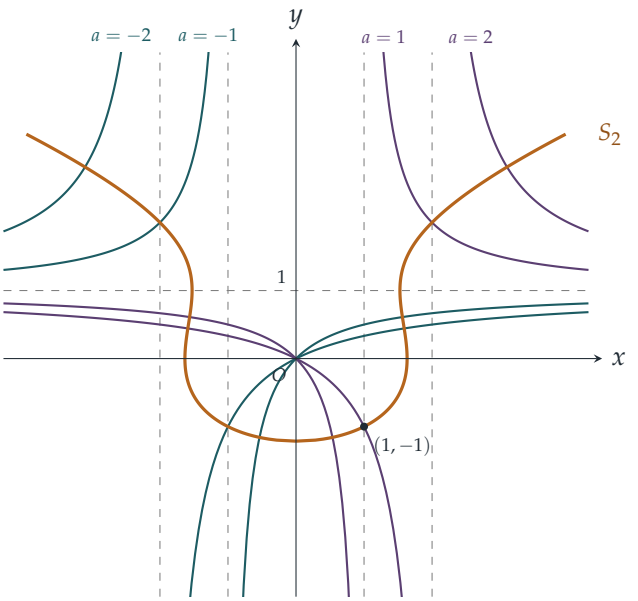
Since $x = (x-a) + a$,

$$y = \frac{x}{x-a} = 1 + \frac{a}{x-a}, \qquad \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{a}{(x-a)^2}.$$

So for $a \neq 0$ the curve is a rectangular hyperbola with vertical asymptote $x = a$ and horizontal asymptote $y = 1$, and it passes through the origin.

- If $a > 0$ the gradient is always negative. The branch with $x < a$ comes down from $y = 1$ (as $x \to -\infty$), passes through O and falls to $-\infty$ as $x \to a^-$; the branch with $x > a$ comes down from $+\infty$ towards $y = 1$.
- If $a < 0$ the gradient is always positive. The branch with $x > a$ rises from $-\infty$ through O towards $y = 1$; the branch with $x < a$ lies above $y = 1$.

Replacing x by $-x$ turns $\frac{x}{x-a}$ into $\frac{x}{x+a}$, so the curve for $-a$ is the reflection in the y-axis of the curve for a. A handy point to plot: $y = 3$ when $x = \frac{3}{2}a$.



The dashed lines are the asymptotes. The amber curve is the member of S_2 through $(1,-1)$, whose shape we justify below.

The differential equation for S_1

From $y(x-a)=x$ we get $x-a=\dfrac{x}{y}$ and $a=x-\dfrac{x}{y}=\dfrac{x(y-1)}{y}$. (On a curve of S_1, $y=0$ only at the origin, so away from that point we may divide by y.) Substituting into the gradient,

$$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{a}{(x-a)^2}=-\frac{x(y-1)}{y}\cdot\frac{y^2}{x^2}=-\frac{y(y-1)}{x},$$

so every member of S_1 satisfies

$$x\frac{\mathrm{d}y}{\mathrm{d}x}=y(1-y).$$

The differential equation for S_2

Two curves meet at right angles exactly when the product of their gradients at the meeting point is -1. So if p_1 is the gradient of the S_1 curve and p_2 that of the S_2 curve at a common point, then $p_1p_2=-1$, i.e. $p_1=-1/p_2$.

> **Ask yourself** The equation we just found is about S_1. Why does it tell us anything about S_2?

Through each point (x,y) (with $x,y\neq 0$) there passes exactly one curve of S_1, the one with $a=x(y-1)/y$, and its gradient there is $-y(y-1)/x$. A curve of S_2 through the

same point must have gradient $p_2 = -1/p_1$, so on S_2

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{-y(y-1)/x} = \frac{x}{y(y-1)}, \qquad \text{that is,} \qquad x = y(y-1)\frac{\mathrm{d}y}{\mathrm{d}x},$$

as required.

The member of S_2 through $(1, -1)$

The equation separates:

$$\int (y^2 - y)\,\mathrm{d}y = \int x\,\mathrm{d}x \quad \Longrightarrow \quad \frac{y^3}{3} - \frac{y^2}{2} = \frac{x^2}{2} + C.$$

At $(1, -1)$: $-\frac{1}{3} - \frac{1}{2} = \frac{1}{2} + C$, so $C = -\frac{4}{3}$. Multiplying by 6, the curve sketched is

$$3x^2 = 2y^3 - 3y^2 + 8.$$

To sketch it, write $g(y) = 2y^3 - 3y^2 + 8$, so $x = \pm\sqrt{g(y)/3}$ and the curve is symmetric in the y-axis. Since $g'(y) = 6y(y-1)$, g has a local maximum $g(0) = 8$ and a local minimum $g(1) = 7$. Both are positive, so g has just one real zero, between -2 and -1 (as $g(-2) = -20$, $g(-1) = 3$); numerically it is $y_0 \approx -1.214$. So the curve:

- has its lowest point at $(0, y_0)$, where $\mathrm{d}y/\mathrm{d}x = 0$;
- has vertical tangents where $y = 0$ (at $x = \pm\sqrt{8/3} \approx \pm 1.63$) and where $y = 1$ (at $x = \pm\sqrt{7/3} \approx \pm 1.53$), since there $y(y-1) = 0$;
- between $y = 0$ and $y = 1$ bends back slightly towards the axis (for $x > 0$ the gradient $x/(y(y-1))$ is negative there), then grows like $y \approx (\frac{3}{2}x^2)^{1/3}$ for large $|x|$.

As a check on the picture, the S_1 curve through $(1, -1)$ has $-1 = \frac{1}{1-a}$, so $a = 2$, with gradient -2 there; the S_2 gradient is $\frac{1}{(-1)(-2)} = \frac{1}{2}$, and indeed $(-2) \cdot \frac{1}{2} = -1$. The vertical tangent at $y = 1$ also makes sense: there the curve crosses $y = 1$, the $a = 0$ member of S_1, at right angles.

Checks

- At $(1, -1)$: $3(1)^2 = 3$ and $2(-1)^3 - 3(-1)^2 + 8 = 3$. ✓
- The S_2 curve meets the $a = 1$ curve at $(2, 2)$: there the S_1 gradient is $-1/(2-1)^2 = -1$ and the S_2 gradient is $2/(2 \cdot 1) = 1$, product -1.
- Numerically, we found all twelve crossings of the amber curve with the four sketched hyperbolas (three each, e.g. $a = -2$ at $(1.5901, 0.4429)$) and the product of gradients was -1.00000000 at each.

Common slips

- Leaving a in the differential equation: the whole point is to eliminate it using $y = x/(x-a)$ itself.

- Using $p_2 = -p_1$ or $p_2 = 1/p_1$ instead of $p_1 p_2 = -1$.
- Sketching the S_2 curve as a simple U: it has vertical tangents at $y = 0$ and $y = 1$ and a small backward bend between them.

Beyond STEP

The curves of S_2 are the *orthogonal trajectories* of S_1. They appear in physics as field lines crossing equipotentials, or streamlines crossing lines of constant velocity potential. The trick here (eliminate the parameter, then replace y' by $-1/y'$) is the standard method, and the resulting first-order equations are a staple of a first ODE course; see Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP III 1992, Question 5

Pure • Vector Geometry in 3D

STEP III 1992, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q5

The key idea

The symmetry does almost all the work. Reflecting in the plane $y+z=2$ maps the tetrahedron to itself and fixes A and D, so it must swap B and C. That tells us BC is perpendicular to the plane and that its midpoint N lies in the plane. After that, everything is Pythagoras and the scalar product, using the fact that all four vertices are the same distance from the centre Y.

Solution

Finding N and the radius

We have $\overrightarrow{AY} = (3,-2,4) - (0,4,-2) = (3,-6,6)$, so the radius of the sphere is

$$YA = \sqrt{9+36+36} = 9.$$

Since $\overrightarrow{AY} = 3\overrightarrow{YN}$, we get $\overrightarrow{YN} = (1,-2,2)$, so $N = (4,-4,6)$ and $YN = 3$.

Notice that A, Y and N all satisfy $y+z=2$: all three lie in the plane of symmetry.

Why BC is perpendicular to the plane

Let Π be the plane $y+z=2$. Reflection in Π maps the tetrahedron onto itself, so it permutes the vertices. It fixes A and D, which lie in Π. Neither B nor C lies in Π: if one did, the reflection would fix it, and then it would have to fix the fourth vertex too, putting all four vertices in Π and leaving no tetrahedron. So the reflection swaps B and C, which means C is the mirror image of B. Hence

- BC is perpendicular to Π, so it is parallel to the normal $(0,1,1)$;
- the midpoint N of BC lies in Π (which matches what we found).

Ask yourself Why is the triangle YNB right-angled at N?

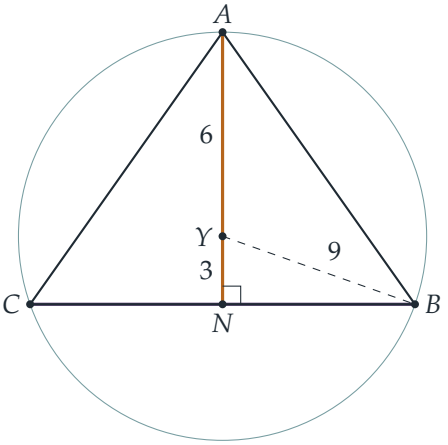
Both Y and N lie in Π, so YN is a line in Π. But NB is along the normal to Π. A normal is perpendicular to every line in the plane, so $\angle YNB = 90°$.

BN and the coordinates of B and C

B lies on the sphere, so $YB = 9$. By Pythagoras in triangle YNB,

$$BN^2 = YB^2 - YN^2 = 81 - 9 = 72, \qquad \text{so} \qquad BN = 6\sqrt{2},$$

as required.



The figure shows the plane ABC, which contains Y (since Y is on AN), drawn to scale. The amber line AN is where it meets Π; the circle is where it cuts the sphere.

A unit vector along BC is $\frac{1}{\sqrt{2}}(0,1,1)$, so

$$\overrightarrow{NB} = \pm 6\sqrt{2} \cdot \tfrac{1}{\sqrt{2}}(0,1,1) = \pm(0,6,6).$$

With $N = (4,-4,6)$, the two ends of the chord are $(4,2,12)$ and $(4,-10,0)$. So

$$B = (4,2,12),\ C = (4,-10,0) \qquad \text{or} \qquad B = (4,-10,0),\ C = (4,2,12).$$

As a check, $(4,2,12) - (3,-2,4) = (1,4,8)$ has length $\sqrt{1+16+64} = 9$, so B really is on the sphere.

The coordinates of D

Write $\overrightarrow{YD} = (p,q,r)$. We have three conditions.

- D and Y both lie in Π, so $\overrightarrow{YD}$ is perpendicular to the normal $(0,1,1)$: $q + r = 0$.
- D is on the sphere: $p^2 + q^2 + r^2 = 81$.
- The angle condition: $\overrightarrow{YA} \cdot \overrightarrow{YD} = YA \cdot YD \cos \angle AYD = 9 \cdot 9 \cdot \frac{1}{3} = 27$.

Ask yourself Three equations in three unknowns, but one is quadratic. How many solutions should we expect?

Up to two, since we will end with a quadratic. Geometrically, D lies on the circle where Π cuts the sphere (a great circle, as $Y \in \Pi$), and the points on that circle making a given angle with YA are the two mirror-image positions either side of A.

With $\overrightarrow{YA} = (-3, 6, -6)$ and $r = -q$, the angle condition gives

$$-3p + 6q + 6q = 27 \quad \Longrightarrow \quad p = 4q - 9.$$

Substituting into the sphere equation, with $r^2 = q^2$:

$$\begin{aligned}(4q-9)^2 + 2q^2 &= 81\\ 18q^2 - 72q &= 0\\ 18q(q-4) &= 0.\end{aligned}$$

- $q = 0$: then $p = -9$, $r = 0$, so $\overrightarrow{YD} = (-9, 0, 0)$ and $D = (-6, -2, 4)$.
- $q = 4$: then $p = 7$, $r = -4$, so $\overrightarrow{YD} = (7, 4, -4)$ and $D = (10, 2, 0)$.

Both give genuine tetrahedra: D is in Π but not on the line AN (the only part of the plane ABC inside Π), so D is not in the plane ABC. Hence

$$D = (-6, -2, 4) \qquad \text{or} \qquad D = (10, 2, 0).$$

Checks

- Each D satisfies $y + z = 2$, and $|\overrightarrow{YD}|^2 = 81$ in both cases (81 and $49 + 16 + 16$).
- For $D = (10, 2, 0)$: $\overrightarrow{YA} \cdot \overrightarrow{YD} = -21 + 24 + 24 = 27 = 81 \cdot \frac{1}{3}$. ✓
- B and C are mirror images: reflecting (x, y, z) in $y + z = 2$ gives $(x,\, 2 - z,\, 2 - y)$, which sends $(4, 2, 12)$ to $(4, -10, 0)$.
- A computer searched every integer point of Π with coordinates up to 20 for points on the sphere with $\cos \angle AYD = \frac{1}{3}$, and found exactly $(-6, -2, 4)$ and $(10, 2, 0)$. The tetrahedron has volume $1728/6 = 288$ in both cases, so it is not flat.

Common slips

- Getting N wrong. $\overrightarrow{AY} = 3\overrightarrow{YN}$ puts Y three-quarters of the way along AN, so $N = Y + \frac{1}{3}\overrightarrow{AY}$, not $A + \frac{1}{3}\overrightarrow{AY}$.
- Using the radius as YN or AN. The radius is $YA = 9$; $YN = 3$ is just the distance from the centre to the chord BC.
- Forgetting that D must lie in Π. Without $q + r = 0$ there is a whole circle of possible points.

Beyond STEP

The argument that the reflection must swap B and C is a small piece of group theory: the symmetries of a shape act by permuting its vertices, and fixed points of a reflection must lie in its mirror. The reflection itself is the linear-algebra map $\mathbf{x} \mapsto \mathbf{x} - 2\frac{(\mathbf{x}\cdot\mathbf{n}-d)}{|\mathbf{n}|^2}\mathbf{n}$, an orthogonal transformation with determinant -1. Strang's *Introduction to Linear Algebra* treats reflections and orthogonal matrices, and Allenby's *Rings, Fields and Groups* develops symmetry groups.

WORKED SOLUTION

STEP III 1992, Question 6

Pure • Trigonometric Reduction Formulae

STEP III 1992, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q6

The key idea

Both reductions rest on turning a difference of trigonometric functions into a product that contains a factor $\sin x$, which then cancels the awkward denominator. For I_n the tool is $\sin^2 A - \sin^2 B = \sin(A+B)\sin(A-B)$; for J_n it is the sum-to-product formula for $\sin P - \sin Q$. Once the denominators have gone, what is left is an elementary integral, and the two reduction formulae telescope.

Solution

First a word on the integrands. Each has $\sin x$ in the denominator, which vanishes at $x = 0$ and $x = \pi$, but the singularities are removable: since $\sin(nx)/\sin x \to n$ as $x \to 0$ and $\sin(nx)/\sin x \to \pm n$ as $x \to \pi$, both integrands stay bounded. So all the integrals below are ordinary integrals of continuous functions.

The relation $I_n - I_{n-1} = J_n$

Using $\sin^2 A - \sin^2 B = (\sin A - \sin B)(\sin A + \sin B)$ and the sum-to-product formulae,

$$\sin^2 A - \sin^2 B = \left(2\cos\tfrac{A+B}{2}\sin\tfrac{A-B}{2}\right)\left(2\sin\tfrac{A+B}{2}\cos\tfrac{A-B}{2}\right) = \sin(A+B)\sin(A-B).$$

With $A = nx$ and $B = (n-1)x$ this gives

$$\sin^2(nx) - \sin^2\big((n-1)x\big) = \sin\big((2n-1)x\big)\sin x.$$

Therefore, for $n \geq 1$,

$$I_n - I_{n-1} = \int_0^\pi \frac{x\left[\sin^2(nx) - \sin^2((n-1)x)\right]}{\sin^2 x}\,\mathrm{d}x = \int_0^\pi \frac{x\sin(2n-1)x}{\sin x}\,\mathrm{d}x = J_n,$$

as required. (For $n = 1$ we read I_0 as the integral of $x\sin^2 0/\sin^2 x = 0$, so $I_0 = 0$.)

A reduction formula for J_n

Now compare J_{n+1} with J_n. By the sum-to-product formula $\sin P - \sin Q = 2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$,

$$\sin(2n+1)x - \sin(2n-1)x = 2\cos(2nx)\sin x,$$

so again the $\sin x$ cancels:

$$J_{n+1} - J_n = 2\int_0^{\pi} x\cos(2nx)\,\mathrm{d}x.$$

Integrating by parts with $u = x$ and $\mathrm{d}v = \cos(2nx)\,\mathrm{d}x$,

$$\begin{aligned}\int_0^{\pi} x\cos(2nx)\,\mathrm{d}x &= \Big[\frac{x\sin(2nx)}{2n}\Big]_0^{\pi} - \frac{1}{2n}\int_0^{\pi}\sin(2nx)\,\mathrm{d}x \\ &= 0 + \frac{1}{4n^2}\Big[\cos(2nx)\Big]_0^{\pi} = \frac{1}{4n^2}(1-1) = 0,\end{aligned}$$

since $\sin(2n\pi) = 0$ and $\cos(2n\pi) = 1$ for integer n. The reduction formula is therefore

$$J_{n+1} = J_n \qquad (n \geq 1).$$

Ask yourself A reduction formula that says $J_{n+1} = J_n$ looks suspiciously trivial. Is that really the answer?

Yes: the $2\cos(2nx)$ that appears integrates against x to zero over a whole number of periods, so nothing changes from one n to the next. To pin the value down we only need the first term:

$$J_1 = \int_0^{\pi}\frac{x\sin x}{\sin x}\,\mathrm{d}x = \int_0^{\pi} x\,\mathrm{d}x = \frac{\pi^2}{2},$$

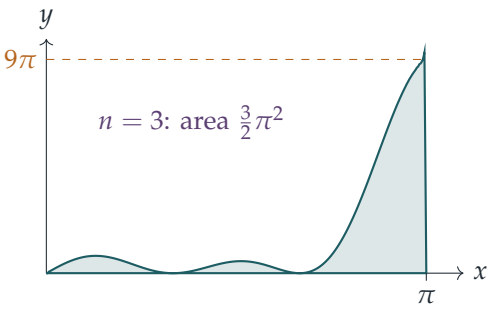
and hence $J_n = \frac{1}{2}\pi^2$ for every positive integer n.

The area under C

Since $x \geq 0$ and $\sin^2(nx)/\sin^2 x \geq 0$ on $(0,\pi)$, the curve never dips below the x-axis, so the area under C is exactly I_n. Summing the first relation (it telescopes) and using $I_0 = 0$,

$$I_n = I_0 + \sum_{k=1}^{n}(I_k - I_{k-1}) = \sum_{k=1}^{n} J_k = n\cdot\frac{\pi^2}{2}.$$

So the area under C is $\frac{1}{2}n\pi^2$, as required.



Checks

- $n = 1$: the curve is $y = x$, and the area of that triangle is $\frac{1}{2}\pi^2$. ✓
- Simpson's rule on the integrals for $n = 1, \ldots, 8$ gives $J_n \approx 4.9348022 = \frac{1}{2}\pi^2$ every time and, for example, $I_3 \approx 14.8044066 = \frac{3}{2}\pi^2$.
- The curve rises to $y = n^2\pi$ at $x = \pi$, yet the area grows only linearly in n: the peaks get taller but narrower (width about π/n), which is consistent.

Common slips

- Forgetting that $I_0 = 0$ (or never checking what I_0 means), so the telescoping sum has no starting value.
- Sign errors in $\int x\cos(2nx)\,\mathrm{d}x$; it is cleaner to note that the boundary term vanishes and the remaining integral of $\sin(2nx)$ over $[0, \pi]$ is zero.

Beyond STEP

The function $\sin^2(nx)/\sin^2 x$ is, up to a factor $1/n$, the *Fejér kernel* of Fourier analysis, and $\sin\big((2n-1)x\big)/\sin x$ is essentially the *Dirichlet kernel*. The fact you found here, that their integrals do not depend on n while the functions concentrate at the ends of the interval, is exactly what makes them useful for proving that Fourier series converge. Stein and Shakarchi's *Fourier Analysis: An Introduction* develops this beautifully; for the real-analysis background, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 1992, Question 7

Pure • Vector Geometry

STEP III 1992, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q7

The key idea

Everything in the question is built from parallelograms and points dividing sides in given ratios, which is exactly what position vectors handle well. Put the origin at A and write every point in terms of the two side vectors $\overrightarrow{AB}$ and $\overrightarrow{AD}$. Then find where two of the lines meet, and check that this point lies on the third line.

Solution

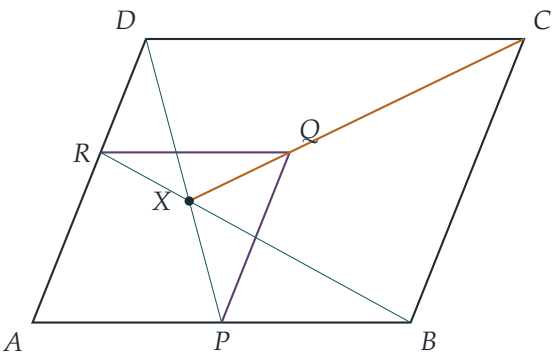
Take A as the origin and let $\mathbf{b} = \overrightarrow{AB}$, $\mathbf{d} = \overrightarrow{AD}$. These are not parallel, because $ABCD$ is a genuine parallelogram. So every point of the plane is $\lambda\mathbf{b} + \mu\mathbf{d}$ for exactly one pair of real numbers (λ, μ), and we can compare coefficients.

Since $ABCD$ is a parallelogram, C has position vector $\mathbf{b} + \mathbf{d}$. Because P lies on side AB and R on side AD, we can write

$$\mathbf{p} = p\,\mathbf{b}, \qquad \mathbf{r} = r\,\mathbf{d}, \qquad 0 \leq p, r \leq 1.$$

Since $APQR$ is a parallelogram, $\overrightarrow{AQ} = \overrightarrow{AP} + \overrightarrow{AR}$, so

$$\mathbf{q} = p\,\mathbf{b} + r\,\mathbf{d}.$$



Where BR meets DP

A general point of line BR is

$$\mathbf{b} + s(r\mathbf{d} - \mathbf{b}) = (1 - s)\,\mathbf{b} + sr\,\mathbf{d},$$

and a general point of line DP is

$$\mathbf{d} + t(p\mathbf{b} - \mathbf{d}) = tp\,\mathbf{b} + (1 - t)\,\mathbf{d}.$$

At a common point the coefficients of $\mathbf{b}$ and of $\mathbf{d}$ must agree:

$$1 - s = tp, \qquad sr = 1 - t.$$

Substituting $s = 1 - tp$ into the second equation gives $r - rpt = 1 - t$, so $t(1 - pr) = 1 - r$.

Ask yourself Can we divide by $1 - pr$?

Since $0 \le p, r \le 1$, we have $pr = 1$ only when $p = r = 1$, that is, when $P = B$ and $R = D$. Then $Q = C$, the "line" CQ is not defined, and there is nothing to prove. So we exclude that case, and then $1 - pr > 0$. This also shows BR and DP are not parallel, so they meet at exactly one point. Then

$$t = \frac{1 - r}{1 - pr}, \qquad s = 1 - tp = \frac{1 - pr - p + pr}{1 - pr} = \frac{1 - p}{1 - pr},$$

and the intersection point X is

$$\mathbf{x} = tp\,\mathbf{b} + sr\,\mathbf{d} = \frac{p(1 - r)\,\mathbf{b} + r(1 - p)\,\mathbf{d}}{1 - pr}.$$

X lies on CQ

A general point of line CQ is

$$\mathbf{c} + u(\mathbf{q} - \mathbf{c}) = (\mathbf{b} + \mathbf{d}) + u\big((p - 1)\mathbf{b} + (r - 1)\mathbf{d}\big) = \big(1 - u(1 - p)\big)\mathbf{b} + \big(1 - u(1 - r)\big)\mathbf{d}.$$

Try $u = \dfrac{1}{1 - pr}$. The coefficient of $\mathbf{b}$ becomes

$$1 - \frac{1 - p}{1 - pr} = \frac{1 - pr - 1 + p}{1 - pr} = \frac{p(1 - r)}{1 - pr},$$

and the coefficient of $\mathbf{d}$ becomes

$$1 - \frac{1 - r}{1 - pr} = \frac{1 - pr - 1 + r}{1 - pr} = \frac{r(1 - p)}{1 - pr}.$$

These are exactly the coefficients of $\mathbf{x}$. So X lies on CQ as well, and the three lines BR, CQ and DP all pass through X. They meet in a point, as required. ■

(The symmetry is reassuring: swapping the roles of B, P, p with D, R, r swaps BR with DP, leaves CQ alone, and leaves the formula for $\mathbf{x}$ unchanged.)

Checks

- If $P = A$ ($p = 0$), then $Q = R$, and $\mathbf{x} = r\mathbf{d}$, which is R. Indeed DP is line DA, which meets BR at R, and CQ is CR, which passes through R.
- If P and R are the midpoints of AB and AD, then $\mathbf{x} = \frac{1}{3}(\mathbf{b} + \mathbf{d})$, one third of the way from A to C. This is the centroid of triangle ABD, as it should be, because BR and DP are two of its medians.
- We checked the concurrency exactly, with fractions, on 5000 random parallelograms and random positions of P and R. Every case matched the formula for $\mathbf{x}$.

Common slips

- Forgetting that the two coefficients can be compared only because $\mathbf{b}$ and $\mathbf{d}$ are not parallel.
- Using the same parameter for two different lines. Each line needs its own (s, t, u).
- Not mentioning the degenerate case $P = B$, $R = D$, where $Q = C$ and the line CQ isn't defined.

Beyond STEP

Notice that the solution never used lengths or angles, only parallel lines and ratios along lines. So the result belongs to *affine geometry*: an affine map (a linear map followed by a translation) sends any parallelogram to a square while keeping concurrency, so it would be enough to prove the result for a square. Questions about lines meeting are also natural in projective geometry, where theorems such as Ceva's and Desargues' live. For a broad, readable tour of these ideas, try Coxeter, *Introduction to Geometry*, or Brannan, Esplen and Gray, *Geometry*.

WORKED SOLUTION

STEP III 1992, Question 8

Pure • Trigonometric Sums via Roots

STEP III 1992, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q8

The key idea

De Moivre's theorem turns $\sin(2n+1)\theta$ into a polynomial in $\cos\theta$ and $\sin\theta$, and dividing out $\sin^{2n+1}\theta$ leaves a polynomial in $\cot^2\theta$. The angles $k\pi/(2n+1)$ make the left-hand side vanish, so their squared cotangents are roots of a known polynomial. Once we know *all* the roots, the three sums are just Vieta's formulas (sums of roots and of reciprocals of roots) in disguise.

Solution

The identity

Write $c = \cos\theta$, $s = \sin\theta$ and $N = 2n+1$. By de Moivre's theorem and the binomial theorem,

$$\cos N\theta + \mathrm{i}\sin N\theta = (c+\mathrm{i}s)^N = \sum_{j=0}^{N}\binom{N}{j}c^{N-j}(\mathrm{i}s)^j.$$

The imaginary terms are those with j odd. Writing $j = 2m+1$, we have $\mathrm{i}^{2m+1} = \mathrm{i}(-1)^m$, so

$$\sin N\theta = \sum_{m=0}^{n}(-1)^m\binom{2n+1}{2m+1}c^{2n-2m}s^{2m+1}.$$

Now reverse the order of summation with $r = n-m$. Then $2m+1 = 2n+1-2r$, and $\binom{2n+1}{2n+1-2r} = \binom{2n+1}{2r}$ by the symmetry of binomial coefficients. So

$$\begin{aligned}\sin(2n+1)\theta &= \sum_{r=0}^{n}(-1)^{n-r}\binom{2n+1}{2r}c^{2r}s^{2n+1-2r}\\ &= \sin^{2n+1}\theta\sum_{r=0}^{n}(-1)^{n-r}\binom{2n+1}{2r}\cot^{2r}\theta,\end{aligned}$$

taking out s^{2n+1} and using $c^{2r}/s^{2r} = \cot^{2r}\theta$. This holds whenever $\sin\theta \neq 0$, as required.

The roots of the polynomial

Let

$$p(x) = \sum_{r=0}^{n} (-1)^r \binom{2n+1}{2r} x^r,$$

and put $\theta_k = k\pi/(2n+1)$ for $k = 1, \dots, n$. Then $(2n+1)\theta_k = k\pi$, so $\sin(2n+1)\theta_k = 0$, while $0 < \theta_k < \pi/2$ gives $\sin\theta_k \neq 0$. Dividing the identity by $\sin^{2n+1}\theta_k$ and multiplying by $(-1)^n$,

$$\sum_{r=0}^{n} (-1)^r \binom{2n+1}{2r} \cot^{2r}\theta_k = 0, \qquad \text{that is,} \qquad p\left(\cot^2\theta_k\right) = 0.$$

Ask yourself How do we know these are *all* the roots? The sums below depend on it.

The angles $\theta_1 < \theta_2 < \cdots < \theta_n$ all lie in $(0, \pi/2)$, where cot is positive and strictly decreasing. So the n numbers $\cot^2\theta_k$ are distinct. The leading coefficient of p is $(-1)^n\binom{2n+1}{2n} = (-1)^n(2n+1) \neq 0$, so p has degree exactly n and so at most n roots. Hence the roots of $p(x) = 0$ are exactly $\cot^2\theta_k$ for $k = 1, \dots, n$, each simple.

The coefficients we need are

$$[x^n]:\ (-1)^n\binom{2n+1}{2n} = (-1)^n(2n+1),$$
$$[x^{n-1}]:\ (-1)^{n-1}\binom{2n+1}{2n-2} = (-1)^{n-1}\binom{2n+1}{3},$$
$$[x^1]:\ -\binom{2n+1}{2} = -n(2n+1), \qquad [x^0]:\ 1.$$

Part (i)

The sum of the roots is minus the ratio of the x^{n-1} and x^n coefficients:

$$\sum_{k=1}^{n} \cot^2\theta_k = -\frac{(-1)^{n-1}\binom{2n+1}{3}}{(-1)^n(2n+1)} = \frac{1}{2n+1}\cdot\frac{(2n+1)(2n)(2n-1)}{6} = \frac{n(2n-1)}{3}.$$

Part (ii)

Since $\tan^2\theta_k = 1/\cot^2\theta_k$, we want the sum of the reciprocals of the roots x_k. None of them is zero (the constant term is 1), and

$$\sum_{k=1}^{n} \frac{1}{x_k} = \frac{\sum_k \prod_{j\neq k} x_j}{\prod_k x_k},$$

whose numerator and denominator are, by Vieta, $(-1)^{n-1}\cdot[x^1]/[x^n]$ and $(-1)^n\cdot[x^0]/[x^n]$. Their ratio is $-[x^1]/[x^0]$, so

$$\sum_{k=1}^{n} \tan^2\theta_k = -\frac{-n(2n+1)}{1} = n(2n+1).$$

(Equivalently: the reciprocals $1/x_k$ are the roots of $x^n p(1/x)$, the polynomial with its coefficients reversed, and we read off its sum of roots.)

Part (iii)

Using $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$ and part (i),

$$\sum_{k=1}^{n} \operatorname{cosec}^2\theta_k = n + \frac{n(2n-1)}{3} = \frac{3n + 2n^2 - n}{3} = \frac{2n(n+1)}{3}.$$

Checks

For $n = 1$ the polynomial is $1 - 3x$, with root $\frac{1}{3} = \cot^2(\pi/3)$. Then (i) gives $\frac{1}{3}$, (ii) gives $3 = \tan^2(\pi/3)$ and (iii) gives $\frac{4}{3} = \operatorname{cosec}^2(\pi/3)$, all correct. For $n = 3$ the angles are $\pi/7, 2\pi/7, 3\pi/7$, and a calculator gives the three sums as 5, 21 and 8, matching $\frac{3\cdot5}{3}$, $3\cdot7$ and $\frac{2\cdot3\cdot4}{3}$. We checked the identity at random θ and n, and all three sums for $n \le 29$, numerically.

Common slips

- Forgetting to argue that the $\cot^2\theta_k$ are distinct and that p has degree exactly n. Without this, Vieta's formulas can't be applied to the $\cot^2\theta_k$.
- Sign errors in the coefficients: the factor $(-1)^n$ cancels in part (i), but only if both coefficients carry it.
- Using k up to $2n$: the angles $k\pi/(2n+1)$ for $k > n$ repeat the same values of $\cot^2$, so only $k \le n$ gives distinct roots.

Beyond STEP

Part (iii), with $\cot^2\theta < 1/\theta^2 < \operatorname{cosec}^2\theta$ for $0 < \theta < \pi/2$, squeezes $\sum_{k=1}^{n} 1/k^2$ between two multiples of the sums found here. Letting $n \to \infty$ gives Euler's famous result $\sum 1/k^2 = \pi^2/6$, a lovely elementary proof that first-year analysis courses often present. The same idea of reading sums off a polynomial's coefficients underlies symmetric functions and Newton's identities. For the analysis side, see Abbott, *Understanding Analysis*; for more problems in this spirit, Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 1992, Question 9

Pure • Polar Equation of Ellipse

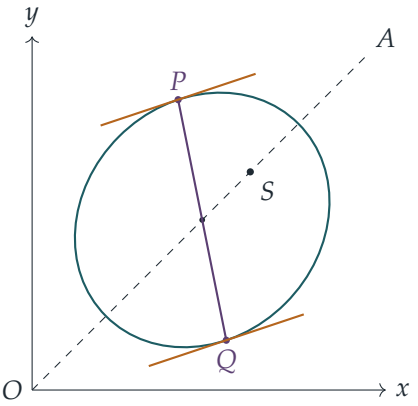
STEP III 1992, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q9

The key idea

Polar coordinates are measured from S with the initial line pointing along $y = x$, so a point at polar angle θ has Cartesian direction $\theta + \frac{\pi}{4}$ from S. Writing x and y as functions of θ and differentiating gives the gradient. For the second part, the property of a diameter we can actually use is that an ellipse is symmetric about its centre: the tangents at the two ends of a diameter are *parallel*. So we set the two gradients equal and let half-angle trigonometry do the rest.



In the sketch ($e = \frac{1}{2}$), PQ is a diameter through the centre, and the tangents at P and Q are parallel.

Solution

The gradient of the tangent

Let S be the point (s, s) with $s > 0$, since S lies on the line $y = x$ in the first quadrant. A unit vector along SA is $\frac{1}{\sqrt{2}}(1, 1)$, at angle $\frac{\pi}{4}$ to the x-axis. So the point with polar

coordinates (r, θ) about S is

$$x = s + r\cos\phi, \qquad y = s + r\sin\phi, \qquad \text{where } \phi = \theta + \tfrac{\pi}{4}.$$

On E we have $r = \ell/(1 + e\cos\theta)$, so

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{\ell e\sin\theta}{(1+e\cos\theta)^2} = \frac{r\,e\sin\theta}{1+e\cos\theta}.$$

Writing r' for $\mathrm{d}r/\mathrm{d}\theta$, and using $\mathrm{d}\phi/\mathrm{d}\theta = 1$,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = r'\cos\phi - r\sin\phi, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = r'\sin\phi + r\cos\phi.$$

Only the ratio matters, so we multiply both by the positive quantity $(1 + e\cos\theta)/r$:

$$\frac{1+e\cos\theta}{r}\frac{\mathrm{d}x}{\mathrm{d}\theta} = e\sin\theta\cos\phi - (1+e\cos\theta)\sin\phi = e\sin(\theta-\phi) - \sin\phi = -\tfrac{e}{\sqrt{2}} - \sin\phi,$$
$$\frac{1+e\cos\theta}{r}\frac{\mathrm{d}y}{\mathrm{d}\theta} = e\sin\theta\sin\phi + (1+e\cos\theta)\cos\phi = e\cos(\theta-\phi) + \cos\phi = \tfrac{e}{\sqrt{2}} + \cos\phi,$$

since $\theta - \phi = -\frac{\pi}{4}$. Now $\sin\phi = \frac{1}{\sqrt{2}}(\sin\theta + \cos\theta)$ and $\cos\phi = \frac{1}{\sqrt{2}}(\cos\theta - \sin\theta)$. Multiplying through by $\sqrt{2}$, a tangent vector at $\theta = \alpha$ is

$$\mathbf{d}(\alpha) = \big(-(\sin\alpha + \cos\alpha + e),\ \cos\alpha - \sin\alpha + e\big). \qquad (*)$$

Hence

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos\alpha - \sin\alpha + e}{-(\sin\alpha + \cos\alpha + e)} = \frac{\sin\alpha - \cos\alpha - e}{\sin\alpha + \cos\alpha + e},$$

as required. (Where $\sin\alpha + \cos\alpha + e = 0$ the tangent is vertical; $(*)$ still holds there.)

Ends of a diameter

Ask yourself How can we turn "PQ passes through the centre" into an equation in α and β, when we haven't found the centre?

We use the first part. An ellipse is symmetric under a half-turn about its centre C. If P and Q are the ends of a diameter, the half-turn swaps them and maps the tangent at P to the tangent at Q. A half-turn sends every line to a parallel line, so *the tangents at P and Q are parallel*.

So, with $\theta = \alpha$ at P and $\theta = \beta$ at Q, the vectors $\mathbf{d}(\alpha)$ and $\mathbf{d}(\beta)$ from $(*)$ are parallel. Working with vectors rather than gradients means we never divide by zero when a tangent is vertical. Write $s_\alpha = \sin\alpha$, $c_\alpha = \cos\alpha$, and so on. The condition is

$$(s_\alpha + c_\alpha + e)(c_\beta - s_\beta + e) = (s_\beta + c_\beta + e)(c_\alpha - s_\alpha + e).$$

Expanding, the terms $c_\alpha c_\beta$, $-s_\alpha s_\beta$, ec_α, ec_β and e^2 appear on both sides and cancel, leaving

$$s_\alpha c_\beta - c_\alpha s_\beta + es_\alpha - es_\beta = s_\beta c_\alpha - c_\beta s_\alpha + es_\beta - es_\alpha,$$

that is,

$$\sin(\alpha-\beta)+e(\sin\alpha-\sin\beta)=0.$$

Now use $\sin(\alpha-\beta)=2\sin\frac{\alpha-\beta}{2}\cos\frac{\alpha-\beta}{2}$ and $\sin\alpha-\sin\beta=2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}$:

$$2\sin\tfrac{\alpha-\beta}{2}\left(\cos\tfrac{\alpha-\beta}{2}+e\cos\tfrac{\alpha+\beta}{2}\right)=0.$$

Ask yourself Can we cancel $\sin\frac{\alpha-\beta}{2}$?

Yes. P and Q are different points, so α and β don't differ by a multiple of 2π, and $\sin\frac{\alpha-\beta}{2}\neq 0$. Hence

$$\cos\tfrac{\alpha-\beta}{2}+e\cos\tfrac{\alpha+\beta}{2}=0.$$

Expanding with $\frac{\alpha}{2}$ and $\frac{\beta}{2}$:

$$(1+e)\cos\tfrac{\alpha}{2}\cos\tfrac{\beta}{2}+(1-e)\sin\tfrac{\alpha}{2}\sin\tfrac{\beta}{2}=0.$$

Dividing by $(1-e)\cos\frac{\alpha}{2}\cos\frac{\beta}{2}$ (here $1-e>0$ because E is an ellipse) gives

$$\tan\tfrac{\alpha}{2}\tan\tfrac{\beta}{2}=-\frac{1+e}{1-e},$$

as required. The division is valid unless $\cos\frac{\alpha}{2}$ or $\cos\frac{\beta}{2}$ is zero. If, say, $\alpha=\pi$, the equation forces $\sin\frac{\beta}{2}=0$, so $\beta=0$. That diameter is the major axis along OSA, and there the tangent product isn't defined. Every other diameter satisfies the result.

Checks

- At $\alpha=0$ (the vertex nearer S) the formula gives gradient $\frac{-1-e}{1+e}=-1$. That's right: the tangent at the end of the major axis is perpendicular to $y=x$.
- With $e=\frac{1}{2}$ the product should be -3. Take $\alpha=\frac{\pi}{2}$, so $\tan\frac{\alpha}{2}=1$. Then $\tan\frac{\beta}{2}=-3$, so $\sin\beta=-\frac{3}{5}$ and $\cos\beta=-\frac{4}{5}$. From $(*)$, $\mathbf{d}(\alpha)=(-\frac{3}{2},-\frac{1}{2})$ and $\mathbf{d}(\beta)=(\frac{9}{10},\frac{3}{10})$, which are parallel. ✓
- Numerically, for 2000 random choices of ℓ, e, S and α, the gradient formula agreed with a finite-difference derivative, and reflecting P in the centre (at distance $\ell e/(1-e^2)$ from S along AS) gave a point Q on E whose β satisfied the identity to 10^{-13}.

Common slips

- Forgetting the $\frac{\pi}{4}$ rotation and using $x=r\cos\theta$, which gives the gradient in the wrong frame.
- Dividing by $\sin\frac{\alpha-\beta}{2}$ without saying why it is non-zero.
- Trying to find the centre and impose "C is the midpoint of PQ" directly. It works but is far messier than using parallel tangents.

Beyond STEP

Parallel tangents at the ends of a diameter are part of the theory of *conjugate diameters*: the midpoints of a family of parallel chords lie on a diameter, and the tangents at its ends are parallel to those chords. Under an affine map that turns the ellipse into a circle, all of this becomes elementary circle geometry, which is one reason affine and projective geometry are so powerful. The half-angle tangents here are also the standard parametrisation used for orbits in celestial mechanics. For a good first-year route into this geometry, try Needham, *Visual Complex Analysis*, or Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 1992, Question 10

Pure • Arc Length and Area

STEP III 1992, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q10

The key idea

Both perimeters are arc-length integrals that can't be done in elementary terms (they are *elliptic integrals*), so the only way to show they are equal is to turn one integral into the other. Write down the arc-length element for each curve, simplify each integrand using $\cos^2 + \sin^2 = 1$, and then a substitution and a little symmetry show that the two are the same integral. The areas are easier because both can be evaluated directly.

(The range printed in the paper ends with $\frac{1}{4}\theta$. This is clearly a misprint for $\frac{1}{4}\pi$, and we take $-\frac{1}{4}\pi < \theta < \frac{1}{4}\pi$.)

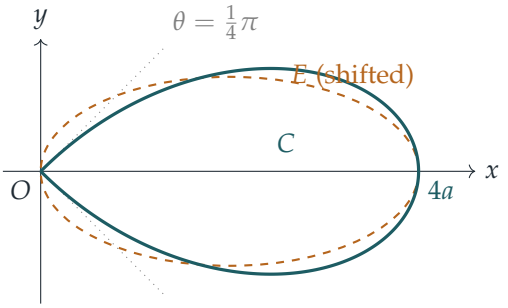
Solution

Sketching C

For $-\frac{1}{4}\pi < \theta < \frac{1}{4}\pi$ we have $-\frac{1}{2}\pi < 2\theta < \frac{1}{2}\pi$, so $r = 4a\cos 2\theta > 0$ throughout (taking $a > 0$). Some features:

- $r(-\theta) = r(\theta)$, so C is symmetric about the initial line.
- The maximum $r = 4a$ is at $\theta = 0$, the point $(4a, 0)$.
- As $\theta \to \pm\frac{1}{4}\pi$, $r \to 0$, so the curve runs into the origin, tangent there to the lines $\theta = \pm\frac{1}{4}\pi$.

So C is a single closed loop, one petal of the four-petalled rose $r = 4a\cos 2\theta$, lying along the positive x-axis. The figure shows it together with a copy of E translated by $2a$ so that it spans the same interval $0 \le x \le 4a$.



Perimeter of C

For a polar curve the element of arc length satisfies $\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right)^2 = r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2$. Here $\frac{\mathrm{d}r}{\mathrm{d}\theta} = -8a\sin 2\theta$, so

$$\begin{aligned} r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 &= 16a^2\cos^2 2\theta + 64a^2\sin^2 2\theta \\ &= 16a^2\left(1 + 3\sin^2 2\theta\right). \end{aligned}$$

Hence, since the loop is traced once as θ runs over $(-\frac{1}{4}\pi, \frac{1}{4}\pi)$,

$$\text{perimeter of } C = \int_{-\pi/4}^{\pi/4} 4a\sqrt{1 + 3\sin^2 2\theta}\,\mathrm{d}\theta.$$

Substitute $\phi = 2\theta$, so $\mathrm{d}\theta = \frac{1}{2}\mathrm{d}\phi$ and the limits become $\pm\frac{1}{2}\pi$:

$$\text{perimeter of } C = 2a\int_{-\pi/2}^{\pi/2}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi = 4a\int_0^{\pi/2}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi,$$

the last step because the integrand is an even function of ϕ.

Perimeter of E

With $x = 2a\cos\phi$, $y = a\sin\phi$,

$$\left(\frac{\mathrm{d}x}{\mathrm{d}\phi}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\phi}\right)^2 = 4a^2\sin^2\phi + a^2\cos^2\phi = a^2\left(1 + 3\sin^2\phi\right),$$

and E is traced once as ϕ runs from 0 to 2π, so

$$\text{perimeter of } E = a\int_0^{2\pi}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi.$$

? **Ask yourself** Why is $\int_0^{2\pi}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi$ four times the integral over $[0, \frac{1}{2}\pi]$?

Write $g(\phi) = \sqrt{1 + 3\sin^2\phi}$. Because $\sin^2(\pi - \phi) = \sin^2\phi$, the graph of g on $[\frac{1}{2}\pi, \pi]$ is the mirror image of its graph on $[0, \frac{1}{2}\pi]$, so $\int_0^{\pi} g = 2\int_0^{\pi/2} g$. Because $\sin^2(\phi + \pi) = \sin^2\phi$, g has period π, so $\int_{\pi}^{2\pi} g = \int_0^{\pi} g$. (Geometrically, the four quadrants of the ellipse have equal length.) Therefore

$$\text{perimeter of } E = 4a\int_0^{\pi/2} \sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi,$$

which is exactly the expression we found for the perimeter of C. So the two perimeters are equal, and we never had to evaluate the integral.

Areas

The area enclosed by a polar curve is $\frac{1}{2}\int r^2\,\mathrm{d}\theta$. Using $\cos^2 2\theta = \frac{1}{2}(1 + \cos 4\theta)$,

$$\begin{aligned}\text{area inside } C &= \frac{1}{2}\int_{-\pi/4}^{\pi/4} 16a^2\cos^2 2\theta\,\mathrm{d}\theta = 4a^2\int_{-\pi/4}^{\pi/4}(1 + \cos 4\theta)\,\mathrm{d}\theta \\ &= 4a^2\Big[\theta + \tfrac{1}{4}\sin 4\theta\Big]_{-\pi/4}^{\pi/4} = 4a^2\cdot\frac{\pi}{2} = 2\pi a^2.\end{aligned}$$

For the ellipse, the area enclosed by a closed curve traced anticlockwise is $\frac{1}{2}\oint\left(x\frac{\mathrm{d}y}{\mathrm{d}\phi} - y\frac{\mathrm{d}x}{\mathrm{d}\phi}\right)\mathrm{d}\phi$, which gives

$$\text{area inside } E = \frac{1}{2}\int_0^{2\pi}\left(2a^2\cos^2\phi + 2a^2\sin^2\phi\right)\mathrm{d}\phi = \frac{1}{2}\cdot 2a^2\cdot 2\pi = 2\pi a^2.$$

(Equally, this is the standard $\pi \times$ semi-axis $\times$ semi-axis $= \pi(2a)(a)$.) So the two areas are equal, both being $2\pi a^2$.

Checks

- With $a = 1.7$, approximating each curve by a polygon of 200 000 sides gave both perimeters as 16.470362, which equals $4a\int_0^{\pi/2}\sqrt{1 + 3\sin^2\phi}\,\mathrm{d}\phi$ by Simpson's rule. So each perimeter is about $9.6884a$.
- The same polygons gave both areas as $18.158406 = 2\pi(1.7)^2$.
- A sanity check on the area of C: it lies inside the rectangle $0 \le x \le 4a$, $|y| \le 1.1a$, which has area about $8.8a^2$, and $2\pi a^2 \approx 6.3a^2$ is a sensible fraction of that.

Common slips

- Forgetting the factor $\frac{1}{2}$ from $\mathrm{d}\theta = \frac{1}{2}\mathrm{d}\phi$ when substituting $\phi = 2\theta$. The factor $4a$ for C becomes $2a$, and only becomes $4a$ again after the even-function step.
- Using $\int r\,\mathrm{d}\theta$ instead of $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ for the polar area.
- Treating the whole rose $r = 4a\cos 2\theta$ for $0 \le \theta < 2\pi$ as C. The given range picks out a single petal.

Beyond STEP

The integral $\int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \phi}\, \mathrm{d}\phi$ is the *complete elliptic integral of the second kind*, and it has no elementary antiderivative. That's why the question forbids evaluating the integrals: the perimeter of an ellipse has no closed form. Our integral is the case $k^2 = -3$. It can be rewritten in the standard form with $0 < k < 1$ by replacing $\sin^2 \phi$ with $1 - \cos^2 \phi$ and then ϕ with $\frac{1}{2}\pi - \phi$. Elliptic integrals led to elliptic functions and elliptic curves, which are central to modern number theory. Siklos's *Advanced Problems in Mathematics* has more STEP questions where arc lengths and areas of polar and parametric curves are compared like this.

WORKED SOLUTION

STEP III 1992, Question 11

Mechanics • Collisions and Restitution

STEP III 1992, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

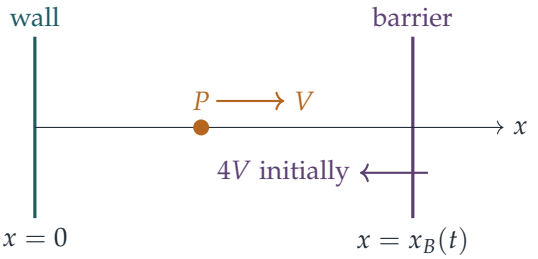
step.maths.org/questions/92-s3-q11

The key idea

Measure every position from the wall, so that the particle and the barrier each have a simple formula for their distance from it. A collision is just a moment when two of these formulas agree, which turns each timing question into a quadratic. At an impact, Newton's law of restitution applies to the *relative* velocity, so the barrier's own velocity at that instant matters.

Solution

Model P as a particle moving on a smooth table along the line through O perpendicular to the wall. Take x as distance from the wall AB, positive towards the barrier, and measure time t from the moment of projection. Velocities are positive when directed away from the wall.



The first impact

The barrier starts at distance $\frac{75}{32}VT$ with velocity $-4V$ and has constant acceleration $+4V/T$, so by the constant-acceleration formulae

$$x_B(t) = \tfrac{75}{32}VT - 4Vt + \frac{2Vt^2}{T}, \qquad v_B(t) = -4V + \frac{4Vt}{T}.$$

Until it first hits anything, P moves at constant speed (the table is smooth), so $x_P = Vt$. They meet when $x_P = x_B$. Writing $s = t/T$,

$$2s^2 - 5s + \tfrac{75}{32} = 0 \quad\Longleftrightarrow\quad 64s^2 - 160s + 75 = 0 \quad\Longleftrightarrow\quad (8s-5)(8s-15) = 0.$$

The first meeting is at the smaller root, so the impact takes place at $t = 5T/8$, as required. (Before then the gap $x_B - x_P$ is positive, since it is positive at $s = 0$ and has no root in between.)

At this moment P is at $x = \frac{5}{8}VT$ and the barrier's velocity is

$$v_B\left(\tfrac{5}{8}T\right) = -4V + \tfrac{5}{2}V = -\tfrac{3}{2}V,$$

so the barrier is still moving towards the wall, at speed $\frac{3}{2}V$.

Speed after the impact

The barrier's motion is unaffected, so only Newton's experimental law is needed. The speed of approach is $V - (-\frac{3}{2}V) = \frac{5}{2}V$. If P has velocity w just after impact, the speed of separation is $-\frac{3}{2}V - w$, and

$$-\tfrac{3}{2}V - w = \tfrac{1}{2}\cdot\tfrac{5}{2}V \quad\Longrightarrow\quad w = -\tfrac{11}{4}V.$$

So P rebounds towards the wall with speed $\dfrac{11V}{4}$.

Ask yourself Why is the rebound speed bigger than the speed V that P had before?

The barrier is moving towards P at the moment of impact, like a bat hitting a ball. Restitution acts on the speed of P *relative to the barrier*: relative to the barrier, P arrives at $\frac{5}{2}V$ and leaves at $\frac{5}{4}V$. Adding back the barrier's own $\frac{3}{2}V$ towards the wall gives $\frac{11}{4}V$.

Bounce off the wall and the second impact

P has distance $\frac{5}{8}VT$ to cover at speed $\frac{11}{4}V$, which takes $\frac{5}{8}\cdot\frac{4}{11}T = \frac{5}{22}T$. It therefore hits AB at

$$t_1 = \tfrac{5}{8}T + \tfrac{5}{22}T = \tfrac{75}{88}T.$$

The barrier cannot have got in the way. It is moving towards the wall more slowly than P throughout this interval (its speed falls from $\frac{3}{2}V$), and it never reaches the wall at all: its closest approach is at $t = T$, where $v_B = 0$ and $x_B = \frac{75}{32}VT - 4VT + 2VT = \frac{11}{32}VT > 0$.

The wall is fixed, so with $e = \frac{1}{2}$ the particle rebounds with speed $\frac{11}{8}V$, and afterwards

$$x_P = \tfrac{11}{8}V\left(t - \tfrac{75}{88}T\right) = \tfrac{11}{8}Vt - \tfrac{75}{64}VT.$$

The next collision with the barrier is at the first time after t_1 when $x_P = x_B$. In terms of $s = t/T$,

$$\tfrac{75}{32} - 4s + 2s^2 = \tfrac{11}{8}s - \tfrac{75}{64} \quad\Longleftrightarrow\quad 128s^2 - 344s + 225 = 0 \quad\Longleftrightarrow\quad (8s-9)(16s-25) = 0.$$

The roots are $s = \frac{9}{8}$ and $s = \frac{25}{16}$. Both are later than t_1, since $\frac{75}{88} < 1 < \frac{9}{8}$. At t_1 the gap $x_B - x_P = x_B(t_1)$ is positive, so the first time it closes is the smaller root. The next collision with the barrier is therefore at time $9T/8$ from the start, as required. (As a check, both are at $x = \frac{3}{8}VT$ then.)

Ask yourself Why must we take the smaller root, and not just any root?

The quadratic describes the two motions as if they could pass through each other. Its second root, $\frac{25}{16}T$, would be a "meeting" after the real collision, when the motion has already changed. The physical collision is the first time the gap closes.

Checks

- At $t = \frac{5}{8}T$: $x_B = \frac{75}{32} - \frac{80}{32} + \frac{25}{32} = \frac{20}{32} = \frac{5}{8}$ (in units of VT), equal to x_P.
- At $t = \frac{9}{8}T$: $x_B = \frac{75}{32} - \frac{144}{32} + \frac{81}{32} = \frac{12}{32} = \frac{3}{8}$, and $x_P = \frac{11}{8} \cdot \frac{24}{88} = \frac{3}{8}$.
- A time-stepping simulation with $V = T = 1$ (step 10^{-6}) found impacts at $t \approx 0.6250$ (rebound velocity -2.750), $0.8523 \approx \frac{75}{88}$ and 1.1250.

Common slips

- Applying restitution to P's speed alone, which gives $\frac{1}{2}V$ instead of $\frac{11}{4}V$. The barrier is moving, so use relative velocities.
- Getting the sign of the acceleration wrong. The barrier *starts* moving towards the wall but is decelerating, and it turns round at $t = T$.
- Forgetting to check that the barrier doesn't reach P (or the wall) before P has bounced off AB.

Beyond STEP

The neat trick here is to work in the barrier's frame at the instant of impact. Because the barrier is so massive, restitution reverses and scales the relative velocity, just as a fixed wall would. Changing frames like this is central to first-year mechanics, and it leads on to Galilean invariance and to centre-of-mass frames in two-body collisions. Kleppner and Kolenkow's *An Introduction to Mechanics* develops these ideas carefully.

WORKED SOLUTION

STEP III 1992, Question 12

Mechanics • Conservation of Momentum

STEP III 1992, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

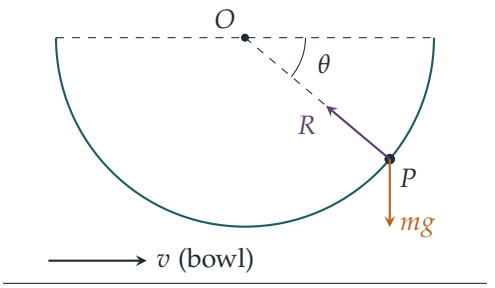
step.maths.org/questions/92-s3-q12

The key idea

Nothing horizontal acts on the bowl-and-particle system from outside, and nothing does work except gravity, so we have two conservation laws: horizontal momentum and energy. Momentum links the bowl's speed to $\dot{\theta}$; energy then gives v^2. For the reaction, the cleanest route is to work in the bowl's (accelerating) frame, where the particle simply moves in a circle about O, and to remember that the reaction is also the only horizontal force on the bowl.

Solution

Model. The bowl is smooth, so the reaction R on P acts along the normal, that is along PO. The carriage is light and the table smooth, so the only horizontal force on the bowl (with its carriage) is the horizontal component of the particle's push. Take P to start at the right-hand end of the rim, so θ increases from 0 as P slides down; the bowl then moves to the right.



Conservation of horizontal momentum

Let the bowl's horizontal position be X, so $\dot{X} = v$. Relative to O, the particle is at $(a\cos\theta,\ -a\sin\theta)$, so its position is $(X + a\cos\theta,\ -a\sin\theta)$ and its velocity is

$$\big(v - a\dot{\theta}\sin\theta,\ -a\dot{\theta}\cos\theta\big).$$

There is no external horizontal force on the system, and it starts at rest, so the total horizontal momentum stays zero:

$$2mv + m\big(v - a\dot{\theta}\sin\theta\big) = 0 \quad\Longrightarrow\quad 3v = a\dot{\theta}\sin\theta,$$

as required. Notice that the particle's horizontal velocity is then $v - 3v = -2v$: it moves left at twice the bowl's speed, as it must to cancel the bowl's momentum $2mv$.

Energy

The reaction does no net work (it is internal and the surface is smooth), and the table does no work, so energy is conserved. When OP is at angle θ the particle has fallen $a\sin\theta$. Using $a\dot{\theta} = 3v/\sin\theta$, the particle's vertical speed is $a\dot{\theta}\cos\theta = 3v\cot\theta$, so

$$\begin{aligned} mga\sin\theta &= \tfrac{1}{2}(2m)v^2 + \tfrac{1}{2}m\big[(2v)^2 + 9v^2\cot^2\theta\big] \\ &= mv^2\Big(3 + \tfrac{9}{2}\cot^2\theta\Big). \end{aligned}$$

Multiplying by $2\sin^2\theta/m$:

$$2ga\sin^3\theta = v^2\big(6\sin^2\theta + 9\cos^2\theta\big) = v^2\big(9 - 3\sin^2\theta\big),$$

so

$$v^2 = \frac{2ga\sin^3\theta}{3(3 - \sin^2\theta)},$$

which is the second result.

The reaction

> **Ask yourself** The bowl is accelerating. How do we write down the particle's equation of motion along PO?

Use two equations. First, for the bowl: the particle pushes on it with force R directed from O to P, whose horizontal component is $R\cos\theta$ to the right. So the bowl's acceleration A satisfies

$$2mA = R\cos\theta. \tag{1}$$

Second, for the particle, work in the bowl's frame. There the particle moves on a circle of radius a about O with speed $a\dot{\theta}$, and as well as R and mg it feels a fictitious force mA to the left. Resolving towards O (direction $(-\cos\theta, \sin\theta)$), gravity contributes $-mg\sin\theta$ and the fictitious force contributes $+mA\cos\theta$, so

$$R - mg\sin\theta + mA\cos\theta = ma\dot{\theta}^2. \tag{2}$$

(Equivalently, in the ground frame the particle's acceleration is the bowl's acceleration A plus its circular acceleration relative to the bowl; resolving that along PO gives the same equation.)

From the first two results,

$$a\dot{\theta}^2 = \frac{9v^2}{a\sin^2\theta} = \frac{6g\sin\theta}{3-\sin^2\theta}.$$

Substituting $mA = \frac{1}{2}R\cos\theta$ from (1) and collecting the terms in R:

$$R\left(1+\tfrac{1}{2}\cos^2\theta\right) = mg\sin\theta + \frac{6mg\sin\theta}{3-\sin^2\theta} = mg\sin\theta\,\frac{9-\sin^2\theta}{3-\sin^2\theta}.$$

Since $1+\frac{1}{2}\cos^2\theta = \frac{1}{2}(3-\sin^2\theta)$,

$$\boxed{R = \frac{2mg\sin\theta\,(9-\sin^2\theta)}{(3-\sin^2\theta)^2} = \frac{2mg\sin\theta\,(8+\cos^2\theta)}{(2+\cos^2\theta)^2}.}$$

Checks

- At the start, $\theta = 0$ gives $R = 0$, as it should: the particle is at rest at the rim with nothing pressing it into the bowl.
- At the bottom, $\theta = \frac{1}{2}\pi$: the bowl's acceleration is zero (by (1)), $a\dot{\theta}^2 = 3g$, and $R - mg = 3mg$, so $R = 4mg$, matching the formula $2mg\cdot 8/2^2$.
- If the bowl were fixed (infinitely heavy), the usual result would be $v_P^2 = 2ga\sin\theta$ and $R = 3mg\sin\theta$; the recoiling bowl makes R smaller near the rim (about $2mg\sin\theta$ rather than $3mg\sin\theta$ for small θ, as the bowl moves away from P) but larger at the bottom ($4mg$ rather than $3mg$).
- Stepping the full equations of motion numerically (with $m = 1.3$, $a = 0.7$, $g = 9.81$) reproduced $3v = a\dot{\theta}\sin\theta$, the formula for v^2 and the formula for R to within 10^{-7} all the way from the rim to the bottom.

Common slips

- Writing $R - mg\sin\theta = ma\dot{\theta}^2$, as if the bowl were fixed. The bowl accelerates, so the particle's acceleration is not just its circular acceleration relative to the bowl.
- Using $\frac{1}{2}m(a\dot{\theta})^2$ for the particle's kinetic energy: $a\dot{\theta}$ is its speed relative to the bowl, not relative to the table.

Beyond STEP

This is a two-degree-of-freedom system, and the tidiest systematic treatment is Lagrangian mechanics: write the kinetic and potential energies in terms of X and θ, and

the momentum conservation drops out because X does not appear in the Lagrangian (it is a "cyclic" coordinate). The fictitious force we used is the simplest case of mechanics in a non-inertial frame, which first-year courses extend to rotating frames with centrifugal and Coriolis forces. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both non-inertial frames and systems of interacting bodies very well.

WORKED SOLUTION

STEP III 1992, Question 13

Mechanics • Rotation About a Fixed Axis

STEP III 1992, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q13

The key idea

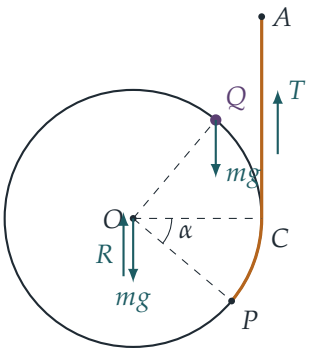
The axis is fixed, so the disc and particle form one rigid body with a single coordinate, the angle turned. Conservation of energy (elastic, gravitational, kinetic) gives the angular speed at every position, and "P reaches C" just means the kinetic energy never falls to zero on the way. Once we have the angular speed at C, the reaction on the axis comes from Newton's second law for the whole system, applied to its centre of mass.

Solution

The model and the geometry

We treat Q as a particle and the string as light. We take the rim as smooth, so the tension T is the same all along the string, and the normal contact forces from the string on the rim pass through O. Hence the string's only moment about O is that of the tension pulling tangentially at P, which is $2bT$. (Exactly the same moment arises if we regard the tension as acting where the string leaves the rim at C: vertically up along AC, at distance $2b$ from O.)

Let $\alpha = \angle POC$, with P below OC. At release $\alpha = \frac{\pi}{2}$, and P reaches C when $\alpha = 0$. Since $\angle POQ = 90°$, OQ makes angle $\frac{\pi}{2} - \alpha$ above OC, so Q is at height $2b\cos\alpha$ above O. In particular Q starts level with O (at C) and is directly above O when P reaches C.



The string's total length is $AC + \text{arc } CP = \pi b + 2b\alpha$, and its natural length is πb, so its extension is $2b\alpha$. By Hooke's law

$$T = \frac{kmg \cdot 2b\alpha}{\pi b} = \frac{2kmg\alpha}{\pi},$$

and the elastic energy stored is

$$E(\alpha) = \frac{kmg\,(2b\alpha)^2}{2\pi b} = \frac{2kmgb\,\alpha^2}{\pi}.$$

The moment of inertia about the axis is that of the disc plus that of Q on the rim:

$$I = \tfrac{1}{2}m(2b)^2 + m(2b)^2 = 6mb^2.$$

P reaches C

As the disc turns so that P rises towards C (so α decreases), the string shortens and gives up elastic energy, while Q rises. The axis is fixed and smooth, so energy is conserved. With the system starting from rest at $\alpha = \frac{\pi}{2}$, where Q is at height 0 relative to O and $E = \frac{1}{2}\pi kmgb$,

$$\begin{aligned}\tfrac{1}{2}I\dot{\alpha}^2 &= E\left(\tfrac{\pi}{2}\right) - E(\alpha) - mg \cdot 2b\cos\alpha \\ &= \frac{2kmgb}{\pi}\left(\frac{\pi^2}{4} - \alpha^2\right) - 2mgb\cos\alpha.\end{aligned}$$

The disc does start to turn the right way: at release the string pulls with $T = kmg$ at radius $2b$, while Q's weight acts at horizontal distance $2b$ the other way, so the net moment is $2(k-1)mgb > 0$, because $k > \frac{4}{\pi} > 1$.

P reaches C provided the kinetic energy stays positive for every α in $[0, \frac{\pi}{2})$, that is, provided

$$\frac{k}{\pi}\left(\frac{\pi^2}{4} - \alpha^2\right) > \cos\alpha \qquad \text{for } 0 \le \alpha < \tfrac{\pi}{2}.$$

Since $k > \frac{4}{\pi}$ and $\frac{\pi^2}{4} - \alpha^2 > 0$, the left-hand side is greater than

$$\frac{4}{\pi^2}\left(\frac{\pi^2}{4} - \alpha^2\right) = 1 - \frac{4\alpha^2}{\pi^2},$$

so it is enough to show that $g(\alpha) = 1 - \dfrac{4\alpha^2}{\pi^2} - \cos\alpha \geq 0$ on $[0, \frac{\pi}{2}]$.

Ask yourself How can we compare a quadratic with $\cos\alpha$ on a whole interval?

Look at the second derivative. We have $g(0) = g(\frac{\pi}{2}) = 0$, $g'(\alpha) = -\frac{8\alpha}{\pi^2} + \sin\alpha$, so $g'(0) = 0$, and $g''(\alpha) = \cos\alpha - \frac{8}{\pi^2}$. Since $\frac{8}{\pi^2} < 1$ and $\cos\alpha$ decreases from 1 to 0, there is a single α_0 with $g'' > 0$ on $[0, \alpha_0)$ and $g'' < 0$ on $(\alpha_0, \frac{\pi}{2}]$.

- On $[0, \alpha_0]$, g' is increasing from $g'(0) = 0$, so $g' \geq 0$ and $g \geq g(0) = 0$.
- On $[\alpha_0, \frac{\pi}{2}]$, g is concave, so its graph lies above the chord joining its endpoint values $g(\alpha_0) \geq 0$ and $g(\frac{\pi}{2}) = 0$; hence $g \geq 0$ there too.

So $g \geq 0$ throughout, and the strict inequality $k > \frac{4}{\pi}$ makes the kinetic energy strictly positive for $0 \leq \alpha < \frac{\pi}{2}$. The disc never comes to rest before $\alpha = 0$, so P reaches C. In particular, at C (put $\alpha = 0$),

$$3mb^2\dot{\alpha}^2 = \frac{\pi k m g b}{2} - 2mgb \quad \Longrightarrow \quad \omega^2 = \frac{(\pi k - 4)g}{6b},$$

where ω is the angular speed when P is at C. (Here we see where the condition $k > \frac{4}{\pi}$ comes from: it is exactly what makes $\omega^2 > 0$.)

The reaction on the axis

When P is at C, two things happen at once.

- The string is at its natural length ($AC = \pi b$ and the arc is gone), so $T = 0$.
- Q is vertically above O, so its weight has no moment about O.

So there is no moment about the axis at that instant, and the angular acceleration is zero. The disc's centre O is fixed, and Q moves on a circle of radius $2b$ with zero angular acceleration, so its acceleration is purely centripetal: $2b\omega^2$ directed from Q towards O, that is, vertically downwards.

Ask yourself Which forces go into the equation for the reaction?

Apply Newton's second law vertically to the whole system (disc plus Q). The external forces are the two weights $2mg$ downwards, the reaction R upwards on the axis, and the string tension, which is zero at this instant. The internal force between Q and the disc cancels. Only Q accelerates, so

$$R - 2mg = m\,(-2b\omega^2),$$

and therefore

$$R = 2mg - 2mb \cdot \frac{(\pi k - 4)g}{6b} = 2mg - \frac{mg(\pi k - 4)}{3} = \frac{mg(10 - \pi k)}{3},$$

as required. (The horizontal component of the reaction is zero at this instant, since Q has no horizontal acceleration.)

Checks

- At the borderline $k = \frac{4}{\pi}$ the system would just creep up to C with $\omega = 0$, and the formula gives $R = 2mg$, the static value. That is consistent.
- A computer integrates the equation of motion $6mb^2\ddot{\psi} = 2bT - 2mgb\cos\psi$ (with $\psi = \frac{\pi}{2} - \alpha$) numerically. With $m = g = b = 1$ and $k = 2$ it reaches C with $\omega^2 = 0.38053$, matching $(2\pi - 4)/6$, and $R = 1.23894 = (10 - 2\pi)/3$. It agrees to six decimal places for $k = 1.29$, 1.5, 3 as well.
- If $k > \frac{10}{\pi}$ the formula gives $R < 0$: the axis must pull down, because Q is whirling over the top fast enough to need more than the system's weight as centripetal force.

Common slips

- Taking the extension as πb throughout. In fact it shrinks to zero as P reaches C, which is why the tension vanishes there.
- Forgetting the disc's own moment of inertia $\frac{1}{2}m(2b)^2$ and using only Q's.
- Showing only that the kinetic energy at C is positive. "P reaches C" needs the kinetic energy to stay positive all the way there.

Beyond STEP

The final step is the rigid-body version of "total external force equals total mass times acceleration of the centre of mass", and the reaction at a pivot is a classic application of it. A first-year rigid-body dynamics course develops this together with the angular-momentum equation $\mathrm{d}\mathbf{L}/\mathrm{d}t = \boldsymbol{\tau}$. The inequality $\cos\alpha \le 1 - 4\alpha^2/\pi^2$ is a cousin of Jordan's inequality $\sin x \ge 2x/\pi$, and it was proved here by the standard convexity argument. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats fixed-axis rotation and pivot reactions very clearly.

WORKED SOLUTION

STEP III 1992, Question 14

Mechanics • Unwinding String Dynamics

STEP III 1992, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q14

The key idea

Everything is governed by the single angle θ. The free part of the string is tangent to the disc at Q, and its length is the arc of string that has come off the disc, so P is easy to locate in terms of θ (the curve traced is an *involute* of the circle). Differentiating, the velocity of P turns out to be perpendicular to the string. So the string's tension does no work, and only friction acts along the direction of motion. Resolving along and perpendicular to PQ then splits the problem neatly in two.

Solution

Position of P

Model P as a particle on a horizontal table whose only resistance is the given friction, and the string as light and inextensible. Gravity and the normal reaction cancel, so the only horizontal forces are the tension T along PQ and the friction.

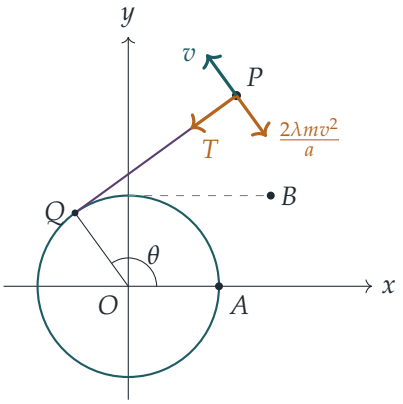
With $A = (a, 0)$ on the x-axis, the point of contact is $Q = (a\cos\theta,\, a\sin\theta)$. When P was at A, all the string was on the disc; since the string is inextensible, the free part QP has length equal to the arc AQ that has unwound, namely $a\theta$. The string is tangent at Q, so QP is perpendicular to OQ, in the direction of the unit vector $(\sin\theta,\, -\cos\theta)$ (at $\theta = \frac{\pi}{2}$ this is $(1, 0)$, pointing from the top of the disc towards B, as in the figure). Hence

$$\overrightarrow{OP} = a(\cos\theta,\, \sin\theta) + a\theta(\sin\theta,\, -\cos\theta),$$

so

$$x = a\cos\theta + a\theta\sin\theta, \qquad y = a\sin\theta - a\theta\cos\theta.$$

The first is the required x-coordinate. At B the free string is parallel to OA, touching the disc at the top, so $\theta = \frac{\pi}{2}$ there and $B = (\frac{\pi a}{2},\, a)$.



Velocity and acceleration

Write $\mathbf{e} = (\cos\theta, \sin\theta)$ for the unit vector along OQ and $\mathbf{f} = (-\sin\theta, \cos\theta)$ for the unit vector from P towards Q (the reverse of the string direction above). Note that $\dot{\mathbf{e}} = \dot{\theta}\,\mathbf{f}$ and $\dot{\mathbf{f}} = -\dot{\theta}\,\mathbf{e}$.

Differentiating x and y,

$$\dot{x} = (-a\sin\theta + a\sin\theta + a\theta\cos\theta)\,\dot{\theta} = a\theta\dot{\theta}\cos\theta,$$
$$\dot{y} = (a\cos\theta - a\cos\theta + a\theta\sin\theta)\,\dot{\theta} = a\theta\dot{\theta}\sin\theta,$$

so the velocity is $a\theta\dot{\theta}\,\mathbf{e}$: parallel to OQ, and so perpendicular to the string. Differentiating again with the product rule,

$$\ddot{\mathbf{r}} = a\left(\dot{\theta}^2 + \theta\ddot{\theta}\right)\mathbf{e} + a\theta\dot{\theta}\,\dot{\mathbf{e}} = a\left(\dot{\theta}^2 + \theta\ddot{\theta}\right)\mathbf{e} + a\theta\dot{\theta}^2\,\mathbf{f}.$$

So the acceleration has component $a\theta\dot{\theta}^2$ along PQ (towards Q) and $a\dot{\theta}^2 + a\theta\ddot{\theta}$ perpendicular to PQ (in the direction of $\mathbf{e}$), as required.

The equation of motion

At B, $\theta = \frac{\pi}{2}$ and the velocity is $a\cdot\frac{\pi}{2}\,\dot{\theta}\,(0,1)$. Projection perpendicular to OA and away from it means velocity V in the positive y-direction, so

$$\dot{\theta} = \frac{2V}{\pi a} > 0 \quad \text{initially.}$$

While $\dot{\theta} > 0$, the speed is $v = a\theta\dot{\theta}$ and the motion is in the direction $+\mathbf{e}$, so friction of magnitude $2\lambda mv^2/a = 2\lambda ma\theta^2\dot{\theta}^2$ acts in the direction $-\mathbf{e}$. The tension acts along $\mathbf{f}$ and has no $\mathbf{e}$-component.

> **Ask yourself** Which of the two acceleration components should we use to get an equation for θ alone?

The perpendicular one: the tension T is unknown, and it appears only in the component along PQ. Newton's second law perpendicular to PQ gives

$$m\left(a\dot{\theta}^2 + a\theta\ddot{\theta}\right) = -2\lambda ma\theta^2\dot{\theta}^2.$$

Dividing by $ma\theta\dot\theta$ (allowed, as $\theta > 0$ and $\dot\theta > 0$),

$$\frac{\ddot\theta}{\dot\theta} = -\left(\frac{1}{\theta} + 2\lambda\theta\right)\dot\theta,$$

which is the required result.

Finding $\dot\theta$

The left-hand side is $\frac{\mathrm{d}}{\mathrm{d}t}\ln\dot\theta$, and by the chain rule the right-hand side is $\frac{\mathrm{d}}{\mathrm{d}t}\left(-\ln\theta - \lambda\theta^2\right)$. Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\ln\left(\theta\dot\theta\right) + \lambda\theta^2\right) = 0, \qquad \text{so} \qquad \theta\dot\theta\,\mathrm{e}^{\lambda\theta^2} = \text{constant}.$$

At B the constant is $\frac{\pi}{2}\cdot\frac{2V}{\pi a}\,\mathrm{e}^{\lambda\pi^2/4} = \frac{V}{a}\,\mathrm{e}^{\lambda\pi^2/4}$. Therefore

$$\dot\theta = \frac{V}{a\theta}\,\exp\!\Big(\lambda\Big(\frac{\pi^2}{4} - \theta^2\Big)\Big).$$

This is positive for every θ, so $\dot\theta$ never vanishes and the assumption $\dot\theta > 0$ used above holds throughout. (The answer necessarily involves V as well as θ, λ and a.) Equivalently, the speed is $v = a\theta\dot\theta = V\mathrm{e}^{\lambda(\pi^2/4-\theta^2)}$.

Tension at $\theta = \pi$

Along PQ the only force is T, towards Q, so $T = ma\theta\dot\theta^2$. At $\theta = \pi$,

$$\dot\theta^2 = \frac{V^2}{a^2\pi^2}\,\mathrm{e}^{2\lambda(\pi^2/4-\pi^2)} = \frac{V^2}{a^2\pi^2}\,\mathrm{e}^{-3\lambda\pi^2/2},$$

so

$$T = ma\pi\cdot\frac{V^2}{a^2\pi^2}\,\mathrm{e}^{-3\lambda\pi^2/2} = \frac{mV^2}{\pi a}\,\mathrm{e}^{-3\lambda\pi^2/2}.$$

Checks

- The free string PQ has length $a\theta$ and is the radius of curvature of the involute, so Q is the centre of curvature and the component towards Q should be $v^2/(a\theta) = a\theta\dot\theta^2$. It is.
- With $\lambda = 0$ (no friction) the speed stays V, as it must, since the tension is perpendicular to the motion and does no work. At $\theta = \frac{\pi}{2}$, $T = mV^2/(\frac{\pi}{2}a)$, which is mv^2/r for the string length $r = \frac{\pi a}{2}$ at B.
- Numerically, with $a = 1.3$, $m = 0.7$, $V = 2.1$, $\lambda = 0.25$, we stepped the equation for θ with a Runge–Kutta method: it matched the formula for $\dot\theta$ to 10^{-14}. Finite-difference accelerations of (x, y) matched both components and the friction, and gave $T \approx 0.018667$ at $\theta = \pi$, agreeing with the formula.

Common slips

- Using $\theta = 0$ as the starting value: the particle starts at B, where $\theta = \frac{\pi}{2}$, not at A.
- Resolving along PQ to find $\dot{\theta}$: that equation contains the unknown tension, and friction has no component there.
- Forgetting that friction opposes the velocity, which is along OQ, not along the string.

Beyond STEP

The path of P is the involute of a circle, the curve traced by the end of a taut string unwinding from a spool, and the shape used for the teeth of most gears. The facts used here (the velocity is perpendicular to the string, and the string's length is the radius of curvature) are special cases of the evolute–involute relationship in the differential geometry of curves. Dynamically, it is a good example of using intrinsic (tangential and normal) components to separate a known force from an unknown constraint force. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats motion in polar and intrinsic coordinates and constrained motion in this spirit.

WORKED SOLUTION

STEP III 1992, Question 15

Probability and Statistics • Geometric Probability

STEP III 1992, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q15

The key idea

Because the goat's position is uniform over the field, every probability is just an area divided by a^2. So the whole question comes down to finding the area of the part of the square within distance r of the corner O. Once r passes a, the quarter-circle spills out of the square, and the clean way to find the area left inside is to cut it into two right-angled triangles and a circular sector. The density then follows by differentiating, and the tethered goat is the same idea with a different region.

Solution

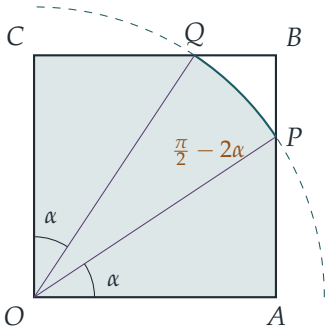
Put O at the origin with OA along the x-axis and OC along the y-axis, so that the field is $0 \le x \le a$, $0 \le y \le a$. Write $F(r) = \mathrm{P}(R < r)$, which is the area of the part of the field with $x^2 + y^2 < r^2$, divided by a^2.

The distribution of R

When $0 < r \le a$. The quarter-disc of radius r centred at O lies entirely inside the field, so

$$F(r) = \frac{\frac{1}{4}\pi r^2}{a^2} = \frac{\pi r^2}{4a^2}.$$

When $a \le r \le a\sqrt{2}$. Now the arc meets the side AB (the line $x = a$) at $P = \left(a, \sqrt{r^2 - a^2}\right)$ and the side CB (the line $y = a$) at $Q = \left(\sqrt{r^2 - a^2}, a\right)$. Let $\alpha = \cos^{-1}(a/r)$, which is the angle AOP (since $\cos \angle AOP = a/r$). By symmetry, angle COQ is also α.



The shaded region (the part of the field within distance r of O) splits into three pieces:

- triangle OAP, a right angle at A, with legs a and $\sqrt{r^2 - a^2}$, so area $\frac{1}{2}a\sqrt{r^2 - a^2}$;
- triangle OCQ, congruent to it, with the same area;
- the sector OPQ of radius r and angle $\frac{\pi}{2} - 2\alpha$, with area $\frac{1}{2}r^2\left(\frac{\pi}{2} - 2\alpha\right) = \frac{\pi r^2}{4} - r^2\alpha$.

Adding these and dividing by a^2,

$$\begin{aligned} F(r) &= \frac{a\sqrt{r^2 - a^2}}{a^2} + \frac{\pi r^2}{4a^2} - \frac{r^2}{a^2}\,\alpha \\ &= \left(\frac{r^2}{a^2} - 1\right)^{\frac{1}{2}} + \frac{\pi r^2}{4a^2} - \frac{r^2}{a^2}\cos^{-1}\left(\frac{a}{r}\right), \end{aligned}$$

which is the required result. (For $r \geq a\sqrt{2}$ the whole field is within distance r of O, so $F(r) = 1$. The formula gives $1 + \frac{\pi}{2} - 2 \cdot \frac{\pi}{4} = 1$ at $r = a\sqrt{2}$, as it should.)

The median

At $r = a$ we have $F(a) = \pi/4 \approx 0.785 > \frac{1}{2}$, so the median m lies in the range $0 < m < a$, where the simple formula holds:

$$\frac{\pi m^2}{4a^2} = \frac{1}{2} \quad \Longrightarrow \quad m = a\sqrt{\frac{2}{\pi}} \approx 0.798a.$$

The probability density function

The density is $f(r) = F'(r)$. For $0 < r < a$ this is $f(r) = \dfrac{\pi r}{2a^2}$.

For $a < r < a\sqrt{2}$ we need the derivative of $\cos^{-1}(a/r)$:

$$\frac{\mathrm{d}}{\mathrm{d}r}\cos^{-1}\left(\frac{a}{r}\right) = -\frac{1}{\sqrt{1 - a^2/r^2}} \cdot \left(-\frac{a}{r^2}\right) = \frac{a}{r\sqrt{r^2 - a^2}}.$$

Then, differentiating each term of F (using the product rule on the last),

$$\begin{aligned} f(r) &= \frac{r}{a\sqrt{r^2-a^2}} + \frac{\pi r}{2a^2} - \frac{2r}{a^2}\cos^{-1}\left(\frac{a}{r}\right) - \frac{r^2}{a^2}\cdot\frac{a}{r\sqrt{r^2-a^2}} \\ &= \frac{r}{a^2}\left(\frac{\pi}{2} - 2\cos^{-1}\left(\frac{a}{r}\right)\right), \end{aligned}$$

since the first and last terms cancel. So

$$f(r) = \begin{cases} \dfrac{\pi r}{2a^2}, & 0 < r \le a, \\ \dfrac{r}{a^2}\left(\dfrac{\pi}{2} - 2\cos^{-1}\left(\dfrac{a}{r}\right)\right), & a < r \le a\sqrt{2}, \\ 0, & \text{otherwise.} \end{cases}$$

This makes geometric sense: $f(r)\,\mathrm{d}r$ is (arc length of the circle of radius r inside the field) $\times\, \mathrm{d}r/a^2$, and that arc subtends an angle $\frac{\pi}{2}$ when $r \le a$ and $\frac{\pi}{2} - 2\alpha$ when $r > a$. The two pieces agree at $r = a$ (both give $\pi/(2a)$), and f falls to 0 at $r = a\sqrt{2}$, where $\alpha = \pi/4$.

The tethered goat

Once tethered, the goat can reach only the points of the field with $R \le a$, the quarter-disc of radius a centred at O, which has area $\frac{1}{4}\pi a^2$. Its position is still uniform, now over this quarter-disc. That is exactly the original model conditioned on the event $R \le a$.

? **Ask yourself** Which coordinate measures the distance from the fence OC?

With OC along the y-axis, the distance from OC is x. So we need the area of the part of the quarter-disc with $x > a/2$:

$$\begin{aligned} \int_{a/2}^{a} \sqrt{a^2-x^2}\,\mathrm{d}x &= \left[\frac{x\sqrt{a^2-x^2}}{2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]_{a/2}^{a} \\ &= \frac{\pi a^2}{4} - \left(\frac{\sqrt{3}\,a^2}{8} + \frac{\pi a^2}{12}\right) = \frac{\pi a^2}{6} - \frac{\sqrt{3}\,a^2}{8}. \end{aligned}$$

(Geometrically, this is a sector of angle $\pi/3$ minus a triangle with sides $a/2$ and $\sqrt{3}a/2$, because the line $x = a/2$ meets the arc at angle $\pi/3$ from OA.) Dividing by the area $\frac{1}{4}\pi a^2$,

$$\mathrm{P}\big(\text{distance from } OC > \tfrac{a}{2} \,\big|\, R \le a\big) = \frac{2}{3} - \frac{\sqrt{3}}{2\pi} \approx 0.391.$$

Checks

- Continuity: both formulas for F give $\pi/4$ at $r = a$, and $F(a\sqrt{2}) = 1$.

- A Monte Carlo simulation with 4×10^5 uniform points in the unit square agreed with $F(r)$ to about three decimal places at $r = 0.3$, 0.7, 1.1, 1.25, 1.4. It gave a sample median of 0.7986 against $\sqrt{2/\pi} = 0.7979$, and a conditional proportion of 0.3914 against $\frac{2}{3} - \frac{\sqrt{3}}{2\pi} = 0.3910$.
- The density integrates numerically to 1 over $[0, a\sqrt{2}]$, and matches a numerical derivative of F at several points.
- The tethered answer is less than $\frac{1}{2}$, as it should be: the quarter-disc is thinner far from OC.

Common slips

- Forgetting that F needs a different formula for $r > a$ when finding the median. It's worth checking $F(a) > \frac{1}{2}$ first, which shows the median lies in the easy range.
- Using the sector angle $\frac{\pi}{2} - \alpha$ instead of $\frac{\pi}{2} - 2\alpha$. Two corner pieces are cut off, one at each side.
- Dividing by a^2 in the tethered part. The conditional probability divides by the area of the quarter-disc, $\frac{1}{4}\pi a^2$.

Beyond STEP

Here R is a function of a point chosen uniformly in a region, and its distribution came from measuring the region where $R < r$. That is the general method for distributions of functions of random variables, and the density formula $f(r) = (\text{arc length})/a^2$ is a first taste of the "coarea" idea of slicing a region into level sets. The same geometric reasoning is behind problems such as Buffon's needle and the distance between two random points in a square. Grimmett and Welsh, *Probability: An Introduction*, and Blitzstein and Hwang, *Introduction to Probability*, both develop this geometric approach to continuous random variables.

WORKED SOLUTION

STEP III 1992, Question 16

Probability and Statistics • Poisson Thinning

STEP III 1992, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s3-q16

The key idea

Once the total number of misprints is fixed, each one is an independent trial that I either spot or miss, so part (i) is a binomial probability. Part (ii) is a conditional probability, and the denominator, the chance that I spot exactly m misprints, needs us to add up over every possible number of missed misprints. That sum turns out to be an exponential series, which is what makes everything collapse to a neat Poisson form.

Solution

Throughout, write $q = 1 - p$. Let N be the total number of misprints in the issue, S the number I spot and $U = N - S$ the number I fail to spot. We are told that

$$\mathrm{P}(N = k) = \frac{\mathrm{e}^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \ldots$$

Part (i)

Given $N = m + n$, each of the $m + n$ misprints is spotted with probability p, independently of the others. So, conditional on $N = m + n$, the number spotted has the binomial distribution $\mathrm{B}(m + n, p)$. Any particular choice of m spotted and n missed misprints has probability $p^m q^n$, and there are $\binom{m+n}{m}$ such choices, so

$$\mathrm{P}(S = m \mid N = m + n) = \binom{m+n}{m} p^m (1-p)^n.$$

Part (ii)

We want $\mathrm{P}(U = n \mid S = m)$. The event $\{S = m,\ U = n\}$ is the same as $\{N = m+n,\ S = m\}$, so by part (i)

$$\begin{aligned}\mathrm{P}(S = m,\ U = n) &= \mathrm{P}(N = m+n)\,\mathrm{P}(S = m \mid N = m+n)\\ &= \frac{\mathrm{e}^{-\lambda}\lambda^{m+n}}{(m+n)!}\cdot\frac{(m+n)!}{m!\,n!}\,p^m q^n\\ &= \mathrm{e}^{-\lambda}\,\frac{(\lambda p)^m}{m!}\cdot\frac{(\lambda q)^n}{n!}.\end{aligned}$$

The factor $(m+n)!$ cancels, which is what makes this question work.

> **Ask yourself** How do we find $\mathrm{P}(S = m)$, when we don't know how many misprints there were?

If I spot exactly m, the number I missed could be any of $0, 1, 2, \ldots$, and these cases are mutually exclusive. So we sum the joint probability over all values j of U:

$$\begin{aligned}\mathrm{P}(S = m) &= \sum_{j=0}^{\infty}\mathrm{e}^{-\lambda}\,\frac{(\lambda p)^m}{m!}\cdot\frac{(\lambda q)^j}{j!} = \mathrm{e}^{-\lambda}\,\frac{(\lambda p)^m}{m!}\,\mathrm{e}^{\lambda q}\\ &= \frac{(\lambda p)^m}{m!}\,\mathrm{e}^{-\lambda p},\end{aligned}$$

using $\sum_{j\geq 0} x^j/j! = \mathrm{e}^x$ and $-\lambda + \lambda q = -\lambda p$. (So the number of misprints I spot is itself Poisson, with mean λp.) This is positive, so we may divide by it. Therefore

$$\begin{aligned}\mathrm{P}(U = n \mid S = m) &= \frac{\mathrm{P}(S = m,\ U = n)}{\mathrm{P}(S = m)} = \frac{\mathrm{e}^{-\lambda}\,\dfrac{(\lambda p)^m}{m!}\cdot\dfrac{(\lambda q)^n}{n!}}{\mathrm{e}^{-\lambda p}\,\dfrac{(\lambda p)^m}{m!}}\\ &= \frac{(\lambda q)^n}{n!}\,\mathrm{e}^{-\lambda+\lambda p} = \frac{(1-p)^n\lambda^n}{n!}\,\mathrm{e}^{-(1-p)\lambda},\end{aligned}$$

as required.

The answer doesn't depend on m at all: knowing how many misprints I spotted tells me nothing about how many I missed. The number missed is Poisson with mean $(1-p)\lambda$ whatever m is.

Checks

- Summing the final answer over $n = 0, 1, 2, \ldots$ gives $\mathrm{e}^{(1-p)\lambda}\mathrm{e}^{-(1-p)\lambda} = 1$, as a conditional distribution must.
- As $p \to 1$ I spot everything, and the answer tends to 1 for $n = 0$ and 0 for $n \geq 1$. Good.

- With $\lambda = 3$, $p = 0.6$, $m = 2$, $n = 1$ the formula gives $1.2\,\mathrm{e}^{-1.2} \approx 0.3614$. A simulation of 400 000 newspapers, keeping those in which exactly two misprints were spotted, gave 0.360.

Common slips

- Forgetting the binomial coefficient $\binom{m+n}{m}$ in part (i): the m spotted misprints can be any m of the $m + n$.
- Taking $\mathrm{P}(S = m)$ to be $\mathrm{P}(N = m)$, or to be the part (i) answer, rather than summing over all possible numbers of missed misprints.

Beyond STEP

This is the *thinning* (or splitting) property of the Poisson distribution: if a Poisson(λ) number of objects are each kept with probability p, the kept and discarded counts are *independent* Poisson variables with means λp and $\lambda(1 - p)$. The joint probability we found factorises into exactly these two pieces. Thinning is a key tool in the theory of Poisson processes, where it lets you split one stream of random events into independent sub-streams. Grimmett and Welsh, *Probability: An Introduction*, covers it, along with the generating-function methods that prove it in one line.

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